

number.<sup>5</sup> In this way, at any information set possessed by player 1 and given any strategies player 1 supposes her opponents might choose, she will always choose the appropriate part of the fixed strategy (Aa in this case) as a strictly dominant choice.

Of course, there is more to the game in figure 13.3(b) than this. First, nature picks between the two versions with probability .7 and .3. This is intended to model player 2's initial assessment whether player 1 is irrational or not. Player 1, at each of his information sets, knows which version was selected; that is, player 1 "knows" whether she is irrational or not. But player 2 at his information set isn't given this information.

The trick, then, is to model irrational behavior of a particular form by creating uncertainty about the payoffs of the player who might be irrational, where in states in which the player is modeled as irrational we simply make up payoffs that force (by reason of dominance) the player to act in the irrational manner. The player is told his own payoffs, so the player "knows" when to act irrationally. And other players are kept in the dark.<sup>6</sup>

This is how we model irrationality of the form: The player takes a particular course of action regardless of what other players do. Of course, we could have multiple forms of irrationality of this sort for a single player in a single game — if player 2 assessed probability .2 that player 1 irrationally plays Aa, .15 that player 1 irrationally plays Ad, .25 that player 1 irrationally plays D, and .5 that player 1 is "rational," then we would look at a game of incomplete information with four versions of the initial game. In three, player 1 would be given payoffs that force her to play according to the particular behavior being modeled, and in the fourth, player 1 would be given her original payoffs.

At the risk of beating the point to death, let me make a further observation about the nature of irrationality that we are modeling here. In this particular game, a rational player 1 will always pick D. Yet in the preceding paragraph, I spoke of a model where player 2 assesses probability .25 that player 1 irrationally plays D. The distinction here is "rationality"

<sup>5</sup> To check your understanding of this rule, consider how it would work if we fixed 1's strategy as Dd. You can check your answer by looking at the bottommost version of the basic game in figure 13.4(c).

<sup>6</sup> A more complicated situation is where there is uncertainty on the part of player 2 about the rationality of player 1, although player 3 does happen to know that player 1 acts rationally. That would be modeled as follows: There is initial uncertainty about player 1's payoffs, as in the example. Player 2 doesn't learn the resolution of that uncertainty and player 1 does, as in the example. And, if we suppose that player 3 knows player 1's "state of mind," player 3 has the uncertainty resolved whenever player 3 has an opportunity to take an action.

is meant to connote behavior that is in accord with the originally given payoffs, taking into account the predicted behavior of one's opponent(s). "Irrationality" connotes, for the time being, behavior that is taken without much thought or introspection. Now in this game it is fairly easy to see that introspection by player 1, given that player 1's payoffs are as originally described, leads to rational play of D. But we can and do want to be able to model situations where player 1 might take action D either because she engages in this sort of thought or because, say, she is averse to the letter A.

Hence, with caveats to be given momentarily, we see how to model the notion that one player assesses positive probability that another player might act in a particular fashion, regardless of what the rationally computed "best interests" of the second player might be. What about other forms of irrationality? Some are easily handled. For example, a player might play to maximize the expected difference between his payoff and that of his rival. It isn't clear that one should call this irrational; perhaps it is better described as play according to a different set of preferences (hence a different set of payoffs). But whatever you choose to call it, this is easily modeled with a game of incomplete information. To take a slightly more complex example, one player might assess positive probability that his opponent plays to maximize the minimum payoff the second player might receive. One could model this in a two-step procedure: First determine what sort of strategy this criterion would lead to; and then enter that into the model "behaviorally," as we have done above. In general, we try to reduce irrationality of any sort to "behavior," so that we can use the sort of model that has been described. In most cases, this will work. (See below for particularly crafty ways to do this.)

The model in figure 13.3(b) implicitly makes other assumptions that should be mentioned. Most importantly, it assumes that player 1 is aware of player 2's uncertainty about player 1's rationality. Player 1, knowing the structure of the game, is aware of the fact that player 2 assesses probability .3 that player 1 will play Aa as a strictly dominant strategy. Player 2 is aware of the fact that player 1 is aware of this, and so on. The model closes with player 2's uncertainty. We can say two things about this:

(a) This knowledge held by player 1 is only conceivably of use to her in the top half of the game in figure 13.3(b). That is, if nature picks the bottom initial node, player 1 will pick Aa no matter what she believes player 2 is doing. So when we say that player 1 is aware of player 2's uncertainties, we could rephrase this as: Player 1, if rational, is aware of 2's uncertainties. The model formally has the irrational type of

player 1 knowing this as well, but the knowledge is irrelevant given 1's payoffs.

(b) We could, of course, close the model further down the chain. That is, we could have player 2 uncertain about 1's rationality, with a rational player 1 uncertain about 2's assessment. Precisely the sort of model given in figure 13.1(b) would be employed if we closed the model with: Player 1's assessment concerning player 2's assessment concerning player 1's rationality is known to player 2, etc. The same comment applies here as applied (more generally) in the previous section: On theoretical grounds, we don't need to close the model after any finite number of stages. But on practical grounds, we will always look at models that do so.

Let us now change the payoffs and probabilities in figure 13.3 just a bit to make a couple of final points. Consider first the game in figure 13.4(a). The difference is in player 1's payoff if player 1 chooses A followed by  $\delta$  by player 2. The payoff 5 in this case means that player 1, playing "rationally" according to the given payoffs, would happily choose A if she thought that player 2 was going to respond with  $\delta$  with high probability. Of course, if player 2 is sure that player 1 is "rational" and will play d at the final node, then player 2 chooses  $\alpha$ . In which case 1 should rationally choose D. So the choice of A by player 1 seems irrational. So perhaps it scares player 2 into  $\delta$ . In which case the choice of A by 1 isn't irrational, and player 2 shouldn't play  $\delta$ . But then A is irrational, and so on.

How do we model this mess? By saying that player 2 isn't completely sure at the outset that player 1 is rational, 2 entertains doubts at the start whether player 1 will choose d at the final node. We could, for example, hypothesize that player 2 assesses probability .1 at the outset that player 1 will "irrationally" play Aa, building the model shown in figure 13.4(b), and proceed to analyze this.

The analysis of the game in figure 13.4(b) is a bit complex, and I don't think one can conclude that the "answer" obtained from equilibrium analysis is a clearly obvious way to play this game. But let us pursue this analysis and see if it makes sense in the end.

To begin, note that in any equilibrium, player 1 in the bottom part of the game will choose Aa by strict dominance. So we need only be concerned with the play of player 1 in the top half of the game. Next, note that the game has a pure strategy Nash equilibrium that isn't very plausible: In this equilibrium, player 1 plays Aa in the top half of the game (as well as in the bottom), and player 2 responds with  $\delta$ . It is clear why player 2 responds with  $\delta$ ; he is going to get -5 if he gives the move to player 1, regardless of which half of the game we are in. And then

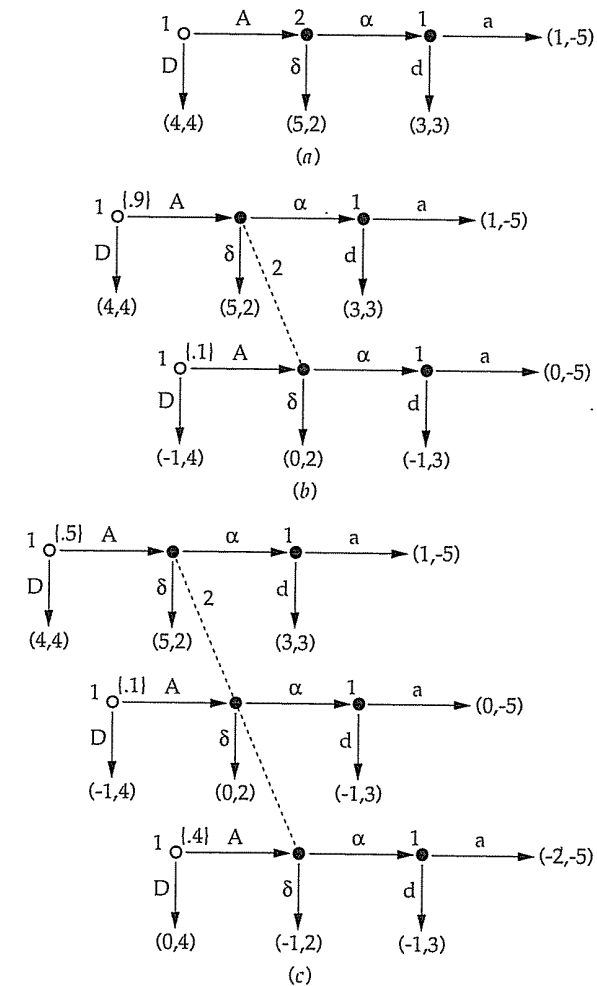


Figure 13.4. Modeling irrationality, continued.

player 1's strategy (in the top half) is optimal; given player 2 will choose  $\delta$ , A is better than D, and the choice of a over d is irrelevant. But this depends on player 1 choosing a in the top half of the game if given the

move, and it is implausible that given the move player 1 would actually do that. Why take 1 when she can have 3? Or, in the language of Nash refinements, the choice of  $\alpha$  in the top half of the game isn't sequentially rational. So we hereafter restrict attention to cases where player 1 plays  $d$  in the top half of the game (and, of course,  $\alpha$  in the bottom half).

Now suppose that player 2, given the move, assesses probability  $\mu$  that he is in the top part of the game. If he chooses  $\delta$ , she gets 2 for sure. Whereas if he chooses  $\alpha$ , he gets 3 with probability  $\mu$  and  $-5$  with probability  $1-\mu$ , for an expected utility of  $3\mu - 5(1-\mu) = 8\mu - 5$ . Comparing this with the 2 he gets for sure from  $\delta$ , player 2 will choose  $\delta$  for sure if  $\mu < 7/8$ , he will choose  $\alpha$  for sure if  $\mu > 7/8$ , and he is just indifferent (and can play any mixed strategy at all) if  $\mu = 7/8$ .

Suppose that the solution of the game is that he plays  $\delta$  for sure. Then player 1, realizing this, plays  $A$  in the top half of the game. This means that player 2 should assess probability .9 that he is in the top half of the game, given the move, which leads him to choose  $\alpha$  for sure. This isn't an equilibrium.

Suppose that the solution is that he plays  $\alpha$  for sure. Then player 1 will optimally choose  $D$  in the top half of the game. But then if player 2 is given the move, he knows he is in the bottom half of the game, and he chooses  $\delta$ . This isn't an equilibrium either.

So the only possible equilibrium (where 1 plays  $d$  in the top half of the game) is where player 2 randomizes between  $\alpha$  and  $\delta$ . For this to be optimal, he must assess probability  $7/8$  that he is in the top half of the tree, given the move. How can this happen? Suppose player 1 in the top half of the game randomizes between  $A$  and  $D$ , playing  $A$  with probability  $\pi$ . Then Bayes' rule tells us that player 2, given the move, assesses probability  $.9\pi / (.9\pi + .1)$  that he is in the top half of the tree. This equals  $7/8$  if  $\pi = 7/9$ .

And what would allow a randomized strategy for 1 in the top half of the tree to be optimal? Suppose that 2's strategy at his information set is to play  $\alpha$  with probability  $\phi$  and  $\delta$  with probability  $1-\phi$ . Then, in the top half of the tree, player 1 nets 4 for sure if she plays  $D$  and 5 with probability  $1-\phi$  or 3 with probability  $\phi$  if she plays  $A$ . If  $\phi = 1/2$ , player 1 in the top half is indifferent between  $A$  and  $D$ .

What we have here is: Assuming 1 plays  $d$  in the top half of the tree, there is no possibility of a pure strategy equilibrium. The only possible equilibrium is where player 1 mixes in the top half between  $A$  and  $D$  just enough so that player 2 is indifferent between  $\alpha$  and  $\delta$ , and player 2 mixes between  $\alpha$  and  $\delta$  so that player 1 is just indifferent between  $A$  and  $D$ . The appropriate mixing probabilities are computed above. You

can decide for yourself whether this qualifies as a "solution." But (always with the reasonable presumption that 1 will play  $d$  in the top half of the tree), it is the only possible candidate.

To help build your intuition, note that we get a very intuitive solution if we increase 2's initial assessment that 1 is irrational to .2, or if we decrease the  $-5$ s at the rightmost terminal nodes to  $-10$ s. With either of these changes, even if player 2 thinks that player 1 will choose  $A$  in the top half of the game for sure, player 2 thinks the odds of a bad outcome combined with the severity of that outcome if he picks  $\alpha$  are too great; he will pick  $\delta$  no matter what he thinks the rational sort of player 1 is doing. And then player 1 in the top half of the game clearly wants to give player 2 the opportunity to pick  $\delta$ .

Let us next pursue a further elaboration on the game of figure 13.3. We have supposed in figure 13.4(b) that player 2 is unsure about the character of player 1 and that player 1 is aware of this (and in the previous analysis tries to take advantage of this). Imagine that player 2 entertains the following more complex initial beliefs: There is probability .1, as before, that player 1 will irrationally choose  $Aa$ . Otherwise, player 1 plays according to the payoffs originally given. But there is an additional marginal probability .4 that player 1 is unaware of the fact that player 2 is uncertain of player 1's character. With this .4 probability, player 1 thinks that player 2 thinks the game is precisely as in figure 13.3(a). And with probability .5, player 1 is aware of player 2's uncertainty, including (now) the fact that player 2 entertains three possibilities for player 1: the .1 chance that 1 is irrational, the .4 chance that 1 is "rational but naive," and the .5 chance that 1 is "rational and sophisticated about all this." How would we model this?

One way to model this is to ask, How will the rationally naive player 1 act? This type of player 1 presumably views the situation as given by figure 13.4(a). If we conclude (and for purposes of argument, please so conclude) that player 1, thinking that 13.4(a) describes the game, finds  $D$  the obvious action to take, then we can build the sort of model shown in figure 13.4(c), where the topmost version of the initial game represents the possibility that player 1 is rational and sophisticated, the middle version represents the possibility of player 1 irrationally playing  $Aa$ , and the bottom version represents the possibility of a rationally naive player 1 — a player who is led by our assignment of payoffs to pick  $D$  as a dominant strategy.

This is how one might consider modeling players who are truly ignorant of the situation — who attach zero probability to certain things that other players think are in fact possible. We first look at the world through the eyes of this type of player to try to discern what this player's perspec-

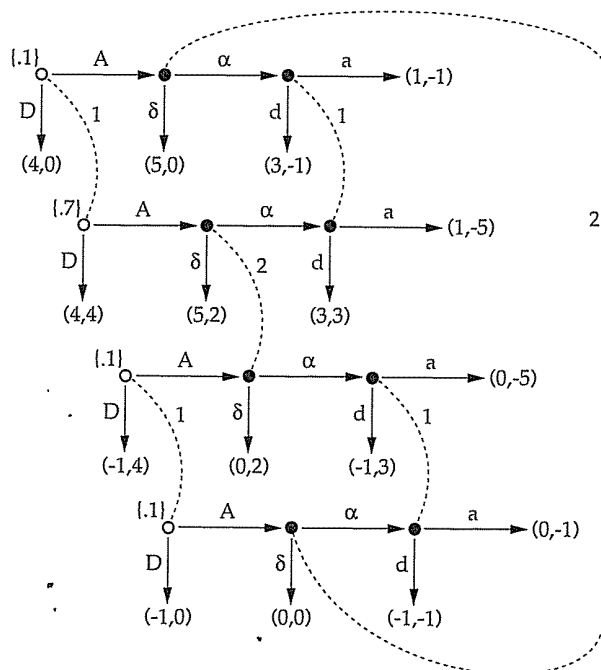


Figure 13.5. An extensive form game of incomplete information.

tive will lead to. And if we can find how that player will act, given his view, we then build that into the larger game in just the way that we build in simpler forms of irrationality.

This two-step (or, rather, multistep) procedure works in a large variety of situations that one wishes to model. But there is another, more complex way to get similar effects, which sometimes permits somewhat more general models. This is to go back to the basic model of games with incomplete information and consider games in which (a) players may have different prior distributions on the set of initial nodes, and (b) those priors may have different supports. That is, one player thinks that certain initial nodes have zero probability, while others think the game might well begin from one of those nodes. Can you see how one would use this sort of model to capture the story just told of the "rational but naive" player 1?

We have looked exclusively at examples where one player is uncertain about the "rationality" of a second, but the second has no questions about

the rationality of the first. This is done to keep matters simple. But it is completely unnecessary. One can easily model situations where player 1 is unsure whether player 2 is rational and, at the same time, player 2 is uncertain about the rationality of player 1. To test your ability to disentangle this sort of model, consider the game depicted in figure 13.5 and try to describe just what is being modeled here. (See problem 5 for some more-pointed questions about this game.)

Note that what is called "rational naivety" is another form of irrationality or, at least, limited rationality. It is a type of player who isn't fully clued in to what might be going on — a type of player who isn't aware of all the possibilities entertained by another player.

### 13.4. More on refinements: Complete theories

*This entire section is optional. Nothing that follows in the book builds on it, and it may prove to be difficult reading for a first time through the subject.*

Consider for a final time the game 12.1(o), which is reproduced here as figure 13.6. What we know about this game is easily summarized: It is dominance solvable. U strictly dominates R, and once R is removed, l weakly dominates r, leading to L-l as the solution. This is, of course, a Nash equilibrium. But U-r is also a Nash equilibrium; U-r is even sequential. To support the U-r equilibrium, 2's beliefs at her out-of-equilibrium information set must put weight 1/4 or more on player 1 having picked R instead of L.

Are such beliefs at all plausible? We gave in chapter 12 two arguments, one for and one against.

(1) Against these beliefs was the argument that R is dominated by U while L is not. Put differently, given that 1 has the option U, she has no business at all playing R; she can't possibly do better with R than with U. But there is a reason she might play L, since there is the chance (if 2 responds with l) that she will do better with L than with U. Hence 2 "ought" to conclude, given the move, that L was probably 1's choice, which mandates a response of l.

(2) In favor of these beliefs was Myerson's properness criterion. One begins by taking the perspective that out-of-equilibrium actions are "mistakes" —

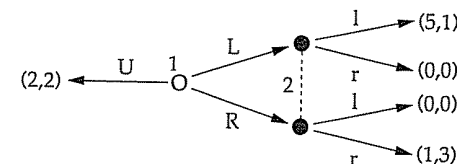


Figure 13.6. An extensive form game.

trembles of the hand. And to this one adds the notion that the relative chance of making a mistake should be related to how severe the mistake will be; players are meant to be more attentive to avoiding mistakes the consequences of which are more substantial. If player 2 is playing  $r$ , then  $L$  by mistake is a bigger mistake than  $R$  by mistake. Hence player 2 should conclude, given the move, that 1 is more likely to have chosen  $R$  by mistake. And this leads to  $r$  as the rational action.<sup>e</sup>

Now these two arguments are not of equal substance. For one thing, the first argument pins down the answer definitively. If you accept this argument,  $L$  is the only possible answer. The second argument is more circular, holding that  $U$ - $r$  can't be disqualified as a "solution" if it is a solution. That is to say, in the second argument,  $L$  is a mistake in the first place only if one accepts the premise that 2 will play  $r$ .

I am more interested, though, in a particular strength of the second argument. That strength is that there is a "story" about why one might see a deviation from the supposed solution, a story that is used to guide what is considered a reasonable conjecture out-of-equilibrium. This story is quite explicit: Deviations from a solution are the result of mistakes (the story of the trembling hand), and, moreover, bigger mistakes are less likely than smaller mistakes.

There isn't a similar story for the first argument, but it isn't hard to create one. For example, note that there is a single Nash equilibrium in which player 1 gives the move to player 2:  $L$ - $l$ . Suppose that we tell the following story. The players in this game are randomly matched. Most of the players come from a society in which it is "understood" that the solution is what the solution is. But a sprinkling of the players come from other societies, where other "solutions" might prevail. Whenever an out-of-equilibrium action is observed, the most likely explanation, if it fits at all, is that this action was taken by someone from another society in which the game has another conventional solution. If no explanation of this sort can be found, then the next most likely explanation is that the out-of-equilibrium action is the result of a mistaken action.

With this story, since  $L$ - $l$  is the only possible "other" conventional solution for this game, when player 2 is put on the move, he assesses as (conditionally) most likely the possibility that this is an action taken by someone from a society where  $L$ - $l$  is the convention. But this then destroys the possibility of  $U$ - $r$  being a solution, by the argument given.

A different story that one can tell for the first argument is that out-of-equilibrium actions are "tests" of out-of-equilibrium responses. Players are assumed to be extremely certain what will happen if they follow convention in situations where there is conventional wisdom; they are pretty sure as well what happens if they deviate. But, just to check, players sometime experiment with deviations from conventional wisdom. These experiments are not picked haphazardly, however. Players are more likely to experiment with deviations

<sup>e</sup> Myerson's notion of properness goes a good deal further than this. It requires that a smaller mistake be asymptotically infinitely more likely than a larger mistake. We don't need quite that much power here.

that have some chance of beating what (they are virtually sure) will happen if they follow conventional wisdom than with deviations that have no chance of doing so. If the solution of this game is  $U$ - $r$ , if player 2 believes that player 1 deviates from  $U$  on occasion to test the conventional wisdom that 2 will play  $r$ , and if player 2 believes that player 1 is more likely to experiment with  $L$  than with  $R$ , since  $L$  holds some possibility of being better than  $U$ , then player 2 will not choose  $r$  given the move. And then  $U$ - $r$  can't be the solution.

What is the point of these stories? In refining the notion of a Nash equilibrium, we are necessarily thinking through what is relatively more likely to have happened (and to happen subsequently) if the "solution" that is proposed is falsified. In any real application, a given "solution" describes what is pretty certain to happen. But it would be foolish (or, at least, heroic) to be completely certain that a given solution applies. There are presumably reasons why "deviations" occur. And if, in a particular context, we understand something about the cause of deviations, we are better able to argue what inferences players will make when confronted with a deviation. This, then, informs the sort of refinement we might be willing to accept.

The term "complete theory" is meant to capture this sort of logic. By this term is meant a theory that rules nothing out a priori; although it may lead to certain things being assessed as very, very unlikely, it nonetheless offers some explanation for why the very unlikely might happen. Trembling-hand perfection is based on the complete theory of mistakes made in the course of play. Alternative complete theories can be built out of stories of randomly matched players from various societies or experimentation by players to check conventional wisdom.<sup>f</sup>

With this as prologue, let us reexamine the logic of subgame perfection and sequential equilibrium. Consider the game in figure 13.7(a), a game of complete and perfect information. By backwards induction, player 1 will choose  $a$  at his second node, hence 2 will choose  $\alpha$ , and hence 1 will choose  $A$ . This is the unique subgame perfect equilibrium. But  $D$ - $\delta$  is another Nash equilibrium. This is an equilibrium, moreover, that is fairly "safe" for player 2. What is the argument one would make against it being the solution of this game?

If player 2 subscribes to the theory that deviations from a given solution are most likely to be simple mistakes, and if he further subscribes to the theory (fairly explicit in trembling-hand perfection) that a mistake at one information set has no bearing on the likelihood of subsequent mistakes at subsequent information sets, then, if  $A$  is played by 1, he should shrug his shoulders and assume that player 1 will (probably) choose action  $a$  given another opportunity to choose, which then rules out  $\delta$  as a reasonable choice for him.

But let me suggest a different thought process for player 2 that leads to a very different analysis. Suppose that he thinks there is a small chance that his opponent is out to get him — say one chance in one million — but enough chance so this possibility, whenever it is consistent with what he

<sup>f</sup> This is the subject of current research, by Suehiro (1989) in the first case and by Fudenberg and Kreps (1989) in the second.

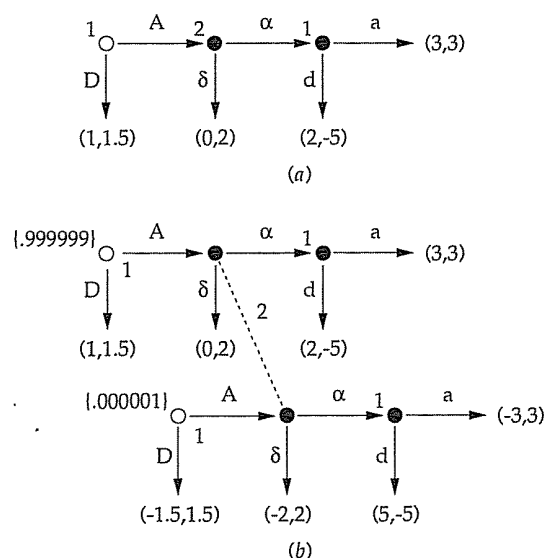


Figure 13.7. Two extensive form games.

observes, is deemed more likely than the chance that his opponent has made an inexplicable error. We can think through the consequences of this sort of thought process by looking at a game of incomplete information, such as the game in figure 13.7(b). Note what we have done here. We've taken the game in figure 13.7(a) and imagined that nature initially selects one of two versions. In the first, player 1's "motivations" are captured by the originally given payoffs. But in the second, which has prior probability .000001, player 1's payoff is simply minus the payoff of player 2; player 1 is "out to get" player 2.

Now suppose that player 2 believes the solution to the game is "probably" that player 1 plays D, player 2 if given the move plays  $\delta$ , and player 1 if given the move by 2 will intend to play a. This is "probably" the solution in the sense that it is meant to describe 1's actions in the very likely case that player 1 isn't out to get player 2. But if player 1 is out to get player 2 (which is very unlikely), then player 1 will play A and, given the chance, d.

If this is how player 2 views the game, his view passes the equilibrium test of internal consistency. In the first place, if he is given the move by player 1, and if he thinks that the one-in-a-million chance that 1 is out to get him is more likely than the chance that 1 has simply erred, then he will assess substantial a posteriori probability that he faces a malicious player 1. (More precisely, he will assess probability exceeding one-half that the bottom part of the game pertains.) Thus his ex post expectations are that player 1, given the

chance to move again, is now more likely to choose d than a. And so he prudently (and optimally) selects  $\delta$ . Moreover, assuming player 1 understands all this, D in the top half and A in the bottom are optimal moves for her.

Of course, the game in figure 13.7(b) possesses another equilibrium. In this equilibrium, player 1 plays A in both halves of the game, player 2 plays  $\alpha$ , and player 1 plays a in the top half and d in the bottom half. Note why it is that player 2 is content to play  $\alpha$  in this case. Since player 1 is meant to play A in both halves of the tree, player 2's assessment upon being given the move is that he is in the top half of the tree with probability .999999. There is a one-in-a-million chance that, if he plays  $\alpha$ , 1 will respond with d. But this isn't nearly a large enough chance to make  $\delta$  a better choice.

I do not mean to say that the equilibrium in which 2 plays  $\delta$  is the obvious solution. But in this game of incomplete information, one would be hard pressed to dismiss the 2-plays- $\delta$  equilibrium as something unreasonable. And so, if this sort of model of incomplete information fleshes out the thought process of player 2 in the game in 13.7(a) concerning where deviations come from, then I don't think one can so easily dismiss the 2-plays- $\delta$  equilibrium in 13.7(a) on grounds that it is subgame imperfect. It comes down to What sort of theory does player 2 have about the source of deviations when/if they occur? The "story" formalized by 13.7(b) is to my mind neither more nor less outrageous than the story of hands that tremble independently at different information sets. And while independently trembling hands lead to refinements such as subgame perfection and sequential equilibrium, other stories can lead to something less.

The sophisticated reader may object to the argument just given on the following grounds: The strategy specified at any information set is meant to be one's prediction about what will happen if that information set is reached. If an information set is off the path of play, then the strategy specified there should be conditional on the deviation having occurred. And as long as the right (conditional) strategy is there, then we can have no grounds for doubting that players should anticipate the play of this strategy. One cannot maintain in the analysis of the game of 13.7(a) that the play of action a at 1's second information set is expected and that player 2 is nonetheless reasonable in the choice of  $\delta$  when it is his move.

To this objection I plead guilty, but only on grounds of expositional ease. What I want to claim is that in the game in figure 13.7(a) it is not so easy to reject the subgame imperfect equilibrium where player 1 chooses D, 2 chooses  $\delta$ , and 1, given the choice again, plays, say, a with probability .1 and d with probability .9. Rejecting this equilibrium is not so easy because the source of "deviations" might be an a priori small chance of a malicious player 1; where the chance that player 1 plays A because she is malicious is nine times more likely than the chance that player 1 plays A by mistake. (The nine times then explains the .1 probability, if we consider a still more complex game of incomplete information in which nature not only gave us a chance of a malicious player 1 but also a nine times smaller chance of a player 1 who mistakenly plays A.)

In this sort of theory, subgame perfection and sequential equilibrium come to grief in the requirement of sequential rationality (or backwards

induction) applied at information sets that follow a deviation from the equilibrium. At least, this is so for the actions of a given player taken subsequent to his own deviation, when one explains the first deviation by the possibility that the player has radically different payoffs from those specified as being likely *ex ante*.

What precisely do we get as a reasonable refinement concept, if we use this sort of story to explain unlikely deviations from equilibrium play when they occur? I will not attempt to give a precise answer here. For one thing, we are now well past what is reasonable for a first course on this subject and to explore this question fully would take us right to the frontiers of current research and beyond. And more importantly, this question doesn't have an answer that I, at least, find fully satisfactory. But let me give a partial answer for which the intuition is not hard to develop. In any game, any equilibrium outcome that is trembling-hand perfect in the normal form will certainly pass if this sort of story is used to explain deviations. Once a player begins to deviate, with this sort of story one can have any *ex post* beliefs about further deviations by the player. In other words, players tremble at the level of entire strategies, which is just normal form perfection. Of course, perfection in the normal form doesn't imply subgame perfection or sequentiality — that is the whole point.<sup>g</sup>

One needn't stop here, however. One can spin "stories" about the causes or sources of out-of-equilibrium play that rationalize equilibrium outcomes that are not even perfect in the normal form. For example, imagine stories as above, where out-of-equilibrium play is manifestation of very different payoffs for the players (so that sequential rationality out-of-equilibrium is not a tenable requirement), and where one adds the possibility that different players' payoffs are correlated. (Why might this be true? Insofar as players come from similar backgrounds, if one player acts in "irrational" fashion, that could indicate that his or her peers are more likely to do so.) Then if we see one player deviating from the "solution," we might increase the odds that others will not conform to the dictates of sequential rationality. Indeed, if we tell stories (admittedly farfetched) about how players don't necessarily have the best possible information about their own payoffs, so that they look to others' behavior for some indication as to what they should do, then virtually any Nash equilibrium can be rationalized.<sup>h</sup>

Or, without telling stories about small chances that players have radically different payoffs, one can tell different stories that lead to correlated conjectures about out-of-equilibrium play. Recall the stories sketched above that ran: A deviation from expected play is taken as an indication that the player deviating is playing some "other" equilibrium. Needless to say, we

<sup>g</sup> I am careful here to say that any equilibrium outcome that is trembling-hand perfect in the normal form can be rationalized. In the game in figure 13.7(a), if we are being precise, we would say that what can be rationalized by this story is  $\delta$  combined with D and a mixture of d and a that puts substantial weight on d. But if we move to the normal form, we don't really distinguish between 1's strategies Da and Dd. So, from the perspective of normal form perfection, it is D- $\delta$  that is perfect.

<sup>h</sup> To see these two stories fleshed out, see Fudenberg, Kreps, and Levine (1988) on the subjects of personal types and general elaborations, respectively.

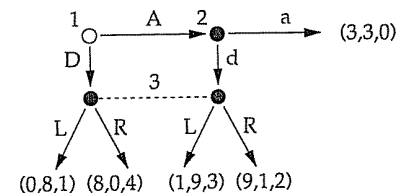


Figure 13.8. An extensive form game.

would expect to see this player continuing to follow the dictates of this other equilibrium, at least until this player gets some indication that his theory of how to play is "wrong." But why should the deviator necessarily be the party with an incorrect theory? In a more-than-two player game, if you see one player deviating from what you suppose is the equilibrium, then perhaps you should entertain the possibility that *your* theory of how people play this game is wrong, and (in consequence) others will play in ways you didn't predict.<sup>i</sup>

It is important in all this that one maintains the hypothesis that players have consensus conjectures concerning the play of others, both in and out of equilibrium. Or, rather, this hypothesis is important if the solution concept to be employed is Nash equilibrium. To take a simple example, consider the game in figure 13.8.

You can quickly verify: No matter what action player 3 takes at her information set, one of players 1 or 2 will choose to give her the move. Hence A-a is not a Nash equilibrium outcome. But if player 1 thinks player 2 will choose a and that player 3 will probably choose L, then 1 prefers A to D. And if player 2 thinks player 1 will choose A and player 3 will probably choose R, then he prefers a. If we allow players to have divergent opinions about what might happen out of equilibrium, we can get "solutions" that aren't even Nash. We saw a related phenomenon in chapter 12, in the example of a game with a single Nash equilibrium that (I assert) is rarely played; "partial solutions" to a given game may be present, and these needn't pass the test of Nash. The point here is that if there is consensus only on what happens along the path of play, then we have only a partial solution, and the Nash criterion is no longer a necessary condition.

So what is the bottom line for refinements of Nash equilibrium? The philosophy espoused here can be paraphrased as: The bottom line is that there is no simple bottom line. In refining Nash equilibria, one is speculating about what is supposed to happen after there is evidence that the going theory is incorrect. Depending on why you (and the players involved) think one sees deviations from a priori likely play and what this portends about future play, one supports or diminishes the relevance of particular formal refinements. Since the appropriate story is apt to be specific to the context (and, especially,

<sup>i</sup> See Hillas (1987) and Suehiro (1989).



to depend on why one thinks that there is a "solution" in the first place), it seems fruitless to try to choose among refinements in the abstract.

The usual rejoinder to this philosophy is that the implicit "criticism" is misplaced. The problem is not with the solution concept, but instead with the model of the game that is used. If there are reasons why unexpected actions might be taken, then the thing to do is to include those reasons within the model, using incomplete information. Once these things are modeled directly, we don't need refinements tuned to particular stories of why unexpected actions take place. And then there is no reason not to think that a "universal" solution concept might be available.

I take the main point of the rejoinder to be that the way to understand the effects of various hypothesized causes of unexpected actions is to build more complete models using incomplete information. I fully agree; that is precisely what we've done here. However "completing the model" in an application may well be impractical, and so it may be of practical value to have handy refinements that are well tuned to various sorts of completions. As a practical matter, one may need to select the refinement used with a view towards what has been omitted from the model for reasons of tractability.

### 13.5. Bibliographic notes

The classic paper that introduces the notion of a game of incomplete information is Harsanyi (1967-1968). Harsanyi deals primarily with models where players begin with commonly held priors. This is certainly as befits the author of the so-called Harsanyi doctrine that priors should always be held in common; in fact this is the application for which that doctrine was first formulated. The notion of a game with incomplete information where players have different priors, or even priors that have different supports, is attributed by Harsanyi to Selten, although I do not have a reference I can give. For the question of how to "close" models with incomplete information but with infinite regresses of assessments, the original reference is Mertens and Zamir (1985). A recent paper by Brandenburger and Dekel (1985) may be relatively more accessible. (The word relative is important to stress; this is a very mathematical subject, and no treatment will be accessible unless your math skills are very high.)

I think the best way to get a feel for how models of games of incomplete information are constructed and used is to look at applications. In section 13.2, we discussed the example of entry deterrence, which is an excellent place to begin. See the seminal work of Milgrom and Roberts (1982) on this. Many other applications fall under the general rubric of market signaling; see chapter 17 for references. Problems in mechanism design also use games of incomplete information; see the references given in chapter 18.

The topic of entry deterrence in general is summarized in Tirole (1988).

As for the philosophy of refinements of Nash equilibrium, please understand that this is a highly contentious issue, and the philosophy I've espoused is far from the consensus point of view. The point of view taken here is further developed in Fudenberg and Kreps (1989), Kreps (1989), and Kreps and Ramey (1987). Fudenberg, Kreps, and Levine (1988) develops the appropriate "mathematics" for stories of radically different payoffs with small probability (and the connection to trembling-hand perfection in the normal form), although the results there are couched in somewhat different language. Reny (1988) independently arrives at this conclusion, in language more directly applicable to the problem as posed here. McLennan (1985), Hillas (1987) and, especially from the point of view of complete theories, Suehiro (1989) develop stories about refinements based on the notion that players may have mistaken theories about how to play. Of course, Selten (1975) and Myerson (1978) began this subject with the development of the "deviations-as-trembles" story. Finally, Kohlberg and Mertens (1986) and Mertens (1987) present a very different view of the entire subject.

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### 13.6. Problems

- 1. (a) Consider the version of the beer-quizche game given in figure 13.1(a). What is its normal form, if we think of it as a two-player game?

*Parts (b), (c), and (d) to be done precisely require you to know the formal definition of Myerson's proper equilibrium, something we have refrained from giving here. You may wish to find that definition in one of the references given here; alternatively, you can get the gist of this problem using only the informal characterization we gave, which states that a player should be relatively much less likely to make a bad mistake than one that is not so bad.*

(b) Following remarks we made in subsection 12.7.4, a "natural" sequential equilibrium for this game is for player 1 to have beer for breakfast whether she is a coward or brave, for player 2 to avoid a fight if he observes 1 having beer for breakfast, and for him to pick a fight if he observes 1 having quizche for breakfast. Using the two-player game normal form you created in part (a) is this equilibrium proper in the sense of Myerson? If we viewed this as a three-player game (the three players being "1 the coward," "1 brave," and 2), would this equilibrium be proper? (To answer

this, you will have to consider what the payoffs are to "1 the coward" if nature selects the bottom half of the game to start. Does your choice of payoffs for "1 the coward" in the bottom half matter to the answer?)

(c) Suppose we rescaled the payoffs of player 1 in the top half of the tree only by multiplying each payoff by a factor of 50. That is, the payoffs if 1 is a coward, 1 has quizche, and 2 fights become (50,1). Note that we don't rescale 2's payoffs, and we don't change 1's payoffs in the bottom half of the tree. Does this sort of rescaling change in any way the strategic nature of the game? (There is no right or wrong answer to this question. The quality of your answer turns on how well you defend the position you take and not necessarily on which position you decide to defend.) After completing this rescaling is the equilibrium described in part (b) proper in the sense of Myerson? (To answer this, you have to decide whether you think of this as a two- or as a three-player game.)

(d) Can you find an equilibrium of this game that gives the same outcome as the equilibrium in part 1, such that no matter how we rescale the payoffs of player 1 in either half of the tree, the equilibrium is and remains proper in the sense of Myerson, viewing this as a two-person game? (When you have the answer to this question, you have an important example to carry around in your head if ever you study stability in the sense of Kohlberg and Mertens.)

*Problem 2 gives a standard sort of story on entry deterrence based on an action taken by the incumbent monopolist that makes her more aggressive postentry. There are many stories of this sort; for others, see Tirole (1988).*

- 2. Consider an industry with an incumbent monopolist, who is currently producing, and a potential entrant, who might or might not choose to enter.

There is a range of possible production technologies for producing this good, which differ in the level of marginal and fixed costs. Each technology is a constant marginal cost technology with marginal cost somewhere between 1 and 3 per unit. In the technology with marginal cost  $c$  (from the range from 1 to 3), fixed costs are  $15 - 4c$ . That is, lower marginal costs bring with them higher fixed costs. Any firm in this industry may choose whatever technology it wishes to employ, but it can make this choice once only. The choice of technology of any firm is known to all potential and actual competitors.

We imagine that there are two periods of production. In each, demand is given by  $D(P) = 9 - P$ . In the first period, the incumbent monopolist

is definitely a monopolist. The monopolist begins this period by selecting her production technology and then producing. In the second period, the entrant decides whether to enter at all. If he does enter, he picks his production technology. And then the two compete Cournot-style, simultaneously picking quantities. If he does not enter, then the monopolist retains her monopoly and picks the quantity she wishes to sell. We assume that the entrant will enter only if he believes he will make positive profits, and the incumbent monopolist acts in a way that maximizes her sum of profits in the two periods. (Adding a discount factor only complicates the arithmetic.)

(a) Suppose there was no threat of entry. What technology would the monopolist select?

(b) Suppose the monopolist selects the technology you got as the answer to part (a). Will the entrant enter? If so, with which technology?

(c) If the monopolist, in the first period, picks a technology that discourages the entrant from entering in the second period, we say that the monopolist has blockaded entry. Is there a technology the monopolist could pick in the first period that would blockade entry? If so, what range of technologies accomplishes this?

(d) Given the threat of entry, what is the optimal course of action for the monopolist? How does the entrant respond to this? (What solution concept are you using?)

■ 3. Analyze the game depicted in figure 13.3(b). Assuming this model describes the situation, do you think you can clearly predict what will happen?

■ 4. (a) Contrast the analysis of the entry deterrence game in section 13.2 with the analysis of the game in figure 13.4(b). In what senses are the two games similar? How are they different?

(b) Analyze the game depicted in figure 13.4(c). Assuming this model describes the situation, do you think you can make any clear prediction concerning what will happen?

■ 5. Consider the extensive form game depicted in figure 13.5.

(a) Would it make any difference to the analysis of this game if we removed the information set of player 2 that links the topmost and bottommost pieces of the tree? If so, what difference? If not, are there information sets of player 1 that are similar candidates for removal?

(b) Analyze the prior distribution over the four possible initial nodes. Suppose we say that player 1 is "rational" if one of the top two of these initial nodes prevails, and player 2 is "rational" if one of the middle two initial nodes prevails. Is there correlation in the rationality of the two players? Is the correlation positive or negative?

(c) Analyze this game. Assuming this model describes a particular situation, do you think you can clearly predict what will happen?

## *chapter fourteen*

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### *Repeated play: Cooperation and reputation*

I begin this chapter with an apology. More than in any other chapter, in this one the text will constantly be shifting between larger and smaller type. The reader who consumes only the larger type should be able to get the flavor of the basic ideas here, but this is a subject that from start to end is best appreciated by seeing at least a bit of how things are done formally and properly. Rather than reserve all the technical material for the back, which would mean repeating a lot of what was said in the front, I have elected to frustrate the nontechnical reader who will often be paging through to find the next group of large type. I hope this frustration will induce some readers to try to consume the entire chapter, which should be possible *if* you understand appendix 2. But to those for whom it doesn't, apologies.

#### *14.1. The prisoners' dilemma*

The one game from figure 12.1 that has yet to be discussed is the game in figure 12.1(m), reproduced here as figure 14.1. This game is called the *prisoners' dilemma*. The story that goes with it runs as follows. Two individuals, player 1 and player 2, are being held by the police in separate cells. The police know that the two (together) committed a crime but lack sufficient evidence to convict. So the police offer to each of them separately the following deal: Each is asked to implicate her partner. If neither does so, then each gets no time in jail. If each implicates the other, then each goes to jail for a while. If one implicates the other but is not implicated, the first gets off (and gets a greater share of the loot), while the one implicated goes to jail for a longer period of time. Each player ranks the four possible outcomes: It is best to implicate your associate and not be implicated (so you get the greater share of the loot). Next best

|          |    | Player 2 |      |
|----------|----|----------|------|
|          |    | t1       | t2   |
| Player 1 | s1 | 5,5      | -3,8 |
|          | s2 | 8,-3     | 0,0  |

Figure 14.1. The prisoners' dilemma.

is not to implicate and not to be implicated (to avoid jail). Then comes implicating and being implicated (a short jail term). And worst of all is to be implicated while not implicating your associate (a long jail term). If we let  $s1$  and  $t1$  be the "do not implicate" strategies and  $s2$  and  $t2$  be the "implicate" strategies, then the normal form game depicted in figure 14.1 is consistent with this story. (Of course, by giving payoffs we have been much more specific about the relative utilities of the various outcomes). In the sequel, we will refer to the strategies  $s1$  and  $t1$  as *cooperation* (with each other and not with the police), while for  $s2$  and  $t2$ , terms synonymous with *noncooperation* will be used.

While the story is fanciful, the basic structure of options and payoffs that characterize this game occur over and over in economics. In this basic structure players can cooperate to greater or to lesser extent. If one player unilaterally decreases the level of her cooperation, she benefits and her rival is made worse off. But if both decrease their level of cooperation equally, both are made worse off. Consider, for example, the case of Cournot duopolists, each (independently) choosing a quantity level to bring to market. Typically, if one firm increases its production (which is a less cooperative strategy), its profits increase, at least for a while, and the profits of its rival decrease. But (past the monopoly level of output) if both firms increase their levels of output, both do worse. We'll get back to this particular economic application in section 14.4.

The analysis of the prisoners' dilemma game using noncooperative game theory seems quite simple:  $s2$  strictly dominates  $s1$ , and  $t2$  strictly dominates  $t1$ . So the solution by application of one round of strict dominance is  $s2$ - $t2$ . Of course, this is also the unique Nash equilibrium. It does seem rather a shame that this happens, because this outcome is quite inefficient. (That is, there is an outcome, namely  $s1$ - $t1$ , that is better for both sides.) It may not be surprising that competition of this sort can lead to inefficient outcomes, but you might still wonder how this particular inefficiency can be avoided.

## 14.2. Repeating games can yield cooperation: The folk theorem

Imagine that the jail sentence involved is quite short (it was only a petty theft), and the two prisoners will be engaged in this circumstance over and over again. That is, imagine that the players play the game not once but repeatedly, and payoffs are added together. This situation has been the subject of experiments. College sophomores (typically) are matched against one another; instead of playing the game once, they are told that they will play the game many times, with no fixed horizon in view. As long as they don't see a horizon looming, the empirical evidence is that such subjects often cooperate (play  $s1$ - $t1$ ). The explanation that is typically given for this lack of short-run optimality is that in repeated play there is the long run to worry about as well. If, say, player 1 tries to take advantage of player 2 in the short run by playing  $s2$  instead of  $s1$ , player 2 might react to this by subsequently playing  $t2$  instead of  $t1$ . Player 1 may feel there is less to gain by optimizing in the short run than to lose in the long run and so chooses to cooperate. If player 2 feels the same way, cooperation ensues. Note well that this is not cooperation borne of altruism or fondness for one's fellow player. This is cooperation arising from a self-interested calculation of the benefits and losses that may accrue from "polite" behavior.

Formal analysis of this point involves what are called repeated games. Imagine two players playing the prisoners' dilemma game over and over again. We could imagine that they play forever and their payoffs for the entire string of games are just the average of the payoffs they get in each stage. That is, if player 1's payoffs in a sequence of rounds are given by  $\{u_1(1), u_1(2), u_1(3), \dots\}$ , where the subscript refers to the player's identity and the argument to the round number, then we could assign overall payoff  $\lim_{t \rightarrow \infty} (1/t) \sum_{i=1}^t u_1(i)$  to this sequence.<sup>a</sup> Or we could imagine that they play forever, and they value a stream of payoffs according to the discounted sum of the payoffs. That is, there is some number  $\alpha \in (0, 1)$  and the overall value to player 1 of the sequence of payoffs  $\{u_1(1), u_1(2), u_1(3), \dots\}$  is  $\sum_{i=1}^{\infty} \alpha^{i-1} u_1(i)$ . Or, to take a third possibility, we could imagine that the game is played a finite but indefinite number of times. For example, we could imagine that after each round of play a

<sup>a</sup> The very careful reader may wonder what happens if the limit doesn't exist. To avoid this difficulty, formal analysis of this problem is based on the criterion that the stream of rewards  $\{u_1(1), u_1(2), u_1(3), \dots\}$  is better than the stream of rewards  $\{u'_1(1), u'_1(2), u'_1(3), \dots\}$  if  $\liminf_t \sum_{i=1}^t [u_1(i) - u'_1(i)] > 0$ .

roulette wheel is spun. If it comes up 0 or 00 (2 chances in 38) we quit; otherwise we play another round and spin the wheel another time, and so on. Being a bit more general, we let  $1 - q$  be the probability that we stop after each round and  $q$  be the probability that we continue for at least one more round. Then, the expected value of the sequence of payoffs  $\{u_1(1), u_1(2), u_1(3), \dots\}$  for player 1 is  $\sum_{i=1}^{\infty} q^{i-1} u_1(i)$ . For  $q \in (0, 1)$ , assuming players maximize their expected sums of payoffs, this is mathematically equivalent to the formulation with discounting where  $q$  takes the place of  $\alpha$ . These are all variations of the general notion of a *supergame*, wherein a particular game called the *stage game* is played over and over by two (or more) players. There is a rich literature concerning supergames for both the first and the second two (equivalent) formulations of the problem. We will deal in this chapter primarily with the second two formulations, referring to the sums that give a player's payoff for an entire game as the *expected/discounted value* of the stream of payoffs.

What happens if we go from a one-round prisoners' dilemma to this sort of repeated formulation? Just what we wanted: Cooperation becomes an equilibrium outcome. Suppose that player 1 announces she will play  $s_1$  as long as player 2 plays  $t_1$ . But if ever player 2 plays  $t_2$ , player 1 will forever after play  $s_2$ . Player 2, if he believes this, finds it easy to formulate a best response. As long as  $t_1$  is played, a payoff of 5 each period will result. This gives a stream of payoffs of 5 extending into the future, which has expected/discounted value  $5/(1-q)$ . But if player 2 begins with  $t_2$  this round, he will get 8 immediately and (at most) 0 forever after. For reasonable values of  $q$  (in this case for anything over  $3/8$ ), player 2 has no incentive to play  $t_2$  in this round. And, a bit of arguing will show, for these values of  $q$  player 2 *never* has any incentive to play  $t_2$ .<sup>b</sup> To have an equilibrium we must also give player 1 the incentive to play the strategy above, but this is easily accomplished. Have player 2 announce that he will play  $t_1$  as long as  $s_1$  is played, but he will play  $t_2$  ever after, if ever player 1 is foolish enough to try  $s_2$ .

The point is that these two strategies by player 1 and player 2 constitute a Nash equilibrium in the repeated game. Each is a best response to the other. And together they give the cooperative outcome.<sup>c</sup>

There are (at least) four problems with this simple story.

<sup>b</sup> The easiest formal proof shows that each strategy is unimprovable in the sense of appendix A2 when played against the other. See the end of the next subsection for a slightly more complex example worked out formally.

<sup>c</sup> These two strategies do not, however, constitute a subgame perfect equilibrium. Starting from the point where one player, say player 1, has just played noncooperatively, player 2 will respond with noncooperation. But according to the strategies given, player 1 will not revert to permanent noncooperation for one round more even though she knows that he is about

### (1) Too many equilibrium outcomes

We have produced one Nash equilibrium for this repeated game. But there are many, many others. Suppose, for example, that player 1 announces the following strategy. She will alternate between  $s_1$  and  $s_2$  as long as player 2 plays  $t_1$  all the time. But if ever player 2 deviates and plays  $t_2$ , player 1 will play  $s_2$  forever after. The effect of this is to offer player 2 the choice of getting payoffs of 5 alternating with  $-3$  forever, or (at best) taking one 8 and 0s afterwards. For high enough  $q$ , the "cooperative" response is better. This allows us to construct an equilibrium as long as player 2 issues a threat that keeps player 1 honest; e.g., player 2 says that he will play  $t_1$  as long as player 1 alternates  $s_1$  and  $s_2$ , but player 2 will play  $t_2$  forever after a deviation from this alternating strategy by player 1. Or consider the following pair of strategies: Player 1 says that she will alternate  $s_1$  and  $s_2$  as long as player 2 does the same, but if ever player 2 deviates, then she will play  $s_2$  forever, and the same for player 2. Again, for  $q$  close enough to one, we have an equilibrium, where the players get alternating 5s and 0s (or, if they are out of sync, 8s and  $-3$ s). Both sides are worse off in this equilibrium than with the first equilibrium we named, but this is nonetheless an equilibrium, as is the situation where both players play  $s_2$ - $t_2$  all the time.<sup>d</sup>

The problem in the repeated game formulation is a profusion of equilibria. Just what is possible as an equilibrium? We know we couldn't have an equilibrium where player 1 or player 2 came away with less than zero expected value. Why? Because by playing  $s_2$  or  $t_2$ , each can guarantee that she or he will get at least that much. But their opponent can hold them to no more and can threaten to use this as a punishment. So, roughly, any pair of payoffs that is feasible and gives each player more than zero apiece can be sustained in an equilibrium. This statement is completely correct in the infinite horizon, undiscounted formulation. For discounting or a finite but indefinite horizon, we need to amend it slightly. A result where one player is left too close to the value zero cannot be sustained in an equilibrium. Each must have enough stake in maintaining the equilibrium so a one-stage defection (followed forever after by zeros) isn't better than carrying through on the agreement.

to play noncooperatively. To make the strategies subgame perfect, amend them as follows: Each player will play cooperatively as long as both have played cooperatively in the past. But if either ever plays noncooperatively, each plays noncooperatively for the rest of the game.

<sup>d</sup> (1) We continue to name strategies that aren't quite subgame perfect, and the technically proficient reader may wish to fix this. (2) We say "for  $q$  close enough to one" repeatedly. How close? See problem 1, and see as well the material at the end of this subsection.

This result, which is fairly obvious when you think about it, has been given for general repeated games. Roughly put, the general result says that any feasible expected payoffs can be sustained in an equilibrium as long as each player has expected payoff at least as large as what that player can guarantee for herself even if all the other players gang up on her. The proof of this proposition is really quite simple. Each player is told by the others to stick to the agreement or everyone will gang up on her. Then no single player, acting alone, has any incentive to deviate; the condition necessary for a Nash equilibrium. This result is known as the *folk theorem*, so-called because its statement belongs to the folk literature of game theory; no one has been brash enough to claim it, since it seems to have been well-known long before it appeared in print.

To be precise, suppose we are given an  $I$  player normal form game as the stage game, where player  $i$  has a finite set  $S_i$  of strategies and where the payoff to player  $i$  at the strategy profile  $(s_1, \dots, s_I)$  is given by  $u_i(s_1, \dots, s_I)$ . Let  $\Sigma_i$  be the space of mixed strategies of player  $i$  and define  $u_i(\sigma_1, \dots, \sigma_I)$  to be the expected payoff to player  $i$  if players play the mixed strategy profile  $(\sigma_1, \dots, \sigma_I)$ . For each player  $i$ , define

$$u_i = \min_{(\sigma_1, \dots, \sigma_{i-1}, \sigma_{i+1}, \dots, \sigma_I)} \max_{s_i} u_i(\sigma_1, \dots, \sigma_{i-1}, s_i, \sigma_{i+1}, \dots, \sigma_I).$$

Note carefully how this is defined. We fix the mixed strategies of all the players except  $i$ , and we find player  $i$ 's best response to those. Then we minimize  $i$ 's payoff over the selection of the strategies of the other players. This is called player  $i$ 's *minmax* value.

Imagine that the players play this normal form game over and over again, where after each round of play the actions chosen by each are known. Roughly put, the folk theorem states that players can achieve in a Nash equilibrium any feasible payoff that leaves each player at or above her minmax value. This is only rough; making it precise requires the following steps.

(a) It is easy to show that in any Nash equilibrium a player can never do worse than her minmax value. If she plays in each round her best response to whatever her opponents are doing, she does at least this well.

(b) Suppose in an undiscounted, average reward formulation we try to implement payoffs that arise from some pure strategy selection by players and that give each player as much or more than her minmax value. This is easy. Everyone is to play this strategy selection as long as everyone else does. If some one player deviates, everyone plays her part in that player's minmaxing strategy forever after. (For a Nash equilibrium, it is irrelevant what we prescribe if more than one deviates, so make any prescription at all.) Note that there is no reason to believe this is a subgame perfect equilibrium — a matter to which we turn in a bit.

(c) If we attempt to implement payoffs that arise from mixed strategy selections, then we can't tell from observed actions whether there have been

deviations. In case we are using an undiscounted, average reward formulation, we can avoid this difficulty by using a pure strategy that "convexifies" with time. If, say, we want a payoff that arises from two-thirds of one pure strategy combination and one-third of another, then we play the first on rounds  $3n+1$  and  $3n+2$  and the second on rounds  $3n$ , for  $n = 1, 2, \dots$ . This is a tiny bit harder to do if the mixing probabilities are irrational, although it is still possible. Note that by this scheme we can get values that result from mixed strategies and also values that arise from correlated mixtures of pure strategies.

(d) If we are using a discounted formulation, we have to worry that a deviation may prove so profitable to a player, relative to what she obtains in equilibrium, that she might deviate even if threatened with minmax punishment. So the statement of the folk theorem becomes: For every feasible payoff above the players' minmax values, there is a value of  $q$  (the discount factor or probability of continuing) sufficiently close to one such that for all larger discount factors/probabilities of continuing, this payoff can be implemented in a Nash equilibrium. And we have to worry about convexifying using time, which is harder in this case.

For detailed arguments and precise statements, see the papers cited in section 14.7.

You might worry that ganging up on a deviator can be quite costly for the other players. One player may be able to inflict a terrible punishment on another, but only at tremendous cost to herself. In this case, the first player might not want to carry out the punishment that is threatened. That is, these punishment strategies may not give a subgame perfect Nash equilibrium. Let us give an example. Consider the normal form game in figure 14.2. In this game, player 2 can threaten player 1 that by playing  $t_2$  he can hold player 1 to a payoff of (no more than) zero. Hence the following is a Nash equilibrium; player 1 plays  $s_1$  all the time. Player 2 plays  $t_1$  as long as player 1 plays  $s_1$ , but moves to  $t_2$  if ever player 1 tries  $s_2$ . The problem is that while player 1 wouldn't want to move to  $s_2$  (if  $q$  is sufficiently large) if she believes that player 2 will carry out this threat, she may find it hard to believe that he would in fact carry out the threat. After all, player 2 will get no more than  $-2$  by playing  $t_2$ , while he will

|          |       | Player 2 |       |
|----------|-------|----------|-------|
|          |       | $t_1$    | $t_2$ |
| Player 1 | $s_1$ | 5,5      | -1,-2 |
|          | $s_2$ | 6,-1     | 0,-3  |

Figure 14.2. A bimatrix game.

get at least  $-1$  with  $t1$ . So player 1 may play  $s2$  with impunity, on the presumption that player 2 will never carry out the threat to punish her.

It has been shown, however, that the conclusion of the folk theorem holds up if one restricts attention to subgame perfect Nash equilibria.<sup>e</sup> The details are fairly complex and won't be repeated here. But it may be instructive to say how we could get the  $s1-t1$  outcome in the game of figure 14.2 in a perfect equilibrium. Suppose that player 1 and player 2 adopt the following pair of strategies. They begin with  $s1$  and  $t1$  and continue with this unless and until there is a deviation. If either (or both) deviate from this, they play  $s2$  and  $t2$  for one round. If both do this, they go back to  $s1$  and  $t1$ . But if either deviates during the "punishment" round, then they have to play  $s2$  and  $t2$  again, and each has to keep this up until they both comply (at which point they can return to  $s1$  and  $t1$ ). Given that player 2 is playing this strategy, will player 1 ever deviate? The answer is no, at least for high enough  $q$ . If play is in the phase where player 1 is supposed to play  $s1$ , by changing to  $s2$  she nets 6 instead of 5, but then she'll have to take a 0 instead of a 5 next time. For high  $q$  this isn't worth it. And in the punishment phase, player 1 prefers both in the short run and the long to get this phase over with and get back to the cooperative phase. As for player 2, when the two are cooperating, he has no incentive whatsoever to move to  $t2$  and trigger a round (or more) of punishment. The question is, Will he carry out the punishment, which is so costly for him? Given player 1's supposed strategy, the answer is yes; player 2 has a choice of either taking a  $-3$  and then getting back to 5s or getting a  $-1$  and delaying by (at least) one round return to cooperation. For large enough  $q$ , it is worth his while to get the punishment phase over with.

Just to show how it is done, let us derive what is "large enough"  $q$  and show formally that this is an equilibrium. In fact, we will show that this equilibrium is subgame perfect, using techniques from appendix 2.

The proper subgames of this game correspond to the start at each time  $t = 1, 2, \dots$  of that period's stage game, given a history of play up to time stage  $t$ , which we write  $h_t$  and which is a sequence of  $t - 1$  pairs of the form  $(sx, ty)$ , where  $x$  and  $y$  are each 1 or 2. Think of play of this game as being at the start of each period in one of two states:  $\phi$  and  $\psi$ , where  $\phi$  represents the state where players are to "cooperate" and  $\psi$  represents the state where players are "punished." Transitions from period to period of this state depend on the state at the start of the period and play in the period. If at the start of a period the state is  $\phi$  and play that period is  $s1-t1$ , then the state is  $\phi$  at the start of the next period. If play in a period begins in state

<sup>e</sup> This is subject to some technical restrictions in the case of discounting; see Fudenberg and Maskin (1986).

$\psi$  and play that period is  $s2-t2$ , then the state next period is  $\phi$ . In all other cases, next period begins in state  $\psi$ .

Note that with this convention, play in period  $t$  according to the supposed equilibrium strategies depends only on the current state. Player 1 is supposed to play  $s1$  in state  $\phi$  and  $s2$  in state  $\psi$ , and similarly for player 2. As noted already, the state in any period depends only on the state in the previous period and play in that period. So we conclude that play from period  $t$  on, according to the presumed strategies, depends on previous history only through the current state. Thus in checking subgame perfection it suffices to show that each player is using a best response beginning in state  $\phi$  and beginning in state  $\psi$ .

We have four things to check. Is player 1 playing a best response by choosing  $s1$  beginning in state  $\phi$ ? Under the equilibrium hypothesis that player 2 conforms to his supposed strategy, if player 1 plays  $s1$  starting in state  $\phi$  and conforms thereafter, the state will stay  $\phi$  forever and player 1 will receive 5 forever, for an expected/present value of  $5 + 5q + 5q^2 + \dots = 5/(1 - q)$ . If, on the other hand, player 1 plays  $s2$  this period and then goes back to following the strategy assigned to her, she will net 6 immediately, the state will change to  $\psi$ , and she will get (conforming thereafter) 0 next round and (after  $\phi$  is thereby restored) 5s for the rest of the game. The expected/present value is  $6 + 0q + 5q^2 + 5q^3 + \dots = 6 + 5q^2/(1 - q)$ . For player 1's strategy in state  $\phi$  to be unimprovable in a single step, it must be that

$$\frac{5}{1 - q} \geq 6 + \frac{5q^2}{1 - q} \text{ or } q \geq \frac{1}{5}. \quad (\phi 1)$$

Next, is player 1 playing a best response by choosing  $s2$  in state  $\psi$ ? By conforming, she nets 0 this round and then 5s for the rest of the game. By failing to conform this round but then returning to conformance, she nets  $-1$  this round, 0 next, and then 5s. This is worse than conformance for all  $q$ . It seems silly, but for the form of things we'll write down this constraint:

$$q \geq 0. \quad (\psi 1)$$

Is player 2 playing a best response by choosing  $t1$  in state  $\phi$ ? You should have no problems with this one — conforming always beats deviating for one step and then conforming:

$$q \geq 0. \quad (\phi 2)$$

Is player 2 playing a best response by choosing  $t2$  in state  $\psi$ ? Conforming forever nets him  $-3$  and then 5s, for an expected/present value of  $-3 + 5q/(1 - q)$ , while deviating to  $t1$  for one stage and then conforming nets him  $-1$ , then  $-3$ , and then 5s, for an expected/present value of  $(-1) + (-3)q + 5q^2/(1 - q)$ .



If you work through the algebra, you'll find that the constraint on  $q$  that ensures that the first is as large as the second is

$$q \geq 1/4. \quad (\psi 2)$$

As long as all four of these constraints are met, which is to say that as long as  $q \geq 1/4$ , neither player ever has an incentive to deviate from her prescribed strategy and then return to conformance. *By proposition 4 in appendix 2, any unimprovable strategy is optimal in an infinite horizon, bounded reward, discounted dynamic programming problem. Thus we know then that for these  $q$ , neither player ever has an incentive to deviate from the prescribed strategy. That is, we have a subgame perfect equilibrium.*

The key here and in much of the formal literature that looks at the folk theorem with discounting is (1) to construct players' strategies in this fashion, using a small number of "states" that evolve from period to period in relatively simple fashion and that determine what happens from any point on and then (2) to use the theory of dynamic programming and especially the result that unimprovable strategies are optimal to verify that a given strategy profile comprises an equilibrium. We will provide you with further examples of this style of analysis later in this chapter.

### (2) Lack of a focal point

Any time there are multiple Nash equilibria, we have to wonder which of them, if any, is the solution. In the story told above, we had one side or the other announcing its strategy and inviting the other to formulate a best response. That is fine if we imagine preplay negotiation between the players, although we might then wonder why one player is able to negotiate from such strength. A good way to interpret the folk theorem, when the players can engage in explicit preplay negotiation, is that it shows how repetition can greatly expand the set of self-enforcing agreements to which the players might come. If the players play the prisoners' dilemma game once (and never meet again, so the game is not embedded into some larger game), then no amount of preplay negotiation will enable them to cooperate. To cooperate, they will need to find some way to form a binding agreement with an enforcement mechanism that is quite outside the game as described. But if this situation is repeated, then there are many self-enforcing agreements they might come to. We don't know which one they will agree to (if any), and we can imagine a rather spirited session of bargaining between them, but we do know that a lot more can (reasonably) be agreed upon.

What if no explicit communication is possible? We said in chapter 12 that the notion of a Nash equilibrium gives a necessary condition for an obvious way to play the game, if one exists, without prejudging

the existence of an obvious way to play the game. Without preplay communication, we rely on convention, or learned behavior, or a focal point, or something equally vague. Can you see any reason to suppose that any one equilibrium is the obvious/intuitive/focal way to play this game?

In the case of the prisoners' dilemma, which is quite simple and symmetric, principles of symmetry and efficiency point us toward  $s1-t1$  as a likely candidate outcome. In other games, not so symmetric, one might also look for focal points. Or one might try to explain the particular equilibrium one sees as conforming to some social norm or convention. The current state of formal theory is that we know, in general, how repetition allows for the possibility of a wide variety of focal point or conventional equilibria. But we certainly don't have many good formal criteria for picking out any particular one.

### (3) Too many out-of-equilibrium threats

The third problem is actually part of the first. Not only is there a profusion of equilibria, but there are many "out of equilibrium" responses that hold them together. In the equilibrium we gave above for the prisoners' dilemma, player 1 told player 2 that if player 2 ever played  $t2$ , then player 1 would resort to  $s2$  forever. But, as long as  $q$  is fairly large, it is enough for player 1 to threaten player 2 with "If you play  $t2$  in any round, then I will respond with  $s2$  in the next round." (How large does  $q$  have to be so that player 2, thinking that player 1 will act in this fashion, will wish to play  $t1$  in every round?) Indeed, player 1 can build this threat into an overall strategy that is very easy to enunciate: "I (player 1) will begin with  $s1$ , and in each subsequent round I will do whatever player 2 did in the previous round." This particular strategy for player 1 is known as "tit-for-tat." The point is that a threat never to cooperate will stop player 2 from trying  $t2$ , but a tit-for-tat threat, and many others besides, will do so as well as long as  $q$  is close enough to 1. Which of these many possible threats will actually occur? It is hard to say, especially when you consider that if the threat works we will never actually see it carried out!

### (4) Games with a definite finite horizon

The fourth problem with the repeated game formulation is that it doesn't work when there is a finite and definite horizon. Consider the repeated prisoners' dilemma and imagine playing that game no more than, say, 100 times. That is, if we do reach round 100, there will be no more

randomizations to determine if there will be a 101st round. Round 100, if we get there, will definitely be the last.

If we get to round 100, the two players know it will be the last, and so player 1 will play  $s_2$  and player 2  $t_2$ ; there is no longer-term loss to weigh against the short-term gain. Now if we get to round 99, there may be one more round. But we know that both sides will play noncooperatively in round 100, so neither side has the incentive to cooperate in round 99; the short-term gain from noncooperation won't be overcome subsequently, since we will have noncooperation subsequently. Thus we have noncooperation in rounds 99 and 100. So what if we reach round 98? Since we will have noncooperation in rounds 99 and 100 no matter what happens in rounds 1 through 98, it follows that we will have noncooperation in 98, and so on. The entire cooperative scheme unravels from the back. To get cooperation, there must always be a future substantial enough to outweigh immediate considerations.

If you are very careful about these things, you will note that what we just proved is that there is a unique subgame perfect equilibrium that involves no cooperation. We claim as well a different result: There is no cooperation along the path of play in any Nash equilibrium.

An argument that proves this other result runs as follows. Suppose that in some Nash equilibrium there is cooperation at some stages with positive probability. Let  $T$  be the last stage at which this is so; that is, there is zero probability of cooperation along the equilibrium path in stages  $T+1, \dots, 100$ . Now examine the incentives of the player who is meant to cooperate along the path of play in stage  $T$ . This player will do no better than zero in the remaining stages by following the equilibrium, since her opponent will not cooperate subsequently along the equilibrium path. By acting noncooperatively in stage  $T$  and in every subsequent stage, the player does better immediately than if she follows the equilibrium prescription, and she can do no worse than zero subsequently. Hence this player will not cooperate in stage  $T$ , a contradiction.

If this sounds vaguely familiar, recall the centipede game from chapter 12. In that game we saw a similar "unravelling from the back." And, indeed, the centipede game is very much like the repeated prisoners' dilemma, except that players take turns moving and the first "noncooperative" action ends the game. Because of these two features (but mostly the first), the centipede game is solvable by backwards induction (i.e., by iterated weak dominance), whereas a slightly more involved argument is needed here. But the same basic idea is at work, and we have the same basic story.

And as with our analysis of the centipede game, the theoretical conclusion that we reach here is not supported either intuitively or empirically.

### 14.3. Noisy observables

Play the prisoners' dilemma, say, 100 times, and in most cases (with college sophomores, or with MBA students) you will get "cooperation" for much of the game. Our theoretical analysis of finitely repeated play of the prisoners' dilemma is missing something. We will get back to this (and to what is missing) in section 14.6.

### 14.3. Noisy observables

Note that in our equilibria a defection from cooperation is noticed by the other side immediately and is dealt with summarily. In applications of these ideas, we might be concerned with cases in which one side isn't quite sure what the other is doing. Observable data might indicate that *perhaps* the other side isn't living up to some cooperative arrangement, but the observable data might be less than perfectly conclusive.

One could model this in a number of ways. Imagine, for example, that we played the prisoners' dilemma game above, but with the following "noise" added. The numbers in figure 14.1 give only the means of the payoffs for the players. The actual payoffs are Normally distributed with those means and with standard deviation, say, 5 units. Each side sees both payoffs but not the move chosen by the other side.<sup>f</sup> For example, if player 1 picks  $s_2$  and player 2 picks  $t_1$  in some round, then in that round player 1 gets a payoff with Normal distribution at mean 8 and standard deviation 5, and player 2's payoff has Normal distribution with mean -3 and standard deviation 5. (The error terms are assumed to independent of one another, both cross-sectionally and serially.)<sup>g</sup>

Suppose you are player 2 playing this game and in the first round you pick  $t_1$ , get back a payoff of 6.2, and your opponent receives 5.2; in the

<sup>f</sup> It is vitally important for what follows that both payoffs are seen, even if this isn't entirely natural. We will say why this is vital below.

<sup>g</sup> A different model of this effect would be to suppose that players work through an intermediary who sometimes fails to follow instructions. That is, imagine that players 1 and 2 on each round send instructions to an intermediary. This intermediary implements the instructions but sometimes does the "wrong" thing; on any given round, the intermediary implements the instructions of a player with probability, say, .8, and with probability .2 takes the "other" action. At the same time, the intermediary may or may not correctly implement the instructions of the other player. Assume that whether the intermediary implements the instructions of a given player is independent of whether he implements the instructions of the second player and independent of what the intermediary does on any other round. And, the key to all this, imagine that players do not see what instructions their rival passes to the intermediary, but only what actions the intermediary takes. This formulation of the problem is of about the same level of technical difficulty of the one we will tackle here, and the reader would do well to see which of the conclusions we derive will work in this somewhat different formulation.

second round you pick  $t1$ , get back 4.8, and your opponent receives 3.9; and in the third you pick  $t1$ , get back  $-1.3$ , and your opponent receives 6.8. (I'll only report your payoff to the level of tenths.) You can't be sure what player 1 has been playing, so you might try maximum likelihood estimation; you would then guess that player 1 has played  $s1, s1, s2$ . But these data are not inconsistent with player 1 choosing  $s1, s1, s1$ . So should you punish player 1? How stiff should your punishment be? If  $-1.3$  is low enough to trigger punishment, what about  $-.3$ ? What about  $1.3$ ? And should your punishment "fit" the crime — more punishment if you get  $-1.3$  than if you get  $.3$ ?

You might consider simply "forgiving" player 1 and not engaging in any punishment. But if you do this, then player 1 has no incentive to play  $s1$  — she can play  $s2$  every time and blame your string of payoffs on bad luck. At the same time, you don't want your punishment to be triggered too often or to be too harsh. Even if player 1 is playing  $s1$  every time, you will see, by the luck of the draw, some bad payoffs occasionally. If you trigger punishment whenever your payoff is, say, 1 or less and/or if that punishment lasts a long time, then you will be spending a lot of time punishing your opponent (and, presumably, she will spend a lot of time punishing you) and the gains from cooperation/collusion will be lost.

We will give an exact analysis of this problem momentarily (in smaller type), but you should have no difficulty seeing the basic trade-off; too little punishment, triggered only rarely, will give your opponent the incentive to try to get away with the noncooperative strategy. You have to punish often enough and harshly enough so that your opponent is motivated to play  $s1$  instead of  $s2$ . But the more often/more harsh is the punishment, the less are the gains from cooperation. And even if you punish in a fashion that leads you to *know* that your opponent is (in her own interests) choosing  $s1$  every time (except when she is punishing), you will have to "punish" in some instances to keep your opponent honest.

Let me make two assertions here (which will be supported in a moment). Recall the third problem listed in the previous section: In the folk theorem there were many threats that could hold together an equilibrium, and we had no reason to choose among them in any fashion. If there are noisy observations, we do have some reason to select one threat over a second; we want our threat (which now will be executed sometimes) to be severe enough to deter the other party from "cheating," but only that severe and no more, because a more severe threat will increase the losses from collusion caused by the inevitable punishments that will be meted.

Second, if we look for equilibria in this setting, and then we vary the level of noise (in terms of the standard deviation of the Normal

distributions, say), then we expect to and do find that the higher the noise the less we can get in an equilibrium for both sides. As the noise (standard deviation) rises, it becomes impossible to disentangle what an opponent is doing, and both sides move to play noncooperative strategies ( $s2$  and  $t2$ ). As the noise falls to zero, we can get closer and closer to the ideal of 5 apiece forever.

We now plunge into a technical analysis of this problem. We will be unable to do full justice to this problem — it sits right at the frontiers of what is known about this form of game — but we can give the technically oriented reader enough of a lead to understand what conclusions the literature comes to. Our analysis comes in two parts: First we look at "trigger equilibria" and then at a more complex form of equilibrium.

### Trigger equilibria

To begin, we restrict attention to a very simple form of strategy for each player. We assume that each player begins by playing cooperatively, that is,  $s1$  and  $t1$ . And they continue to do so until either of the two payoffs falls to a level at or below some critical *trigger level*  $T$ . If in a given round either payoff is less than or equal to  $T$ , then for the next  $N$  rounds the two play noncooperatively, that is,  $s2$  and  $t2$ . After  $N$  rounds, whatever happens in those rounds, the two revert to cooperative play, until the next time that a payoff is less or equal to  $T$ .

There are two parameters in these strategies, the trigger level  $T$  and the length of the punishment phase  $N$ . We are interested first in the question *For which values of  $N$  and  $T$  are the given strategies an equilibrium?*

To answer this question, we begin by computing the expected values to the players of playing these strategies. (Then we will employ the usual dynamic programming technique of seeing whether the strategies are unimprovable.) To be very precise in our technique, we would imagine a state space with  $N + 1$  states:  $\{\phi, \psi_1, \psi_2, \dots, \psi_N\}$ , where  $\phi$  is the state where the two cooperate and  $\psi_n$  is the state where the two are not cooperating and are meant to do so for  $N - n + 1$  more periods before returning to cooperation. Then we begin in state  $\phi$  and remain there until the first payoff is seen less or equal to  $T$ , at which point we transfer to state  $\psi_1$  and then deterministically through the  $\psi_n$  in order, until we get to  $\psi_N$  from which we move back to  $\phi$ . We won't try to carry that level of formality around, but the reader should check that what we say can be made this formal.

When cooperating, the two are meant to play  $s1$  and  $t1$ . Their immediate rewards are random but have expected values of 5 apiece. What is uncertain is whether they will be cooperating next period or not; if we let  $\epsilon_i, i = 1, 2$  be two independent Normal random variables with means zero and standard deviations 5, then the chance that cooperation will continue to reign is

$$\pi_e = \text{Prob}(5 + \epsilon_1 > T, 5 + \epsilon_2 > T),$$

and the chance that noncooperation will take over is  $1 - \pi_e$ . (We will work in terms of the discounted formulation, so  $q$  is a discount factor and not a

|               |       |       |       |       |
|---------------|-------|-------|-------|-------|
| $N =$         | 1     | 2     | 3     | 4     |
| $T = 0: v =$  | 39.59 | 33.34 | 29.20 | 26.26 |
| $T = -1: v =$ | 41.83 | 36.47 | 32.70 | 29.92 |

Table 14.1. Values for following the given strategies.

chance that the game ends altogether.) We claim, then, that the value to either player of following the strategy given above, assuming her opponent does so as well, is the solution  $v$  of the equation

$$v = 5 + q[\pi_e v + (1 - \pi_e)q^N v].$$

Let us explain: The 5 out in front is the immediate expected reward; the  $q$  is the discount factor, applied to expected future payoffs. And expected future payoffs are  $\pi_e$  times  $v$ , which is the chance that cooperation continues to ensue times the expected value of being in a cooperative phase, plus  $(1 - \pi_e)$  times  $q^N v$ , which is the probability that noncooperation begins times the value starting at a noncooperative phase. This last value requires explanation. For  $N$  periods each player receives an expected payoff of zero, and then play reverts to cooperation. Hence the value of cooperation is simply delayed for  $N$  periods, hence discounted by  $q^N$ .

We can invert this recursion to find

$$v = \frac{5}{1 - q[\pi_e + q^N(1 - \pi_e)]}.$$

Note that as  $\pi_e$  implicitly depends on  $T$ ,  $v$  is a function of  $T$  (implicitly) and  $N$  (explicitly). Because  $\pi_e$  decreases with  $T$ ,  $v$  is decreasing in  $T$  and  $N$  (these take some showing); the quicker the trigger or the longer the punishment phase, the lower the value to the players.

In general, we need to compute expected values to the players starting in every possible state, in this case, in the states  $\psi_1, \dots, \psi_N$  as well as in  $\phi$ . But this is easy in this case: The value of starting in  $\psi_1$  is  $q^N v$ ; in  $\psi_2$  is  $q^{N-1} v$ , and so on.

What sort of values do we get? Consider the parametrization  $q = .9$ . For  $T = 0$  and  $T = -1$ , and for  $N = 1, 2, 3, 4$ , the corresponding values of  $v$  are given in table 14.1 (rounded to the nearest .01). Note that, as expected, values fall with rising  $T$  (quicker to punish) and with rising  $N$  (heavier punishment).

Is each strategy unimprovable given the other? The reader may wonder whether this is a useful question to answer. Since the payoffs are now unbounded, we cannot simply plug into our result that unimprovable equals optimal. But the result is valid here. One can replace the actual payoff in any round with its expectation conditional on the actions chosen, and those conditional expectations are bounded. So the question is useful and is equivalent to asking whether each strategy is optimal given the other, that is, whether we have an equilibrium.

We assert that it is clearly optimal to play noncooperatively when in a punishment phase; cooperation returns no more swiftly and one's immediate expected payoff is less, if one plays cooperatively at these times. But there is a real trade-off when cooperation is supposed to be played. By playing noncooperatively, you increase your immediate expected reward (from 5 to 8), and you also increase the chance that you trigger a bout of punishment. Specifically, if you play noncooperatively for one period and then revert to the strategies prescribed, your expected payoff is computed as follows. Let

$$\pi_d = \text{Prob}(8 + \epsilon_1 > T, -3 + \epsilon_2 > T).$$

Then  $\pi_d$  is the chance that you will see a payoff that doesn't trigger a punishment phase if you don't cooperate. Your expected payoff is

$$v' = 8 + q[\pi_d v + (1 - \pi_d)q^N v].$$

Note well: This computation assumes you immediately go back to playing according to the strategies prescribed; the only effects your "deviant" play has is to increase your immediate payoff and to change the probabilities that you don't and do trigger punishment. Hence we use  $v$  and  $q^N v$  for the "continuation" payoffs.

We have an equilibrium, then, if  $v'$  as computed above is less or equal to  $v$ , since this implies that the strategies are each unimprovable against each other, hence optimal. In table 14.2 we give  $v$  and  $v'$  for  $q = .9$ ,  $T = 0$ , and  $N = 1, 2, 3, 4$ , and we see that we have equilibria for  $N = 3$  and 4 but not for  $N = 1$  or 2: At a trigger level of  $T = 0$ , it takes three rounds of punishment to deter players from playing noncooperatively in a presumably cooperative phase.

We can ask now, What is the best combination of  $N$  and  $T$ ? "Best" here has a clear meaning; we are looking at symmetric equilibria, and so we are concerned with finding that pair  $N$  and  $T$  that gives the highest expected payoff to both players. The scheme for finding the best  $N$  and  $T$ , at least numerically, is clear: Fix  $T$ . For  $N = 1$ , compute the corresponding values of  $v$  and  $v'$ . Find whether  $v' \leq v$  — whether we have an equilibrium. If not, try  $N = 2$ , and so on, until you find for the given  $T$  the smallest  $N$  (hence the best  $N$ ) that supports an equilibrium. Call this  $N(T)$ . (You may find that for the given  $T$  no  $N$  gives an equilibrium.) Do this as you search for the optimal level of  $T$  by some numerical procedure.

Table 14.3 provides partial results for such a search, for the case  $q = .9$ . Each row corresponds to a different level of  $T$ . The first column reports  $T$ ,

|        |       |       |       |       |
|--------|-------|-------|-------|-------|
| $N =$  | 1     | 2     | 3     | 4     |
| $v =$  | 39.59 | 33.34 | 29.20 | 26.26 |
| $v' =$ | 40.99 | 33.79 | 29.00 | 25.61 |

Table 14.2. Checking for an equilibrium:  $T = 0$ .

| $T$    | $N(T)$ | $v$     |
|--------|--------|---------|
| 2      | 11     | 12.76   |
| 1      | 4      | 23.01   |
| 0      | 3      | 29.29   |
| -1     | 3      | 32.70   |
| -2     | 3      | 36.28   |
| -3     | 3      | 39.68   |
| -4     | 3      | 42.66   |
| -5     | 3      | 45.05   |
| -6     | 4      | 46.06   |
| -7     | 5      | 47.16   |
| -8     | 6      | 48.11   |
| -9     | 10     | 48.55   |
| -10    | 20     | 48.96   |
| -10.1  | 23     | 48.99   |
| -10.2  | 26     | 49.02   |
| -10.3  | 34     | 49.05   |
| -10.33 | 38     | 49.06   |
| -10.35 | 43     | 49.069  |
| -10.37 | 52     | 49.0716 |
| -10.38 | 67     | 49.0746 |

Table 14.3. Finding the efficient trigger equilibrium.

the second  $N(T)$ , and the third column the expected value  $v$  at  $T$  and  $N(T)$ . The search for an  $N$  that supported an equilibrium was stopped at  $N = 100$ , and for values of  $T$  outside the range shown (i.e., for  $T = 3$  and  $T = -10.39$ ), at  $N = 100$  an equilibrium had not been found. Note well: For some values of  $T$  no  $N$ , no matter how large, will support an equilibrium. This is easiest to see for very small  $T$ . As  $T$  becomes very small, the chance of triggering punishment is virtually zero, thus there is no penalty for a player to take 8 instead of 5 immediately by playing noncooperatively.

Two things are striking. First,  $N(T)$  falls and then rises with falling  $T$ . The rise as  $T$  gets very small is fairly intuitive. For small  $T$ , the chance of triggering punishment is very small, and so it takes a larger and larger punishment to keep players from playing noncooperatively in the cooperative phase. But the fall in  $N(T)$  as  $T$  falls for large  $T$  is a bit surprising. The intuition here is that for large  $T$  punishment is likely to be triggered with high probability in either case, so it takes a fair amount of punishment to keep the players cooperating. Put differently, the ratio of the likelihoods of triggering punishment if you cooperate and if you don't is quite high for large  $T$ , so the marginal impact on the probability of triggering punishment by playing noncooperatively is quite low. Hence, to keep players honest, punishment must be severe.<sup>h</sup>

<sup>h</sup> For readers who think well in terms of relative likelihoods, it may help to note that

The second striking thing is that the value  $v$  rises with falling  $T$ . (Actually, if we decreased  $T$  continuously, it would jump down at values of  $T$  where  $N(T)$  jumps up by one. But the overall effect is a rise in  $v$  with falls in  $T$ .) The optimal  $T$  is the lowest possible value consistent with equilibrium, which means (although our numbers are too discrete to show this)  $N = \infty$ .

This is not a fluke result. Given our restriction in this subsection to these sorts of trigger strategy equilibria, one can prove that the efficient equilibrium from this restricted class is the smallest  $T$  consistent with equilibrium, which has  $N(T) = \infty$ . To show this takes very advanced methods; the required methodology (and, once you make a few clever changes in the formulation, the result itself) is found in Abreu, Pearce, and Stacchetti (1989). The technically very adept will find it an interesting exercise to (1) prove what has just been asserted, using their results and (2) show that one can get somewhat better (higher  $v$ ) results using "trigger equilibria" a bit more sophisticated than the one we've employed here, equilibria where the region of payoffs that triggers punishment is not simply whenever either payoff falls below some single value.

### More sophisticated and efficient equilibria

For the parameterization with which we have been working, we've pushed the value of a collusive equilibrium to 49.0746. That is pretty good, especially considering that we know 50 is unattainable. (Why do we know this?) But we can do better still with more sophisticated equilibria.

Consider the following sort of strategy profile parameterized by two numbers,  $a$  and  $b$ . There are three phases or regimes of play, labeled  $\phi$ ,  $\psi_1$ , and  $\psi_2$ . Phase  $\phi$  is "cooperative"; players choose  $s_1$  and  $t_1$ , respectively, in this phase. Phase  $\psi_1$  is the "punish player 1 and reward player 2" phase, in which player 1 is supposed to play  $s_1$  while player 2 plays  $t_2$ . Note that given these choices player 1 will have an expected immediate payoff of -3 and player 2 a payoff of 8. Phase  $\psi_2$  is symmetric with  $\psi_1$ ; we reward player 1 and punish 2.

Transitions in the phases from one stage to the next depend on the phase at the start of the stage and the payoffs received by the two players.

*Starting a stage in phase  $\phi$ ,* if player 1 receives more than  $a$  and player 2 receives less than  $a$ , we transfer to phase  $\psi_1$ . (The idea is that player 1 did too well relative to player 2 and must be punished.) If player 1 receives less than  $a$  and player 2 receives more, we transfer to  $\psi_2$ . Otherwise we remain in phase  $\phi$ .

*Starting a stage in phase  $\psi_1$ ,* all that matters is how player 1 does. If she gets more than  $b$ , we remain in phase  $\psi_1$ . (She did too well and has to be punished again.) If she gets less than  $b$ , we go back to stage  $\phi$ .

*Starting in phase  $\psi_2$ ,* everything is symmetric with starting in  $\psi_1$ .

the ratio of the likelihoods of triggering punishment if a player cooperates to if she doesn't approaches zero as  $T \rightarrow -\infty$  because of properties of the tails of Normal distributions.

Play begins in phase  $\phi$ . With this start and the rules given above, players are always, for every point in the game tree, able to figure out what they are meant to do. This recursive structure specifies, in fairly opaque form, strategies for the two players.

Do these strategies constitute an equilibrium, for a fixed  $a$  and  $b$ ? By now the technology for finding out should be clear to you. First we compute the expected payoffs to the players, starting in phase  $\phi$ , in  $\psi_1$ , and in  $\psi_2$ , if they follow the strategies described. Let us call these values for player 1  $v(\phi)$ ,  $v(\psi_1)$ , and  $v(\psi_2)$ . These three numbers satisfy three simultaneous equations:

$$\begin{aligned} v(\phi) &= 5 + q[\pi_1 v(\psi_1) + \pi_2 v(\psi_2) + (1 - \pi_1 - \pi_2)v(\phi)], \\ v(\psi_1) &= -3 + q[\rho v(\phi) + (1 - \rho)v(\psi_1)], \text{ and} \\ v(\psi_2) &= 8 + q[\mu v(\phi) + (1 - \mu)v(\psi_2)], \end{aligned}$$

where  $\pi_1$  is the probability that if players follow these strategies we transfer from  $\phi$  to  $\psi_1$ , and so on. (Note that in the second two equations only two states are involved; given the strategies and transition rules, there is no chance that we transfer from  $\psi_1$  directly to  $\psi_2$  or vice versa.) The transition probabilities are, in the case of  $\pi_1$  and  $\pi_2$ , products of Normal cumulants, while  $\rho$  and  $\mu$  are simple Normal cumulants. For example,

$$\pi_1 = \text{Prob}(5 + \epsilon_1 > a) \times \text{Prob}(5 + \epsilon_2 \leq a).$$

So, for example, if  $a = -9.4$  and  $b = -8.2$ , you can work out (for  $q = .9$ ) that  $v(\phi) = 49.6245$ ,  $v(\psi_1) = 15.6337$ , and  $v(\psi_2) = 62.5915$ .

We check for an equilibrium by checking if each strategy is unimprovable given the other. For player 1, we have to check *Does she prefer s1 to s2 in phase  $\phi$ ?* In the now usual fashion, this involves checking whether

$$v(\phi) \geq 8 + q[\pi'_1 v(\psi_1) + \pi'_2 v(\psi_2) + (1 - \pi'_1 - \pi'_2)v(\phi)],$$

where  $\pi'_1$  and  $\pi'_2$  are the transition probabilities if player 1 plays s2 instead of s1 while player 2 sticks to t1. For example,

$$\pi'_1 = \text{Prob}(8 + \epsilon_1 > a) \times \text{Prob}(-3 + \epsilon_2 \leq a).$$

*Does she prefer s2 to s1 in phase  $\psi_2$ ?* This one is trivial — of course she does. (Why of course?) *Does she prefer s1 to s2 in phase  $\psi_1$ ?* This involves checking whether

$$v(\psi_1) \geq 0 + q[\rho' v(\phi) + (1 - \rho')v(\psi_1)],$$

where  $\rho' = \text{Prob}(0 + \epsilon_1 \leq b)$ .

For example, if  $a = -9.4$  and  $b = -8.2$ , both tests are passed. (In case you are interested, deviating for one stage in phase  $\phi$  gives 49.597 and deviating for one stage in phase  $\psi_1$  gives 15.61.) Since everything here is symmetric in players 1 and 2, we do have an equilibrium for  $a = -9.4$  and  $b = -8.2$ , one that is better than the best trigger strategy equilibrium.

What is going on here? In the trigger strategy equilibria, punishment is very inefficient; one punishes the seeming transgressor (although we know that, in equilibrium, no one has transgressed), but at the same time the seeming transgressed-against is also punished. It is more efficient to punish the seeming transgressor and reward the transgressed-against. You can do this, if you are careful to be sure that the seeming transgressor will go along with her punishment.

The equilibrium offered above is certainly not the best one can do. So let me offer a challenge to readers: How much higher can you push the expected value of playing this game for each player in a symmetric equilibrium?

We close with two comments. First, one can, in this exercise, parametrically vary the amount of noise by varying the standard deviation in the noise terms. As that standard deviation rises, any possible gains from this sort of cooperation vanish, since sooner and sooner comes the descent into punishment. And as that standard deviation falls, gains from this sort of cooperation approach those where there is no noise (a theoretical high of  $5/.1 = 50$  in this case of  $q = .9$ ), which should be clear since the descent into punishment is pushed further and further into the future. (These things have formal proofs as well.)

Second, we stated at the outset that it was important that both sides saw both payoffs. If each side saw only its own payoff, one imagines that equilibria that support collusion can be constructed, but these equilibria will be quite complex — too complex for us to compute, at least. This is so because one player, receiving a low payoff, may initiate punishment of the other, and the other will not know that punishment has begun; she will have to infer that it probably has done so. Put another way, transitions from one phase to another by both players in the clean and crisp fashion of the equilibria above will not be available. It isn't quite clear what will be.

### Forgiveness

In all these equilibria, we speak of punishment as if there is some bad behavior to punish. But there is none in equilibrium. There is "punishment," to be sure, when there is noise that makes it impossible to tell whether a bad outcome is due to bad behavior by one's opponent or to bad luck. But in equilibrium both players "know" it wasn't bad behavior. So why punish? Why not "forgive" your opponent and get on with collusive behavior, especially when you know there is nothing really to forgive?

The reason is simple: If you do this, and if your opponent anticipates that you do this, then she will engage in bad behavior.<sup>i</sup>

<sup>i</sup> Another answer, for those who followed the small type discussion, is that in some

### 14.4. Implicit collusion in oligopoly

The notion that oligopolists collude has been around since the virtual inception of economics. In 1776, Adam Smith (1976, I:144) wrote that "people of the same trade seldom meet together, even for merriment and diversion, but the conversation ends in a conspiracy against the public, or in some contrivance to raise prices."

The theoretical case for how such a conspiracy might be enforced is obvious from developments in the preceding sections. Consider the Cournot game for a duopoly, in which each of two firms simultaneously names a quantity that it will supply to the market. As we saw, the equilibrium in such a situation does not give the firms as much as they could get through collusive action. With collusion, identical firms could each supply half the monopoly quantity, and together they would obtain the monopoly profits. But this isn't an equilibrium; if one side provides half the monopoly quantity, the other side has the incentive to supply more: This lowers overall profits, but raises one's own profits (while lowering the profits of the other side more quickly than one's own are going up). This isn't identical to the prisoners' dilemma game, since there we had a single strictly dominant strategy for each side. But here, as there, we have (in equilibrium) each side taking actions that leave both worse off than if they could collude.

Now imagine that the Cournot game is repeated with profits discounted and summed. A simple application of the folk theorem guarantees (for large-enough discount rate) that we can sustain the collusive outcome. Each side communicates to the other, somehow, its intention to play the strategy: Supply half the monopoly quantity as long as the other side does likewise; revert to the Cournot quantities in the face of any deviation. As long as the discounted present value (or expected value) of half the monopoly profits exceeds the best one can get from a deviation in a single round plus the Cournot profits thereafter, we have an equilibrium in which both sides collude.

In a sense, this is a simple variation on the story of the kinked demand curve. In the story of kinked demand, the response of one firm to the other raising its quantity supplied (or lowering its price) was instantaneous.

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equilibria, such as the last one discussed, punishment for one player means good times for the other. Rather than call it punishment, we might think of it as compensation for the unwitting victim of circumstance. It isn't the fault of one's rival, but isn't it reasonable that your rival should give something up to help compensate you for your bad luck? Of course, this story doesn't go too well with what it takes to get back to collusion in our example (bad luck by your rival, to balance your own). And it won't help in those cases where punishment of one side necessarily entails a loss for the others, as in the story we tell next.

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Here the response is delayed for one round. But as long as the delay isn't too long, which we model mathematically with a discount factor close to one, we get the same outcome, collusion, as a potential equilibrium.

Of course, in the story of the kinked demand curve we had a problem with a multiplicity of "equilibria"; we had no theory about where the status quo point would be. And, although we didn't speak of it at the time (except in problem 10.2), we could have told kinked demand curve stories where the outcomes are not symmetric. Likewise here, the folk theorem tells us that there are many equilibria: Asymmetric equilibria in which one side insists on a share of the market in excess of 50 percent; equilibria in which firms settle on profits less than they get from the most collusive outcome. For two symmetric firms, the "half-the-monopoly-quantity" equilibrium seems a rather strong focal point, but in cases of asymmetric firms, we have no particular theory about which equilibrium will arise, if any.

In a sense, all the classic models of oligopoly can be rooted in the folk theorem. The folk theorem holds that any outcome from the collusive outcome down to the competitive outcome, including all manner of asymmetric outcomes, are outcomes of a Nash equilibrium, if the discount factor is close enough to one. And note well that the fact we (formally) imagine players choosing quantities in a given round is nearly irrelevant. If we imagined them choosing prices, we would have much the same range of possible equilibrium outcomes. Collusion can be sustained if the firms are picking prices. Each firm picks the collusive price, with the implicit threat that it will drop its price precipitously if the other side deviates. And if the firms are picking prices, the same threat to drop prices precipitously will sustain an equilibrium where firms name the Cournot equilibrium prices (and sell the Cournot quantities). Similarly, prices arbitrarily close to Bertrand equilibrium prices can be sustained as an equilibrium (for high-enough discount factor) even if the rivals are simultaneously choosing quantities, although the argument here is a bit complex.<sup>1</sup>

Which of these many equilibria (if any one) is the "solution?" That depends, of course, on how each side imagines the other will react to how each side acts today; it is a matter of equilibrium expectations. This isn't quite the same as the classic story of Cournot, or Bertrand, or von Stackelberg conjectures. But it isn't far removed.

Of course, a model of repeated play is vastly more descriptive of the world of real oligopolists than are the stories we gave at the end of chapter 12, where firms simultaneously pick quantities to output or prices

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<sup>1</sup> For a general treatment of this problem, see Abreu (1986).



they will charge, *once and for all*. Most oligopolistic competition is played out over time, and the competitors do retain some ability to react to how their opponents act, even if they are not quite as free to react as we might suppose in a repeated game formulation. From this perspective, becoming specific about the "institutional details" of competition does not seem to have paid such large dividends in getting us to definite predictions after all.

### *Implicit collusion and noise*

We can recover some value from the general approach of adding institutional details by adding a second element of realism to our model of oligopolistic competition — noisy observables. Imagine playing the repeated Cournot game with the following variation: Instead of seeing how much the other side puts on the market, all you see is the market clearing price. (Of course, you know how much you put on the market.) No problem so far — if you know your own quantity and the market clearing price, then you know how much your opponent(s) put on the market. You have enough information to detect any defection from the implicit agreement, and, once detected, you can punish that defection. Hence we get collusion as an equilibrium outcome. But now suppose that the demand curve for this product is subject to unobservable noise. Suppose, for example, that demand in period  $t$  is  $P = A(t) - X$ , where  $X$  is the total quantity supplied, and  $A(t)$  is a constant whose value you don't know; imagine that it has some probability distribution, and the  $A(t)$  are independently and identically distributed through time. Then if you put on the market the agreed-upon (equilibrium) quantity and you find a lower than expected market clearing price, it could be because your opponent put more on the market than she was supposed to or because  $A(t)$  is lower than its average. Should you punish your opponent? It may not be her fault. But if you never engage in punishment, then your opponent has the incentive to chisel on the deal and put a higher quantity on the market. To keep both sides honest, punishment is called for at least sometimes. Note well: Even punishment of finite duration will cause a diminution in what can be accomplished collusively because it is sometimes set off by "mistake." But, on the other hand, you cannot go too far in lessening the severity of the punishment, making it of shorter duration, say, or only triggering punishment when prices fall very far. This would encourage each side to chisel somewhat, thereby lessening the gains from collusion.

The following notions are suggested:

- (a) It is necessary in any collusive arrangement that each side is able to monitor the actions of others.

- (b) Sometimes such monitoring is difficult (or, at least, is costly).
- (c) The parties to the agreement have an incentive to chisel on the agreement and blame the results on whatever noise is in the system.
- (d) Punishment for implied cheating must, therefore, be imposed, even when the "cheating" is certain to have accidental causes, to prevent chiselling from getting entirely out of hand.
- (e) The loss of collusion from the necessary punishment phases must be balanced with the loss of collusion from the chiselling.
- (f) The requisite costs of monitoring and punishment increase with the number of participants, hence collusion is more difficult to achieve the more firms there are in the industry.
- (g) Collusive schemes work best when they are based on less noisy observables.

These ideas about oligopolistic collusion in noisy environments all predate the sort of formal models we discussed in section 14.3; see, for example, Stigler (1964). But they have recently been addressed with models of the style of section 14.3 in papers by Green and Porter (1984) and Abreu, Pearce, and Stacchetti (1986), among others.

The reader is left to discover the details of the formal treatments of these ideas from the original papers. But it may be useful for the more technically minded reader to see a sketch of how business of this sort is conducted.

We begin with a sketch of Green and Porter. They deal with an infinite repetition, discounted formulation of the problem with  $I$  identical firms producing a homogenous product. In the model they use, price in period  $t$  is given by  $\theta_t(A - X)$ , where  $\theta_t$  is a random disturbance term and  $X$  is total output. They look for equilibria of the following sort: A trigger price  $P'$  is set. As long as the equilibrium price stays above  $P'$ , everyone produces at some level  $x'$ . If the equilibrium price falls below  $P'$ , then a "reversionary" punishment period of set duration ensues: Each firm produces the Cournot quantities for  $N$  periods. After these  $N$  periods, firms go back to producing  $x'$  until the next time they find an equilibrium price below  $P'$ . Fix exogenously  $N$  and  $P'$  and consider the production decision of a given firm. Suppose every other firm produces at level  $x'$ . Then for every level  $x$  produced by the given firm, we can work out from the distribution of  $\theta_t$  the distribution of  $P(t) = \theta_t(A - (I - 1)x' - x)$ . And, therefore, we can find the probability that  $P(t)$  will be less than  $P'$ . This allows us to compute the expected length of time between reversionary periods. And it allows us to compute as a function of  $x$  (and parameters  $x'$ ,  $P'$  and  $N$ ) the discounted profits our single firm will receive if it produces  $x$  in the collusive periods instead of  $x'$ . Optimizing over  $x$ , we find this firm's best



response to everyone else producing  $x'$ . We have an equilibrium when the optimal  $x$  is equal to  $x'$ . The parameters  $P'$  and  $N$  are fixed throughout the analysis so far; we proceed to vary these parameters to find the "most efficient" equilibrium of this sort — the one that gives all the firms, since they are treated symmetrically in the equilibrium, the highest possible average profits.

The results obtained by Green and Porter are just what we would expect: Lowering  $P'$  and/or  $N$ , in essence lowering the amount of "punishment," raises the equilibrium value of  $x'$ ; less punishment means more chiselling. The best equilibrium from the point of view of the firms balances the cost of high  $P'$  and/or  $N$ , coming from the cost of reversionary periods caused by low values of  $\theta_t$ , with the benefit in preventing chiselling. As we add more firms (increase  $I$ ), things get worse; each firm has a smaller impact on  $P(t)$ , hence is more willing to chisel, and the level of collusion decreases. (With perfect monitoring, there is no theoretical reason why adding firms should lower the amount of collusion.)

Abreu, Pearce, and Stacchetti (1986) pick up this analysis and using techniques from dynamic programming and optimization theory are able to give the structure of symmetric equilibria that give the oligopolists the highest possible expected profits.

We asserted that by adding noisy observables we recovered some value from the general approach of being specific about the institutional details of competition. Why is this? Point (g) above is the key. One expects that oligopolistic firms will tend to collude as much as circumstances permit. At least, they will tend to act more collusively, everything else held equal, when there is less noise than when there is more. And some institutional arrangements for competition are inherently more noisy than others. So we would expect greater levels of collusion in industries where institutions allow for more precise monitoring of rivals.

A classic case of this sort drawn from Sultan (1975) and Porter (1983), concerns the competition between General Electric and Westinghouse in the manufacture and sale of large turbine generators. In 1963, the majority of orders that were placed for these very large pieces of capital equipment were placed (with one of the two firms) after protracted "secret negotiations." An electric utility would negotiate first with GE (say), then with Westinghouse, then with GE again, and so on, without either of the two manufacturers ever being certain what the other was bidding. When an order was placed, there still was no clear indication (to the losing firm) what the winning firm had bid. In this environment, prices were quite low and competition quite severe. GE, however, found a way to "restructure" competition with Westinghouse in a manner that made GE's bids predictable and its winning bids public knowledge. You should consult the original sources to see the extremely clever way in which this was done. But the

essential idea was that GE provided themselves with contractual obligations to reveal this information, with auditing and verification provided by a public accounting firm. Westinghouse went along with this restructuring and the result was quite successful implicit collusion between the two firms — collusion so successful that the Antitrust Division of the U.S. Justice Department stepped in and put a stop to the institutions that supported GE's scheme.

### *What's wrong with the game theoretic approach?*

The discussion above centers on the (in)ability to monitor. But there is another problem, intuitively similar, to contend with in applications. What do you monitor? When we have symmetric firms, it seems "obvious" that we should look for symmetric equilibria wherein every firm acts in the same way.<sup>2</sup> But in an oligopoly of asymmetric firms, it won't be crystal clear just what is the "rule" each is supposed to follow. (This is so if we rule out secret negotiations in hotel rooms — something at which many governments look askance.) When firm A finds that its profits are lower than expected, it must decide whether this is due to (a) something completely outside the control of A's rivals, (b) an "honest mistake" by its rivals concerning the nature of the "agreement," or (c) chiselling by its rivals. Stigler and Green and Porter discuss (a) versus (c). But (b) versus (c) is at least as important in real life — especially if one's rivals might chisel and then claim that this was just an honest mistake.

We have returned to two closely related weaknesses of this approach: The twin problems of too many equilibria and the selection of one of them. Compare the model of this section with the Cournot, or Bertrand, or von Stackelberg model. In the three classic models, a definite prediction is made about what will happen. One can (and we do) argue that this definiteness arises speciously. Insofar as the repeated play model is the right model, the classic models make an ad hoc assumption about rivals' conjectures, which is virtually equivalent to assuming which equilibrium

<sup>2</sup> But even this is too pat an answer. Imagine that Coca Cola and Pepsi are in all respects identical. (Of course, they aren't.) Should they then collude by each having a presence in every market in the U.S.? Or should they collude by implicitly segmenting U.S. domestic markets, with Coke the dominant brand in the South and West and Pepsi strong in the Northeast and Midwest? Complicating this: Insofar as there are potential entrants, which scheme is better for preventing entry? Does this inform discussion of how they might collude differently in overseas markets? If you think you've worked out the softdrink business, consider (and compare with) passenger airlines. Since this is a book on microeconomic theory, we will leave such enticing issues in industrial organization and competitive strategy at this point in the hope that the tools you are learning here will help you study such questions.

pertains. Unless there is some reason given for the ad hoc assumption, nothing has been gained except perhaps to explain what sort of conjectures lead to certain equilibrium outcomes. (But at least the classic models make predictions definite enough to be refutable.)

A better comparison is with general equilibrium and, in the context of partial equilibrium, with the usual picture of upward sloping supply and downward sloping demand. In the latter class of models, a definite equilibrium prediction is made. In general equilibrium, one mightn't get a single specific prediction. But as noted in chapter 5, with some high-powered math, results are available establishing the "local uniqueness" of equilibria for most economies. In these contexts, then, one is able to do local comparative statics — jiggle this parameter or that a bit, and see how the predicted equilibrium can be expected to react. This adds utility to the model and it gives a handle for testing the model empirically.

Imagine, in contrast, trying to do local comparative statics in a situation where the folk theorem applies. Only if you have an idea how equilibrium expectations move with changes in parameters can you hope to get anywhere. Since the theory is silent on where equilibrium expectations come from, the theory is of limited help (to put it charitably). What is needed is some theory about how and when equilibrium expectations arise, and how they then adapt to changes in the environment.

A second lacuna in the theory is apparent when we examine closely the claim that specifying institutions helps us to understand outcomes, at least in noisy environments. This may be true as far as it goes. (It takes empirical verification before "may be" can be replaced by "is.") But it leaves open the question, Where did the institutions come from? With the benefit of hindsight, we might "predict" that GE and Westinghouse would achieve greater collusion after GE changed the "rules" than before; we might even make this into a legitimate empirical test by looking cross-sectionally at a number of industries. But how do we explain why, suddenly, GE saw how to change the rules to its own benefit? Why weren't those rules changed before? Is it the changed *rules* that led to greater collusion? Or did collusion increase because of the rule *changes*? More generally as we look across industries, do variations in the "rules" reflect uncontrollable environmental differences, or are institutions the result of some conscious or unconscious choices by participants? The answer is almost always going to be that both types of factor intrude. But this leads us to wonder whether there isn't some connection between the chosen parts of the institutional framework and the equilibrium expectations held by individuals. And even if we don't find such a connection, having a theory about how institutions arise and evolve could be more informative

than theories of equilibrium within the context of a given set of institutions.

We will not resolve these puzzles at this point. In fact, we won't *resolve* them in this book at all; their full resolutions are over the horizon of economic theory at this point. We will return to them, however, especially in part V and the postscript. For now, we proceed with noncooperative game theory, having noted some fairly substantial limitations and weaknesses.

## 14.5. Reputation

We have dealt so far with the power of repeated play when two (or more) parties interact repeatedly. Imagine the following alternative story: A monopolist sells a particular product, which it is able to make either high quality or low. Demand for this good depends on its quality: If the good is high quality, demand is  $P = 10 - X$ . If it is low quality, demand is  $P = 4 - X$ . The cost of making high-quality goods is \$2 per unit, while low-quality goods cost \$1 per unit to make. Consumers are unable to observe the level of quality when they purchase the good, although they learn shortly afterwards just what they bought.

If we think of this situation as a one-time market, we get the extensive form representation shown in figure 14.3. And we get a very inefficient equilibrium. No matter what strategy consumers will use, the firm is better off making and selling low-quality goods; making high-quality goods won't change consumer buying decisions since consumers don't know the

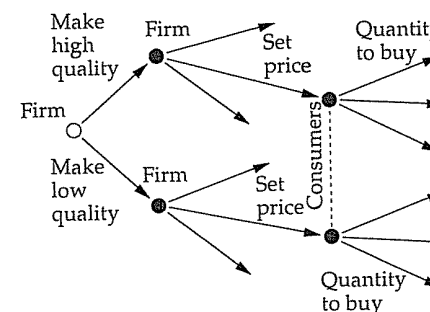


Figure 14.3. A quality choice game.

The firm chooses quality level and sets a price. Consumers purchase knowing the price, but not knowing for sure what quality the firm chose.

quality of the goods. Consumers realize this and anticipate that the firm will produce shoddy goods. So the firm sets price at \$2.50 apiece, sells 1.5 units of the good (which it makes low quality) and reaps profits of \$2.25.

Now imagine that the firm sells the good not once but repeatedly through time. To be formal, imagine there is a market in this good at the sequence of dates  $t = 1, 2, \dots$ . We can imagine that the same consumers continue to buy this good or that there is a fresh batch of consumers each period, as long as we maintain one important assumption: Consumers who might buy at date  $t$  are aware of the quality of goods sold previously at dates  $t-1, t-2, \dots, 1$ . Finally, the firm seeks to maximize its discounted profits, using discount factor  $\alpha$ .

In this new situation, it is still an equilibrium for the firm always to set price at \$2.50, make shoddy goods, and sell 1.5 units per period. But there are other, more interesting equilibria as well.

Imagine that the consumers adopt the rule: Assume the firm is producing high-quality goods as long as the goods produced in the previous three periods were all of high quality. Otherwise, assume that the goods produced this period will be shoddy. (For the first three periods, give the firm the benefit of the doubt. Assume the firm is producing high-quality goods unless it has produced shoddy goods previously.) Then the firm by always producing high-quality goods can set a price of \$6 each period, sell four units, and realize profits of \$16 per period. Of course, the firm could take advantage of consumers in any given period by producing shoddy goods. That would increase the firm's profits from \$16 to \$20 during that period. But the firm would pay for this subsequently; the most it could make in the next three periods would be \$2.25, and if it ever wanted to get back to those \$16 profits, it would have to do worse than that. At reasonable values of  $\alpha$  the firm will continue to produce high-quality goods. And the consumers' assumption that the firm is producing high-quality goods works out.

There is certainly nothing profound or complex in this. We are simply reinventing in the formal language of game theory the notion of reputation. Our firm can have either a reputation for selling high-quality goods or for selling shoddy merchandise. Consumers form their expectations by looking at recent production by the firm. And, in this case, carrying a reputation for selling high-quality goods is viable as an equilibrium. The value of the reputation in the future is always more than what can be had by taking advantage of it in the short run. Indeed, for  $\alpha$  sufficiently close to one, a firm in this situation would want to invest in reputation. It would produce high-quality goods for a few periods even if this cost it money to convince consumers that it produces high-quality goods.

It is crucial here that the firm believes its current actions affect its reputation and, thus, its future demand. If the quality of goods sold at date  $t$  never became known to future consumers, or only became known with a substantial lag, then having a reputation for high quality might not be viable as an equilibrium in the sense that a firm with a reputation for high quality would prefer to exploit that reputation in the short run.

This is very much like the construction of the folk theorem, and it suffers from similar problems. First, there are typically many possible "reputational equilibria." It is not hard to imagine complicating the simple story above so that there are many possible levels of quality among which the firm must choose, and it isn't hard to parameterize the problem so that the firm could carry an (equilibrium) reputation for many of those levels. (See problem 3.) It will simply depend on what the consumers anticipate and how they react to the quality levels the firm has previously produced.

Second, given a multiplicity of equilibria, we worry as before about the lack of a focal point.<sup>3</sup>

Third, if there is a fixed and finite horizon, these reputation constructions can collapse. In our simple example, consumers in the last period rationally assume that the firm will end with a run of shoddy goods, no matter what the firm has produced prior to this. Hence in the next-to-last period, the firm has no incentive to produce high-quality goods, and consumers in that period assume as much, and so on, just as before.

Fourth, noisy observables complicate matters. If our firm cannot perfectly control the quality of its output, we run into difficulties. Imagine, for example, that when the firm is trying to produce shoddy goods, each item produced is low quality with probability .9 and high quality with probability .1, independently of the others. Whereas if the firm is trying to produce high-quality goods, each item is high quality with probability .8 and low quality with probability .2. Then if the firm produces and sells four items and two are shoddy and two are high quality, what do consumers do? If they don't sometimes "punish" the firm, the firm will certainly always produce shoddy goods, blaming it all on an incredible string of bad luck. But even if the firm is trying to produce high-quality goods, every so often it will have an unlucky day. Considerations very much like those in the previous sections apply.

You shouldn't imagine from our example that reputation can always do just as much as can be done with the standard model of repeated play. If we imagined that one player played against a sequence of opponents in

<sup>3</sup> Although when the party carrying the reputation interacts with a sequence of myopic opponents, there can be quite natural focal points; see Fudenberg and Levine (1988).

the prisoners' dilemma game, and if those opponents had only short-run interests (either because they played only once or because they otherwise weren't concerned with their own reputations), then the fact that the one player carries a reputation will get us nothing. All her opponents will play noncooperatively whatever her reputation is, since noncooperative play in the static prisoners' dilemma is strictly dominant. And so there is no reason for this one player to carry a reputation for acting cooperatively.

This raises the question, Is there a neat characterization of what can be obtained in a reputational equilibrium analogous to the folk theorem? One needs to be a bit more precise about the setting and about what is meant by a reputational equilibrium, but results of this sort have been obtained; see, for example, Fudenberg, Kreps, and Maskin (1989).

This sort of reputation construction is used in many ways in economics. While the reputation sketched above is "beneficial" to both sides, there are other contexts in which one side cultivates a reputation which is harmful to others. A neat example (because it illustrates the limitations of reputation vis à vis the folk theorem) is the following somewhat artificial story concerning market leadership. Imagine a long-lived firm that at a sequence of dates  $t = 1, 2, \dots$  is engaged in duopolistic competition with a sequence of rivals. That is, at date  $t$  this firm and a rival compete in a duopoly market; then at date  $t + 1$  this firm competes against a new rival. Imagine that at each date  $t$  the two firms engage in Cournot style competition. You should think of our standard model of a market for an undifferentiated good with linear demand  $P = A - X$  and constant marginal cost  $c$ .

If the two firms competed at date  $t$  and  $t + 1$  and forever into the future (discounting their profits), then the folk theorem shows how with sufficiently high discount factors the two could sustain monopoly profits between themselves. But if one of the two firms competes only at date  $t$ , then it isn't concerned with the future when setting its quantity; its only concern is with the quantity that the long-lived firm might set. If in the equilibrium the long-lived firm chooses quantity  $x_L$ , then the short-lived (date  $t$ ) firm will choose its quantity optimally in response, using the response function  $x_t^*(x_L) = (A - c - x_L)/2$ .<sup>4</sup> So we can never get an equilibrium where the long-lived firm and its rivals share the monopoly profits unless the long-lived firm produces nothing. You can easily confirm that  $x_L + x_t^*(x_L) > (A - c)/2$  for  $x_L > 0$ .

<sup>4</sup> In this problem, even if the long-lived firm sets quantity randomly, the same response function works with  $x_L$  now the expected value of the long-lived firm's quantity as long as the total quantity supplied is always less or equal to  $A - c$ .

But now consider the following equilibrium: The long-lived firm produces  $(A - c)/2$  in each period, as long as it has produced  $(A - c)/2$  in the past. If ever the long-lived firm deviates from this production plan, it will produce  $(A - c)/3$  thereafter. The short-lived firms each produce  $(A - c)/4$ , as long as the long-lived firm in the past has produced  $(A - c)/2$ . But if ever the long-lived firm produces something other than  $(A - c)/2$ , the short-lived firms all subsequently produce  $(A - c)/3$ .

You can check that for a discount factor  $\alpha$  that is sufficiently close to one ( $\geq 9/17$ , if you work it out) this describes an equilibrium. A colorful story goes with this equilibrium. Our long-lived firm has the reputation of being a Stackelberg leader; it sets its quantity at  $(A - c)/2$ , expecting its rivals to reply with  $(A - c)/4$ . Of course, in any single period given that it expects its rival to produce  $(A - c)/4$ , it would prefer (in this round) to set its quantity at  $3(A - c)/8$ . But if it does this, it loses its reputation for being a Stackelberg leader, after which all its (subsequent) rivals assume it will produce the standard Cournot output  $(A - c)/3$ , to which they optimally reply with the same level of output. Note two things here. First, in this equilibrium we have the short-lived firms always producing on their short-run optimal reaction function to what the long-lived firm does. This is a quite general property of this sort of equilibrium. Second, the short-lived firms are worse off in this equilibrium than they would be if the long-lived firm couldn't carry a reputation.

A number of simple variations on this story can be played. We can turn it into a story about entry deterrence if we imagine that all the firms involved have fixed costs. For example, consider the parameterization we have used in tales of entry deterrence: Demand is given by  $P = 9 - X$ , marginal costs are  $c = 1$ , and fixed costs are \$2.25. And consider the following equilibrium: The long-lived firm produces 5 units, as long as it has produced 5 units in the past. If ever it produces something other than 5 units, it produces  $8/3$  thereafter. The short-lived firms produce nothing (which we will assume avoids the fixed cost) as long as the long-lived firm has in the past produced 5 units each period, but if ever the long-lived firm deviates from this, short-lived firms subsequently enter and produce  $8/3$  apiece. You should have no trouble telling the "story" that goes with this equilibrium.

Or consider the original story, but now with noise in the market price as in Green and Porter. This one is left to you as a problem, but try to guess before thinking about it. Should the long-lived firm be "punished" if observed prices are higher than expected or lower? Also left to you is the following very easy exercise. What happens to our basic story if the long-lived and short-lived firms engage in Bertrand style competition?

We will see other, very important applications of this reputation construction in chapter 20.

### 14.6. Reputation redux: Incomplete information

*The details of this section are difficult, and nothing that follows in the book builds on this section. This can be regarded as optional material for a first reading.*

One of the deficiencies of the constructions explored so far in this chapter is that they depend heavily on there always being a future that can be used to influence current behavior. It isn't necessary to have infinitely many periods; in the variation where there is always a probability  $q$  of at least one more round, there are only finitely many periods before the game ends — but it can't be that players can predict that the game will end by some definite time in the future. For once there is a definite terminus to the game; the sorts of cooperative and reputational equilibria that we've been constructing tend to unravel. Behavior is fixed in the final period, which then fixes it in the penultimate, and so on.<sup>5</sup>

The theory, then, seems to imply that these constructions rely crucially on a horizon that never does loom up. But, as we've noted several times now, this doesn't seem right intuitively. And it certainly isn't right empirically. When students play, say, the prisoners' dilemma twenty times, where it is understood from the start that there will be only twenty rounds, there is usually some cooperation early on.

Or consider the game between players 1 and 2 depicted in figure 14.4. This game goes with the following story. Player 2 is an incumbent monopolist in some market. Player 1 is a potential entrant. First player 1 must decide whether or not to enter player 2's market. If player 1 enters, then player 2 must decide whether to acquiesce to this entry or to fight the entry (with slashed prices, etc.). This game has two Nash equilibria in pure strategies. In the first, player 1 enters and player 2 acquiesces. In the second, player 1 doesn't enter and player 2 fights entry if it comes (which it doesn't). By now you will recognize that only the first of these is subgame perfect, and it is the usual prediction that would be made if player 1 and 2 play this game once.

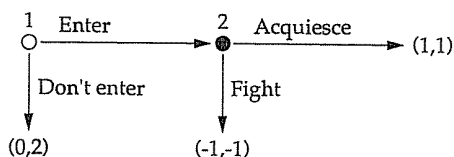


Figure 14.4. The chain-store stage game.

<sup>5</sup> Actually, this happens only if there is a single equilibrium in the static stage game. If there are multiple equilibria in the stage game, even with a finite horizon one can construct cooperative or reputational equilibria. See Benoit and Krishna (1985) for details.

But now add the following complication. Player 2 will play this game not once but, say, twenty times in sequence, each time against a different opponent. Each of 2's opponents plays the game once and once only. The order of play is set, and players 1 who play later observe what happened in earlier rounds of play. The story that goes with this is: Player 2 is a chain-store monopolist with twenty stores in as many towns. In each town, there is the threat of entry from a firm that will if it enters be active in that town only. Moreover, there is a set sequence of actions: The entrant in some given town must decide whether to enter or not and the chain store must react to that entry; then in a second given town the entry-reaction decisions must be taken, and so on.

The theory dictates that this complication doesn't really change matters. In the last round, if the entrant enters, the only subgame perfect reaction for the chain store is to acquiesce. But then in round nineteen, the same is true, and so on; the unique subgame perfect equilibrium has every entrant entering in turn and the chain store always acquiescing.

If we string out infinitely many of these stage games, we can get a different equilibrium using a reputation construction as in the previous section. One equilibrium is: No entrant enters as long as the chain store has never acquiesced. Every entrant enters subsequent to the chain store ever acquiescing. The chain store fights any entry, as long as it never previously acquiesced. And the chain store acquiesces if ever, by chance, it had acquiesced previously. You can verify that this equilibrium is subgame perfect. Roughly put, the chain store is willing to fight any entry because, although this is costly in the short run, it is more costly to acquiesce and invite entry from all subsequent entrants.

As with the finitely and infinitely repeated prisoners' dilemma, the theoretical dichotomy between the finite and the infinite repetition versions of the chain store game does not have much intuitive appeal. Indeed, in the original paper on this subject, Selten (1978) finds this dichotomy so counterintuitive that he refers to it as the *chain-store paradox*.

We saw yet another example of the same sort of thing in the centipede game of chapter 12. This game, reproduced here in figure 14.5(a), has a unique subgame perfect equilibrium in which the two players pick D at every node. It has some subgame imperfect equilibria as well, but every Nash equilibrium necessarily begins with player 1 choosing D. Nonetheless, this doesn't seem a very intuitive solution to this game, and this game, played by students and other experimental subjects, certainly doesn't always end with player 1 choosing D at the first opportunity. We say this is another example of the same sort of thing because the argument that the game has a unique subgame perfect equilibrium involves the same sort of unravelling from the back: Player 2 will surely choose D at his last node, so player 1 will surely choose D at the next-to-last node, and so on.

I next present an argument for why this equilibrium seems unintuitive, which I will later tie back to the notion of reputation. The argument involves the incomplete information variation on the basic game depicted in figure 14.5(b). In this game of incomplete information, we suppose that player 1's payoffs are determined by nature. With probability .999 they are as in the

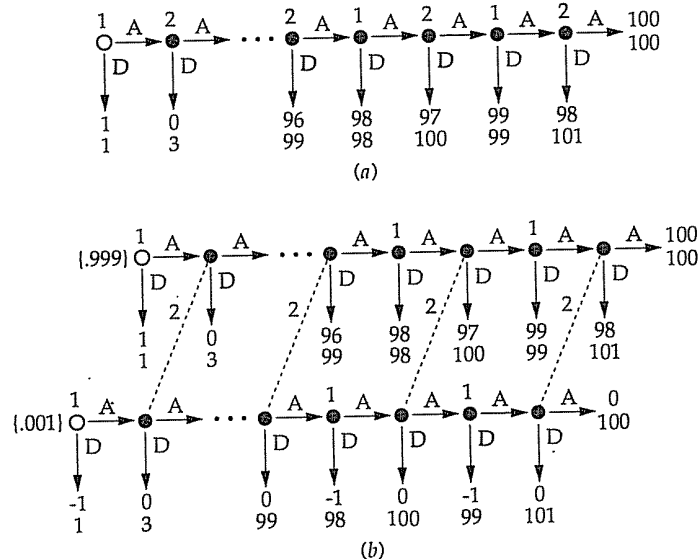


Figure 14.5. The centipede game and a variation.

original centipede game, while with probability .001, these payoffs make playing A until the end of the game a dominant strategy. Think of the following story. Player 2 assesses, at the outset, a one-in-one-thousand chance that his opponent overflows with the milk of human kindness and will play A at every opportunity. Player 1 is aware that player 2 entertains this hypothesis, and player 1 is aware that player 2 doesn't put much store by it. Player 2 only gives it a one-in-one-thousand chance.

It certainly isn't part of an equilibrium in this variation for player 1 to play D at the outset (and thereafter) in the top half of the game. Why? Because if player 1 plays this way in the top half of the game, then player 2, given the move, will believe (by applying Bayes' rule) that he is in the bottom half of the game. Believing this, he plays A until the last node (when he plays D) in anticipation that player 1 will play A at each of her nodes. But then it is in player 1's interests in the top half of the game to play A until the next-to-last node, beating 2 to the punch by one. Which then causes our logic to go haywire, as player 2 won't conclude on the basis of A at the first node that he is in the bottom half of the game.

What does the equilibrium of this game look like? What follows is, unhappily, not entirely transparent.<sup>j</sup> But I know of no way around this.

<sup>j</sup> It is time to remind you that you were specifically warned about this example in

The game has a unique sequential equilibrium, which can be obtained by reasoning back from the last round.<sup>6</sup> First, let me label the nodes  $1x$ ,  $2x$ ,  $3x$ , ...,  $198x$ , where  $x$  is  $t$  for top and  $b$  for bottom, and where  $1x$  is the last node in the sequence,  $2x$  is the second to last, and so on. Note that at nodes with an odd number for an index player 2 moves, while at even numbered nodes player 1 chooses. (Sorry about that.) Since player 2's information sets contain pairs of nodes  $(2n+1)t$  and  $(2n+1)b$ , his actions at these nodes must be the same; we will speak of his move at information set  $2n+1$ .

Because player 1 chooses A for sure in the bottom half of the tree and there is prior probability .001 that the bottom half prevails, player 2 is never completely surprised to be given the move back if he chooses A. This implies that 2's beliefs at each of his information sets can be computed by Bayes' rule from 1's strategy, whatever that strategy is. We will use  $\mu_{2n+1}$  to denote the probability 2 assesses that he is in the top half of the tree, given that information set  $2n+1$  has been reached. Note that  $\mu_{2n+1} \leq .999$  for all  $n$  (since player 1 chooses A for sure in the bottom part of the tree).

At information set 1, player 2 chooses D for sure by dominance. Hence at node  $2t$ , player 1 chooses D.

Now move back to information set 3. Player 2 computes: If he plays D, he will get 100. If he plays A, he will get 101 if he is in the bottom half of the tree and 99 if he is in the top half. Letting  $\mu_3$  denote his probability assessment that he is in the top half of the tree, he nets expected value  $99\mu_3 + 101(1 - \mu_3)$  by playing A, and so he chooses A if  $\mu_3 < 1/2$ , D if  $\mu_3 > 1/2$ , and he is indifferent if  $\mu_3 = 1/2$ .

Now go back to node  $4t$ . Player 1 nets 98 at node  $4t$  by choosing D. If she chooses A, she nets 97 if player 2 follows up with D and 99 if player 2 follows with A. So what she will do depends, in equilibrium, on 2's beliefs at information set 3: If he will believe  $\mu_3 < 1/2$ , then he will play A, in which case she plays A. If he will believe  $\mu_3 > 1/2$ , then he will play D, in which case she will play D. If he will believe  $\mu_3 = 1/2$  and he chooses to randomize exactly 50-50 between A and D, she is indifferent at node  $4t$ .

(a) Since  $\mu_3 < 1$ , it can't be part of an equilibrium for him to believe  $\mu_3 > 1/2$ . Because if we suppose these are going to be his beliefs, she will choose D at  $4t$ , and Bayes' rule will give him beliefs of  $\mu_3 = 0$ , contradicting  $\mu_3 > 1/2$ .

(b) Could it be that  $\mu_3 < 1/2$ ? This is possible if and only if  $\mu_5 < 1/2$  because, if  $\mu_3$  is going to be less than  $1/2$ , he will play A at information set 3. Thus she will play A at node  $4t$ . Thus, by Bayes' rule,  $\mu_5 = \mu_3$ . So if  $\mu_3$  is to be less than  $1/2$ , then  $\mu_5 < 1/2$  in an equilibrium. Conversely, if  $\mu_5 < 1/2$ , then Bayes' rule implies that  $\mu_3 \leq \mu_5 < 1/2$  and he will play A at information set 3, so she will play A at  $4t$  (and  $\mu_3 = \mu_5$ ).

(c) This leaves the case where  $\mu_5 \geq 1/2$ . We can't have  $\mu_3 < 1/2$ , and we can't have  $\mu_3 > 1/2$ , so we will have to have  $\mu_3 = 1/2$ . Denote by  $\pi_4$  the

chapter 1.

<sup>6</sup> In fact, the game has a unique Nash equilibrium; this is left as an exercise.

equilibrium probability that 1 chooses A at 4t. Then given  $\pi_4$  and  $\mu_5$ , Bayes' rule is used to compute  $\mu_3$  as

$$\mu_3 = \frac{\mu_5 \pi_4}{\mu_5 \pi_4 + (1 - \mu_5)}.$$

Since we need  $\mu_3 = 1/2$ , this gives us  $\pi_4 = (1 - \mu_5)/\mu_5$ . This tells us a few things. First, at information set 5, player 2 assesses probability  $1 - \mu_5 + \pi_4 \mu_5 = 2(1 - \mu_5)$  that player 1 will play A next time. So if  $\mu_5 < 3/4$ , he definitely wishes to play A at information set 5. If, on the other hand,  $\mu_5 > 3/4$ , he definitely wishes to play D at information set 5. And he is indifferent if  $\mu_5 = 3/4$ . (Recall that if  $\mu_3 = 1/2$ , he has expected payoff of 100 starting from information set 3.) Next, unless  $\mu_5 = 1/2$ , the only way we will get  $\mu_3 = 1/2$  is if  $0 < \pi_4 < 1$ . This requires that player 1 is indifferent between A and D at node 4t, which in turn requires that player 2 at information set 3 randomizes precisely 50-50 between A and D.

This nails everything down (in terms of  $\mu_5$ ) at nodes 4t and at information set 3 except for the knife-edge case where  $\mu_5 = 1/2$ . In this case,  $\mu_3 = 1/2$  only if player 1 plays A for sure at 4t. And player 1 will play A for sure as long as player 2 chooses A with probability  $1/2$  or more at information set 3. So in this knife-edge case, we do have a bit of freedom in what happens.<sup>7</sup>

Now back to information set 5. We already know rather a lot. If  $\mu_5 > 3/4$ , player 2 will choose D for sure. If  $\mu_5 < 3/4$ , he will choose A for sure. If  $\mu_5 = 3/4$ , he is indifferent, and his expected payoff is 99. All of which we use when we ...

... move back to node 6t. The analysis here is a lot like the analysis of 6t, except the role of  $\mu_3 = 1/2$  is replaced by  $\mu_5 = 3/4$ . Essentially, if  $\mu_7 < 3/4$ , then player 2 is sure to play A at information set 5, so 1 plays A at 6t and  $\mu_5 = \mu_7$ . If  $\mu_7 \geq 3/4$ , then the only way to have an equilibrium is to get  $\mu_5 = 3/4$  — this determines what player 1 must do at 6t. If  $\mu_7 = 3/4$ , there is a bit of latitude in what player 2 must do at information set 5; while if  $\mu_7 > 3/4$ , player 1 must randomize at 6t, which fixes the needed randomization of 2 at information set 5. And when we compute what 1 must do at 6t to get us from  $\mu_7 \geq 3/4$  to  $\mu_5 = 3/4$ , we have begun to ...

... move back to information set 7. If  $\mu_7 < 7/8$ , player 2 will play A for sure; if  $\mu_7 > 7/8$ , he will play D for sure; and if  $\mu_7 = 7/8$ , he is indifferent.

The pattern emerges. The crucial thing on which to focus is that if  $\mu_5 < 15/16$ , player 2 will definitely play A at information set 9, and so on. In particular, if  $\mu_{2n+1} < 1023/1024$  for  $n \geq 10$ , player 2 will definitely play A at information set  $2n+1$ . And then, prior to this, player 1 will play A at node  $(2n)t$ , for  $n > 11$ . But  $\mu_{2n+1} \leq .999 < 1023/1024$  for every  $n$ , since .999

<sup>7</sup>In defense of our claim that the game has a unique sequential equilibrium, we later argue that this knife-edge case doesn't happen given the initial conditions specified for the game.

is the prior. For most of the game, until the last 22 or so moves (out of 198), player 1 (in both halves of the game) and player 2 both choose A for sure, which is completely the reverse of what they are meant to do according to subgame perfection in the simple centipede game.

Of course, this is rather complex. The final 22 or so moves involve a complicated dance of mixed strategies that keep player 2 willing to randomize so that player 1 is willing to randomize so that ...; I am not claiming that the precise equilibrium is very intuitive. But, setting aside those final twenty-two moves, the "main effect" does have a good deal of intuition to it.

Player 2 risks one unit of payoff by choosing A instead of D. The potential gains are substantial. If there is a moderate chance of reaping those gains, it would be worthwhile to give A a try. But that isn't all there is to it. Note that the potential gains are on the order of 100. And the chance that player 1 will always play A is only .001. So if player 1 in the top half of the game were always to play D, this wouldn't be a worthwhile investment for player 2. What makes it worthwhile is that it is in the interests of 1 in the top half of the game to convince 2 to keep trying A. Hence it is in 1's interests to play A. And 2, recognizing this, has it in her own interests to play A.

Do not be misled by this argument into thinking that this is simple or straightforward. Exactly the same argument could be made for the original centipede game, and in the original game this argument isn't consistent with any equilibrium. Perhaps the best way to put it is: It is in the interests of both 1 and 2 to play A until near the end of the game. But each one will do so (in equilibrium) only if she believes that she can, on average, beat the other player to the punch. In an equilibrium, their beliefs have to be consistent, and it is inconsistent in the original game for each to think that she will outguess the other. But when we add a little incomplete information about player 1, player 2 can believe that he will "outguess" player 1 because, with small probability, player 1 will permit herself to be outguessed. From this small wedge, we are able to get them both to play A happily until near the end of the game.

How tied is this to the precise specification of incomplete information that we used? Certainly the number of rounds back and forth plays no role at all, once that number gets beyond the twenty-two nodes needed to reach a point where each plays A for sure. If you look over the proof, it doesn't matter whether there are 198 nodes, or 28, or 199,998; except for the last twenty-two, each side plays A with certainty. The size of the probability that 1 is benevolent isn't so important. If we changed the .001 to .0001 (one in ten thousand!), it would take four more pairs of rounds from the back to get to the point where both sides choose A with probability one. The nature of 1's "irrationality" isn't so crucial; the same basic conclusion would be derived if player 2 assesses small probability that player 1 will play A until sometime in the last ten rounds. And it isn't even necessary that 2 have any real doubts about 1's rationality. Imagine a game of incomplete information in which there are three possible initial nodes. The first has probability .998, and player 1 has precisely the payoffs as in the original centipede game. The second has probability .001, and player 1 has precisely the payoffs as in the original game. And the third has probability .001, and player 1 is "benevolent." Moreover,



player 2 knows when the first starting point prevails, but is unsure when one of the second two prevails which it is. And player 1 is unsure when one of the first two starting points prevail which it is. In other words, with probability .998 player 1 is "rational" and player 2 is aware of that. But in that event, player 1 isn't sure that player 2 is aware of that; player 1 thinks there is a small chance that player 2 thinks there is a good chance that player 1 is benevolent. Player 1, in this case, will give the move to player 2 to "see" if player 2 is perhaps confused (which player 2 indicates by giving the move back to player 1). And player 2, even if he isn't confused, gives the move back to player 1; the same sort of construction works.

These are very complex games with very complex equilibria. I do not want to suggest that the equilibria are in any sense obvious solutions. Whether people put in the position of having to play these games exhibit behavior that conforms to these equilibria is a nontrivial empirical question. For some initial investigation into this, see Camerer and Weigelt (1988).

At the same time, games like this demonstrate, I think, why the theoretical solution of the original centipede game isn't very intuitive. A one-in-one-thousand chance that one's opponent is generous, or that one's opponent assesses a substantial probability that oneself is benevolent, isn't much of a "change" in the game. Yet it completely changes the theoretical prediction. In any game where such a small change in the formulation can result in such a large change in the theoretical prediction, one will want to be very wary of the theoretical prediction. At the risk of confusing you unnecessarily, we should backreference the discussion of chapter 13 on subgame perfection at this point. The argument there was that if deviations can be attributed to an a priori small chance that players have very different payoffs from those in the model of the game, the motivation behind subgame perfection and sequential rationality might be missing. Here we see something worse: For games with this sort of back and forth structure, such attributions can lead to behavior that is very different from Nash equilibrium behavior in the original game, for reasonably sized perturbations. Let me put it this way: Before we attacked subgame perfection on grounds that it wasn't robust to this sort of perturbation. Now we are attacking the notion of Nash equilibrium, when one has a game like the centipede game where the set of Nash equilibria is not robust to these perturbations.<sup>k</sup>

And to tie this back to reputation and finitely repeated games: In the finitely repeated prisoners' dilemma, suppose one player assesses a small probability that the second will "irrationally" play the strategy of tit-for-tat:

<sup>k</sup> There is a theoretical difference between the two points. In our constructions at the end of chapter 13, used to show why a subgame imperfect equilibrium might be reasonable prediction, we can make the small probability that a player is "irrational" as small as we wish (if still strictly positive), and our argument is unaffected. In the construction here, if the probability that player 1 is benevolent is, say, smaller than  $1/2^{2^{10}}$ , then there is a substantial chance that player 1 will play D to start the game. (Put very mathematically, the Nash equilibrium correspondence is upper semi-continuous in these sorts of perturbations, for a fixed extensive form.) But this distinction is not of practical significance. If a  $1/1000$  chance of very different payoffs is enough to upset the analysis for a given game, surely the conclusions of the analysis are not altogether trustworthy.

Do to your opponent whatever your opponent did to you on the previous round. In a long-enough (but still finite) repetition of the game, if you think your opponent plays tit-for-tat, you will want to give cooperation a try. And even if your opponent isn't irrational in this fashion, the "rational" thing for her to do is to mimic this sort of behavior to keep cooperation alive. Or, in the chain-store game, imagine that the long-lived player 2 with small probability will fight any entry into any of his markets, even if it is financially costly, because he just doesn't like entrants. A "rational" player 2 would mimic this behavior to convince entrants that he will act in this fashion to keep them from entering. And so entrants, afraid of being fought either because 2 is really irrational or wishes to appear irrational, choose not to enter.

We can think of these effects as "reputation" in the sense that a player will act "irrationally" to keep alive the possibility that she is irrational, if being thought of as irrational will later be to that player's benefit through its influence on how others play. That is, in this sort of situation, players must weigh against the short-run benefits of certain actions the long-run consequences of their actions. I take this short- versus long-run trade-off to be the hallmark of reputation, thus qualifying for this sort of model as an example of the general phenomenon of reputation.

## 14.7. Bibliographic notes

When structuring a map of the literature on this topic, the following categories are useful. Divide at the broadest level between (I) constructions in the style of sections 14.2 through 14.6, which depend on an infinite horizon, and (II) those that use a bit of incomplete information and so can work in finite horizon models. At the next step, divide between (A) models where two or more players play the game repeatedly and (B) those where one or more of the players are short-lived. Then divide into three: (1) analysis of the average payoff criterion (and more sensitive criteria related to it); (2) analysis of the discounted payoff criterion, for a fixed discount rate less than one; and (3) analysis of the limit set of equilibria in the discounted payoff formulation, as the discount factor goes to one. Finally, we can divide between formulations (a) without and (b) with noisy observables.

On repeated games and the folk theorem with an average payoff criterion and no noise, or category (IA1a), Rubinstein (1979) is a very good place to begin. The seminal references for the case of noise (category [IA1b]) are Radner (1985), Rubinstein (1979), and Rubinstein and Yaari (1983).

When we move to discounted payoffs, most of the action concerns the structure of efficient equilibria. For the analysis without noisy observables (IA2a), consult Abreu (1988). See also Abreu (1986) on applica-



tions to Cournot competition, following on the seminal work of Friedman (1971). When noise is added (IA2b), see Abreu, Pearce, and Stacchetti (1989). Green and Porter (1984) and Abreu, Pearce, and Stacchetti (1986) concern noisy observables in repeated Cournot competition. We should also mention the classic, pregame theoretic paper on oligopolistic collusion by Stigler (1964).

The limit set of equilibria as the discount factor goes to one, without noise or category (IA3a), is the subject of the first half of Fudenberg and Maskin (1986). With noise added (category [IA3b]), see Fudenberg, Levine, and Maskin (1989) and Matsushima (1988).

Concerning models where there are short-run players, which we introduced under the rubric of reputation, the basic notion goes back at least as far as Simon (1951), who works with an example that we will discuss in chapter 20. A number of applications have appeared since, but (to my knowledge) the first formalization of a folk theorem for this situation is Fudenberg, Kreps, and Maskin (1989), which covers categories (IB1) and (IB3a). Fudenberg and Levine (1989) concerns category (IB3b).

Concerning cooperation and reputation in finite horizon models, Selten (1978) is the classic reference pointing to the problem, and Rosenthal (1981) contains a very important commentary and prospectus for later theoretical developments.<sup>8</sup> Benoit and Krishna (1985) show how, when the stage game has multiple equilibria, one can make folk-theorem-like constructions even in a finite horizon model. Models using incomplete information were first explored in Kreps, Milgrom, Roberts, and Wilson (1982) (cooperation in the prisoners' dilemma for two long-lived players), and Kreps and Wilson (1982) and Milgrom and Roberts (1982) (entry deterrence in a case with one long-lived player playing against a sequence of opponents). The back half of Fudenberg and Maskin (1986) concerns folk-theorem-like results for this sort of model, when all players are long-lived, while Fudenberg and Levine (1988) contains general analysis of the problem with one long-lived player. Fudenberg and Kreps (1987) concerns models of reputation where one player plays against many opponents simultaneously instead of in sequence. Hart (1985) analyzes the general case of repeated games with one-sided incomplete information and no discounting.

There is a large literature on repeated play of constant-sum games with incomplete information about which we have said nothing. Aumann (1985) provides a survey and introduction.

<sup>8</sup> This paper is also the origin of the centipede game.

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## 14.8. Problems

■ 1. In a number of places in this chapter, I used phrases like "XYZ is an equilibrium if  $q$  (the probability of going on to the next round) or  $\alpha$  (the discount factor) is large enough." In this problem, you are asked to puzzle through, What is large enough?

(a) In the prisoners' dilemma game, we said that as long as  $q \geq 3/8$ , the threat of any player to meet noncooperation with noncooperation forever would sustain cooperation. Give the calculations that support the  $3/8$ . We also said that for large enough  $q$ , it would be enough to threaten to meet noncooperation with one round of noncooperation. To be precise, assume each player adopts the strategy: Play cooperatively unless and until one's opponent plays noncooperatively. Then play noncooperatively for one round. Then go back to playing cooperatively even if your opponent didn't cooperate in the single round of "punishment." And if, by chance or otherwise, a player happens not to cooperate at a time when punishment isn't called for, then the player does not cooperate for the one stage of punishment. For which values of  $q$  is this an equilibrium?

(b) In the discussion of Stackelberg leadership arising from reputation (near the end of section 14.5), a "large-enough"  $\alpha$  was claimed to be one greater than  $9/17$ . Give calculations that lead to this particular number.

■ 2. In the discussion of noisy observables in section 14.3, we proposed in passing the following sort of model for repeated play of the prisoners' dilemma. Players choose their intended strategies on each round, passing instructions to a referee for implementation. The referee then implements those intentions "noisily": If told to play a noncooperative strategy, the referee plays this strategy with probability .8, but with probability .2, the referee plays cooperatively. If told to play cooperatively, the referee plays cooperatively with probability .8 and noncooperatively with probability .2. Players know what instructions they sent, and at the end of each round the referee's implementation of both strategies are revealed to both players. But players don't know what instructions their opponents sent. Assume that  $q = .9$  and that the random ending of the game in any round and the referee's random implementations of strategies are all mutually and serially independent as random variables.

(a) Prove that it is a Nash equilibrium for players always to send instructions to play noncooperatively. What is the expected payoff to the players in this equilibrium?

(b) Suppose players adopt the strategy: Instruct the referee to play cooperatively until either player fails to cooperate (in terms of the implemented strategy) and thereafter instruct the referee to play noncooperatively. Is this an equilibrium? If so, what are the expected payoffs to the players in this equilibrium?

(c) Suppose players adopt the strategy: Instruct the referee to play coop-

eratively until either player fails to cooperate. Then instruct the referee to play noncooperatively for  $N$  periods. Then instruct the referee to play cooperatively again until the next incident of noncooperation. For which  $N$  is this an equilibrium? For the smallest such  $N$ , what are the expected payoffs to the players in this equilibrium?

(d) Now suppose we have a less capricious referee — one that follows instructions with probability .95. Redo parts (a), (b), and (c).

■ 3. In terms of reputation for quality, imagine that we have a firm that can produce a good at any quality level in the interval  $[0, 1]$ , where 0 means a good of low quality, and 1 means a good of exceptionally high quality. Demand for the good depends on consumer perceptions of the good's quality. If consumers anticipate that the good is of quality  $q$ , their demand is given by  $P = 4 + 6q - X$ , where  $X$  is the quantity demanded. Costs of manufacture depend on the quality level of the good being produced; it costs a constant marginal  $2 + 6q^2$  to produce a unit of quality level  $q$ . Consider the repeated setting described in section 14.5: In each period, the firm chooses a quality level and price. Consumers see the price but not (until after they have bought) the quality level. But consumers do know the level of qualities the firm has produced previously. Assume the firm discounts profits with a discount factor  $\alpha = .9$ . For which levels of quality  $q$  is there a viable reputational equilibrium, that is, an equilibrium where (along the path) the firm produces in each period at quality level  $q$ ?

■ 4. Recall our discussion of the prisoners' dilemma game in circumstances where one distinguished player plays a sequence of opponents. Each opponent plays only once, but is able to observe what the one player did earlier. We remarked that in this case even though the one player can carry a reputation, it doesn't help because her opponents, who have no reputation, will always play noncooperatively no matter what she does or what she has done previously. In light of this, suppose we change the rules of the prisoners' dilemma so that first the opponent moves in full view of the distinguished player, and then that player responds. Will reputation help in this case?

■ 5. (a) What would happen to the Stackelberg leadership story at the end of section 14.5 if competition in each period were Bertrand style instead of Cournot style?

(b) What would happen to this story if we added noise as in Green and Porter? That is, firms pick their output quantities, and price is then a random function of the sum of the quantities. This is a very open-ended

question, but at least consider: Suppose we try to construct a "trigger price" equilibrium, where if price is observed beyond the trigger, we revert to the standard Cournot equilibrium for a while. Should the trigger be above or below the "average" equilibrium price?

■ 6. Prove that in the variation on the centipede game given in figure 14.5(b) the unique sequential equilibrium described is, in fact, the unique Nash equilibrium. (Hint: Take some presumed Nash equilibrium and suppose information set  $2n+1$  [for player 2] is the first unreached information set. Derive an immediate contradiction. Then suppose that node  $(2n)t$  is the first unreached information set and derive a contradiction that is one degree removed from immediate.)

■ 7. The purpose of this problem is to explore what happens in duopoly when the duopolists are able to use two-part tariffs.

(a) Imagine two commodities — snaffle and manna. A single consumer has a utility function of the form  $u(x, z) = x - bx^2 + z$ , where  $x$  is the amount of snaffle consumed, and  $z$  is the amount of manna. The price of manna is always set to be 1. This utility function applies for all levels of snaffle, which just goes to show that too much snaffle can be a bad thing.

Two firms sell snaffle, called Ace Snaffle Vendors (ASV) and United Zeus Snaffle (UZS). Although it won't be relevant until the next part of the problem, both of these firms have a simple production technology; they can produce as much snaffle as they want at a variable unit cost of .1 per unit. In addition, they each face a fixed cost  $K$  if they decide to produce any snaffle at all; we don't preclude the possibility that  $K = 0$ . Each firm charges a two-part tariff for snaffle; if a consumer buys any snaffle from the firm at all, the consumer must pay  $k_i$  up front, and in addition the consumer must pay  $c_i$  per unit snaffle consumed. Here,  $i = 1$  for ASV and  $i = 2$  for UZS. (You may assume that both  $k_i$  and  $c_i$  must be nonnegative.)

Suppose our single consumer faces prices  $k_1$  and  $c_1$  from ASV and  $k_2$  and  $c_2$  from UZS. Snaffle is snaffle — it doesn't matter from whom it comes. Characterize as best you can the demand of the consumer.

(b) There is a large number of identical consumers in this economy — say, 100,000. Suppose the snaffle market operates as follows. First, ASV and UZS must declare (simultaneously and independently) whether they wish to produce any snaffle. If they decide to do so, then they incur the fixed cost  $K$ . Each sees whether the other has decided to be active or not, and then the active firm(s) (simultaneously and independently) declares its two-part tariff scheme  $k_i$  and  $c_i$ . Consumers see those two-part tariffs

and then decide whether to order any snaffle at all, and from whom. Consumers do this to maximize their utility. Firms act to maximize their (expected) profits.

We are looking for all the subgame-perfect, pure strategy equilibria of this game. As a function of  $K$ , what are they?

(c) Now suppose that this sort of game is repeated indefinitely. (N.B., in each period, firms decide whether to "be in" or not [and thus incur the fixed cost that period]; a firm can be out in one period and in the next.) Consumers discount their utility in each period at discount rate .9 per period. Firms discount their profits at the same rate. For  $K = 1$ , which of the following describe the outcome of a subgame perfect equilibrium. Support your contentions.

(i) Firms are both active in every period, and each makes half of what a monopoly firm would make.

(ii) Only one firm (UZS, say) is active in every period, and this firm makes what a monopoly firm would make.

(iii) Only one firm (ASV, say) is active in every period, and this firm makes zero profits in each period.

## chapter fifteen

### Bilateral bargaining

After discussing monopoly, many principles and intermediate microeconomics books move on to the cases of *monopsony* (one buyer and many sellers) and *bilateral monopoly* (one buyer and one seller). The analysis of monopsony is straightforward after monopoly; the monopsonist, taking his supply curve  $S(x)$  as given, equates the marginal value of the factor to him with the "marginal factor cost," or  $S(x) + xS'(x)$ . Then there is a paragraph or two about bilateral monopoly. Being unjust to many fine textbooks, an abridged version of what is typically said follows:

In cases of *bilateral monopoly*, sharp predictions as to what will happen cannot be offered. An efficient exchange between the buyer and seller may be presumed where the marginal cost to the seller is equated to the marginal value of the factor to the buyer. [Many books will be more sophisticated than this and indicate that inefficient trades may take place, owing to the bargaining process.] But the price at which this exchange takes place is indeterminate. The buyer and the seller will, we can safely assume, not agree to a price where either is worse off than if no exchange at all is made. But in general, many prices will meet that requirement and which one will be agreed to is determined by psychological factors; things that determine the bargaining abilities of the two parties, things that are outside the ken of economic science.

This sort of discussion is not limited to bilateral monopoly; in texts that introduce the Edgeworth box and take it seriously as a model of exchange between two individuals, the contract curve (locus of efficient points) will be identified, and there will be a general presumption that some exchange will take place on the contract curve between the indifference curves through the endowment, but typically no prediction will be offered about which point will be chosen.

In this short chapter we will see a bit of what noncooperative game theory has to offer to this rather dismal state of affairs. One might hope

that by being more precise about the protocol of bargaining — the rules by which offers are made and accepted or rejected — we can get some more definite idea about what will be the outcome. We will see that this is a case where institutions matter (according to the theory), and rather more than seems sensible, although the extreme and unintuitive sensitivity to institutional form that we see in the theory may be attributable to the starkness of our models in terms of what players know about each other.

### 15.1. Simultaneous offers and indeterminacy

To make the problem of bilateral bargaining as simple as possible, consider the following one-dimensional version. Two parties, called 1 and 2, are put into a room and told to negotiate on the simple problem of splitting one dollar. If they can agree on a division (within a specified time period), the division they agree to will be implemented. If they fail to agree, each gets nothing.

In fact, this is no more or less stark than the general bargaining problem of an Edgeworth box, if you think about the original endowment as the “no agreement” point, and splits of the dollar representing points along the contract curve inside the lens defined by the two indifference curves through the endowment point. If we allowed the two parties to split the dollar in ways that leave money on the table — for example, 1 gets fifty cents and 2 gets twenty-five — then we would even have representatives of points inside the lens that are off the contract curve.

What will happen here? To sharpen the question, but not the answer, let us be more specific about the rules of this bargaining “game.” Suppose that bargaining is to proceed as follows: The two parties simultaneously and independently name a demand. Ms. 1 says, for example, that she demands seventy-five cents, while Mr. 2 demands fifty cents. If these two demands are compatible (which includes the possibility that they leave money on the table), they are implemented and the game ends. If not, each party, knowing what the other party demanded, decides between *standing firm* on his original demand or *acceding* to the other’s demand. These decisions are made simultaneously and independently. If both stand firm, both get nothing and the game is over. If one stands firm and the other accedes, the demand of the first person is implemented. If both accede, each gets his or her part of the other person’s original demand, which necessarily leaves money on the table.

What we have done here is to move from an ambiguous description of bargaining to very specific, formal rules of negotiation. You should have

no trouble depicting the game just described as an extensive form game, especially if we insist that the first round of demands must be in multiples of a penny (so the game has finitely many strategies for each player). This allows us to analyze bargaining using the tools of noncooperative game theory.

For example, we can ask, What are the possible (Nash) equilibria of the game described? Imagine that 1 is playing the strategy: Demand  $n$  cents on the first round and stick to that demand on the second, if there is a second. Player 2’s best response to this is to demand  $100 - n$  or more on the first round and to accede on the second if he demands more than  $100 - n$  on the first. Let us choose one of these for his strategy: He will demand  $100 - n$  on the first round, and he will stick to that demand on the second. Then 1’s strategy is also a best response to 2’s, and we have a Nash equilibrium with the split  $n$  to player 1 and  $100 - n$  to 2.<sup>a</sup> Since  $n$  is completely arbitrary here (it even works if  $n = 0$  or  $100!$ ), we see that *any* division of the dollar is a possible equilibrium. (We can also construct equilibria where the split isn’t achieved until the second round.)

So we have offered a more precise description of the bargaining process, but we have gotten back the basic indeterminacy of the answer. Given the very simple nature of this bargaining protocol, we probably can’t say that the result will depend on the players’ “relative bargaining abilities.” But assuming they play to an equilibrium, the players’ equilibrium expectations are all-important.

This is not the only bargaining protocol that leads to this sort of indeterminacy. Some variations with the same basic property include:

- (a) Instead of the rules given above, each side simultaneously makes a demand. If demands are compatible, the dollar is split. If not, each side again makes a demand, and so on, for  $M$  rounds of demands. If no agreement is reached by the end of  $M$  rounds, the game is over and each side gets nothing.
- (b) The same as (a), except there is no last round; they can make demands for as long as it takes to reach an agreement. If no agreement is reached, each side gets nothing. Or, instead, at the end of each round, either side can simply quit.
- (c) The same as (a), except that after each round one penny is removed from the pot. In the first round, the players bargain to split a dollar; in

<sup>a</sup> This equilibrium is not subgame perfect. To make it perfect, add: 1 accedes to 2 if 1 asks for more than  $n$  on the first round; 2 accedes to 1 if 2 asks for more than  $100 - n$  on the first round and 1 asks for  $n$  or less. Do you see why these modifications are needed?

the second round, they bargain to split ninety-nine cents; and so on, until all the money is removed (at which point the game is over).

Under any of the rules given, any split, achieved in the first round, is a Nash equilibrium. You should be able to show that in the originally named protocol or in the protocols given in (a) and (b) above, pure strategy Nash equilibria that result in an agreement are always efficient, and the only pure strategy equilibria that don't result in agreement are very special. Protocol (c) opens other possibilities. Consider, for example, the strategies:

Demand sixty cents on the first two rounds. If your opponent demands sixty cents on the first two rounds, demand forty-nine cents on the third round. If your opponent does anything else on the first two rounds, demand whatever is left in the pot less one cent until all the money is gone.

You should be able to see that this strategy is a best response against itself, leading to a split of ninety-eight cents (only) on the third round.<sup>b</sup>

## 15.2. Focal equilibria

If you sat actual players down to play exactly according to the rules given above (one round of demands, then one round of "stand" or "accede"), you would find a large number of pairs playing the equilibrium where the dollar is split fifty-fifty. The symmetry of this equilibrium makes it focal. In a long series of experiments, Alvin Roth and his associates have attempted to see just what makes for a good focal point. This series is summarized in Roth (1983); one of his experiments will be reported here (this one in collaboration with Schoumaker, reported in Roth and Schoumaker [1983]).

To begin, we change the rules a little bit. Instead of bargaining over splitting a dollar, we imagine that the two parties are bargaining over one hundred poker chips. The rules for bargaining are the same: One round of simultaneous demands and, if necessary, one round of simultaneous "stand" or "accede." But prizes are awarded a bit differently. At the start, each party is given a prize that he or she might win. Suppose, for example, that 1 might win \$20, and 2 might win \$5. Each side knows what the two prizes are, each side knows that the other side knows this,

<sup>b</sup> Does this strategy against itself constitute a subgame perfect equilibrium? Why not? How could you modify the pair of strategies so that they give a subgame perfect equilibrium while retaining the basic character of the outcome?

and so on. (In other experiments, Roth tried variations in which one party knew the value of both prizes but one party only knew the value of his own.) Suppose the two agree to split the chips: sixty for 1 and thirty-five for 2, leaving five on the table. Then 1 is given a .6 chance of winning her prize of \$20, and 2 is given a .35 chance of winning his prize of \$5. In other words, each chip is worth a one percent chance of winning a prize, whatever that prize is. (If no agreement is reached, neither side is given any chance at his or her respective prize.)

Just as in the simple split-the-dollar game, any division of the poker chips is a Nash equilibrium of the game; you should have no trouble adapting the old argument to this new setting. But now a fifty-fifty split is no longer so obviously "focal." Now there are at least two competitors:

(a) Give all the chips to 1, since she has the bigger prize.

(b) Split the chips twenty to 1 and eighty to 2. The virtue of this split is that it gives player 1 an expected prize of  $(.2)(\$20) = \$4$  and it gives player 2 an expected prize of  $(.8)(\$5) = \$4$ . That is, this split equalizes the expected prizes of the two.

In earlier experiments, both the fifty-fifty focal equilibrium and the twenty-eighty focal equilibrium had been observed. (Giving all the chips to 1 was not observed.) So Roth and Schoumaker tried the following experiment. All bargaining was done via a computer, so that players never saw each other. Thus Roth and Schoumaker could "condition" their subjects by having them bargain with computers for a while. Subjects played the bargaining game (via a computer) a number of times. Unbeknownst to them, in the first few rounds their opponent was the computer. For some subjects, the computer was programmed to play the fifty-fifty split. For others, the computer was programmed to play the twenty-eighty split. In this way, some subjects were conditioned to believe that the fifty-fifty split was fair or appropriate, while others were conditioned to believe in the twenty-eighty split. And then conditioned subjects were mixed, with results that you can no doubt anticipate: When two subjects who had been conditioned in compatible fashion were paired, they happily went along with the equilibrium they had been taught. When two subjects were paired, one of whom thought he deserved only twenty chips, while the other was willing to settle for fifty, money was left on the table, at least for a while. And when someone who had been trained to expect eighty was paired with someone who had been conditioned to think that fifty-fifty was right, disagreement outcomes were reached.

This experiment reinforces the notion that the bargaining outcomes depend on individual's expectations as to what the outcomes should be. In fact, it makes this point vividly by showing how those expectations can be manipulated through experience. Insofar as noncooperative game theory is silent (or nearly so) on the subject of where equilibrium expectations come from, it seems that the basic textbook commentary on bilateral monopoly and bargaining had it right.

### 15.3. Rubinstein's model

Before giving up on noncooperative game theory as applied to bargaining, however, let us look at one more bargaining protocol, back in the context of splitting a dollar. Imagine that the two players take turns making offers and counteroffers. Player 1 goes first, say; she proposes a split. Player 2 may accept that split immediately or, after a five second delay, make a counteroffer. Player 1, in response to a counteroffer, may accept immediately or, after a five second delay, make a counter-counteroffer, and so forth.

One further feature is added: Players would rather have the money sooner than later. We suppose that both players discount any money they receive at a rate of, say, 10% per year. So if player 1 obtains  $n$  immediately after her first offer, this is worth  $n$  to her. If she obtains  $n$  after player 2's first counteroffer, this is worth slightly less; it is worth  $\delta n$ , where, by the rules of continuous compounding,  $\delta = .999999985$  approximately. If she obtains  $n$  on the third round (on her second offer), it is worth only  $\delta^2 n$  to her, and so on. The same is true for player 2.

This, you might think, changes very little. Suppose, for example, player 1 plays the strategy: Demand  $n$  the first round and refuse to accept any offer or make any offer that gives her less than  $n$ . It would seem that 2's best response to this, no matter what  $n$  is, is to give in immediately. But this doesn't constitute a subgame perfect equilibrium for, say,  $n = 60$ . To see why, note first that

$$\frac{40}{\delta} + 60\delta < 100.$$

In case this isn't obvious, we give a proof. Consider the function  $f(x) = x/\delta + (100 - x)\delta$ . Its derivative is  $1/\delta - \delta$  which, for  $\delta < 1$  is greater than zero. Now let  $n^* = 100\delta/(1 + \delta)$ . Note that  $n^*$  is a tiny bit less than 50, for

the particular  $\delta$  with which we are working; in particular,  $n^* > 40$ . As a matter of simple algebra,

$$\frac{n^*}{\delta} + (100 - n^*)\delta = 100.$$

(Do it!) So

$$\frac{40}{\delta} + 60\delta = f(40) < f(n^*) = \frac{n^*}{\delta} + (100 - n^*)\delta = 100.$$

Let  $\epsilon$  be half the difference between 100 and  $40/\delta + 60\delta$ . And consider the following strategy by player 2. Refuse to take only 40 on the first round. Offer instead to take  $40/\delta + \epsilon$  on the second round. If this offer is accepted, player 2 is better off by  $\delta\epsilon$  for having made the offer instead of accepting 40. Will player 1 accept? *If she thinks that by turning this offer down she will net 60 on the third round of offers, player 1 does better to accept player 2's offer.* By accepting player 2's offer, player 1 obtains  $\delta(100 - 40/\delta - \epsilon) = \delta(60\delta + \epsilon)$  in units that are discounted back to the start of the game. If she spurns player 2's offer, and if she expects to get 60 when it is again her turn to offer, she nets  $60\delta^2$  discounted back to the start of the game. So she is better off accepting 2's offer, and 2 is better off making this offer than to accept 40 only on the first round.

#### The basic result, with proofs

The only place that 60-40 played a role in this argument came where we stated that  $40/\delta + 60\delta < 100$ . That will be true of any split where player 1 gets more than  $100/(1 + \delta)$ . And it suggests the following remarkable result:

**Proposition 1 (Rubinstein, 1982).** *The unique subgame perfect equilibrium in the alternating offers model just given is for the player making the offer always to offer to take  $100/(1 + \delta)$  for himself and leave  $100\delta/(1 + \delta)$  for the other party, and for the other party always to accept this offer or any better and to spurn any offer that is worse.*

We have to show that this is a subgame perfect equilibrium and that it is unique. Showing that it is a subgame perfect equilibrium is not difficult. Imagine first that it is your turn to make the offer. If you offer to take  $100/(1 + \delta)$ , your offer will be accepted. There is no reason, then, to offer to take anything less. If you ask for more, you will be turned down, and



your rival next time will offer you only  $100 - 100/(1 + \delta) = 100\delta/(1 + \delta)$ . You are meant to accept this, so from the point where you are making the offer, the strategy described above is unimprovable for you in a single step.

It is more interesting to consider whether your strategy is unimprovable when you are being made an offer. Suppose your rival offers to take  $n$ , leaving you with  $100 - n$ . If you reject this offer and then conform to the equilibrium above, you will get  $100/(1 + \delta)$  next time. In current value units, this is worth  $100\delta/(1 + \delta)$  to you today. So you should accept whenever  $100 - n > 100\delta/(1 + \delta)$ , reject when the inequality is strict the other way, and do either when there is equality. But if you work through the algebra, this is precisely the strategy you are meant to be carrying out. We have equality when  $n = 100/(1 + \delta)$ , and we resolve the knife-edge case of equality by having you accept.

Thus we have a subgame perfect equilibrium. The remarkable part of the proposition is that this is the unique subgame perfect equilibrium. We will sketch an argument due to Shaked and Sutton (1984).

*Step 1. Let  $\nu$  be the largest amount that any player can achieve in any subgame perfect equilibrium when the player is making the offer. We claim that  $\nu \leq 100/(1 + \delta)$ . The proof essentially uses the argument we gave just before the proposition. First, if  $\nu$  is the largest amount that a player can achieve in any subgame perfect equilibrium when it is his turn to make the offer, then an offer to give a player  $\delta\nu + \epsilon$  for any  $\epsilon > 0$  is sure to be accepted; the best that a player receiving this offer can do by rejecting it is  $\nu$  a period later, which nets  $\delta\nu$  in current value terms, which is less than the  $\delta\nu + \epsilon$  that is offered.*

Second, if  $\nu$  is the largest amount that a player can achieve in any subgame perfect equilibrium when it is his turn to make the offer, it must be achieved by making an acceptable offer on the first round. If the first offer is rejected, then the second player can offer, say,  $(\delta\nu + \nu)/2$ , which the first player is bound to accept by the argument just given (since  $(\delta\nu + \nu)/2 > \delta\nu$ ). But  $(\delta\nu + \nu)/2 < \nu$ , and since it is delayed for a period, it is worth  $\delta(\delta\nu + \nu)/2$ , which is even less.

So  $\nu$  must be achieved by making an initial offer that is accepted.<sup>c</sup> Suppose  $\nu > 100/(1 + \delta)$  and the first player demands  $\nu$ . The second player can reject this and offer  $\delta\nu + \epsilon$ , for  $\epsilon$  equal to half the difference

<sup>c</sup> This is a bit sloppy, in that it presumes that the least upper bound on subgame perfect payoffs is achieved in a subgame perfect equilibrium. If we were being careful, we would have  $\nu$  be the supremum over all subgame perfect equilibrium payoffs and then show that any demand sufficiently close to  $\nu$  will be rejected if  $\nu > 100/(1 + \delta)$ .

between  $100$  and  $\delta\nu + (100 - \nu)/\delta$ , which is positive if  $\nu > 100/(1 + \delta)$ . This is sure to be accepted by the first paragraph. And it gives the second player a payoff of  $100 - \delta\nu - \epsilon = (100 - \nu)/\delta + \epsilon$  next period, or  $100 - \nu + \delta\epsilon$  in present value terms, which is better than the  $100 - \nu$  obtained by acceding to the initial demand of the first player. Hence a demand of  $\nu$  will be rejected if  $\nu > 100/(1 + \delta)$ , and  $\nu > 100/(1 + \delta)$  cannot be a subgame perfect equilibrium payoff for the first player.

*Step 2. Given this, any player can be certain that any offer to give more than  $100\delta/(1 + \delta)$  will be taken. By rejecting this offer, the best one's rival can do is  $100/(1 + \delta)$  starting with her next offer, which is strictly worse than  $100\delta/(1 + \delta) + \epsilon$  for any  $\epsilon > 0$ , however small, immediately.*

*Step 3. The party to whom the first offer is made can never have a payoff exceeding  $100\delta/(1 + \delta)$ . For if this party had a payoff exceeding this level, the payoff to the party making the first offer would have to be less than  $100/(1 + \delta)$ , which contradicts step 2.*

*Step 4. In equilibrium, any offer to give less than  $100\delta/(1 + \delta)$  will certainly be rejected. In fact, the party to whom the first offer is made is assured of a payoff (in present value terms) of at least  $100\delta/(1 + \delta)$ . If offered less than  $100\delta/(1 + \delta)$ , one can reject and offer a bit more than  $100\delta/(1 + \delta)$  to the other side next time. This is sure to be taken, by step 2. But this means you can be sure to get any amount strictly less than  $100/(1 + \delta)$  next time, some one of which will be better than taking something less than  $100\delta/(1 + \delta)$  this time.*

*Step 5. In the unique subgame perfect equilibrium, the party making the first offer proposes to take  $100/(1 + \delta)$  and to give  $100\delta/(1 + \delta)$ , and this is accepted with certainty. No offer to give more can be part of an equilibrium, by step 2. No offer to give less can be part of an equilibrium. It will certainly be rejected by step 4, and then the best one can do, starting from the next period, is less than  $100\delta/(1 + \delta)$  (by step 3). This is worse than taking something a bit less than  $100/(1 + \delta)$ , which is certain to be accepted, by step 2. So the only possible equilibrium offer is  $100/(1 + \delta)$ . And that must be taken with certainty in any equilibrium: If not, the party making the first offer would do better to offer slightly less than  $100/(1 + \delta)$  and be certain of acceptance.*

Where, you may ask, did we use subgame perfection? We used it very directly in steps 1 and 2, when we said that an offer would certainly be taken, even if the offer is out of equilibrium, because the person to whom it was made could not do better by rejecting it. If we looked only

at Nash equilibria, players' responses to counteroffers that are made out of equilibrium are not bound by "rationality" constraints of this sort.

At the risk of overdoing it, we claim a further result: The equilibrium described above is the only equilibrium that survives iterated elimination of dominated strategies.<sup>1</sup> We will only get the argument started and leave it to you to finish. (a) Suppose player 1 offers to player 2 that he (player 2) may have more than  $100\delta$ . Then player 2 is bound to accept; if he rejects, he must wait a period (that is, five seconds) and then the best he could conceivably do is to get the entire dollar, which is worth (in present value terms) no more than  $100\delta$ . That is, we can remove from consideration any strategy that includes rejection of such a generous offer. (b) Suppose player 1 offers to player 2 something less than  $(100 - 100\delta)\delta$ . Then player 2 should definitely turn this down, because he can, in his turn, offer to player 1 that she take  $100\delta$  and be sure to be accepted. This would (next period) leave him with no worse than  $(100 - 100\delta)$ , which in present value terms is no worse than  $(100 - 100\delta)\delta$ . There is no reason to take less today. Thus we can remove from consideration any strategy that includes acceptance of such a miserly offer. (c) The argument in (b) means that a player can never hope to get more than  $100(1 - \delta + \delta^2)$ . Suppose, then, one player offers something more than  $100\delta(1 - \delta + \delta^2)$  to his rival. If she turns it down, she can hope to do no better than  $100(1 - \delta + \delta^2)$  starting next period, which is worse. So she is sure to accept. We can remove strategies that include rejection of such generous (but less generous than in [a]) offers. (d) Now that you know a slightly wider band of offers that are sure to be accepted, return to part (b) and expand the set of offers that are sure to be rejected, and so on. In the end, all strategies that accept less or reject more than  $100\delta/(1 + \delta)$  are removed, and the equilibrium is forced.

### *Bargaining over general issues*

The result just given extends far beyond splitting a dollar. Imagine two players bargaining over many different commodities or many different clauses in a contract. In general, we might imagine that some abstract set  $X$  represents the range of possible agreements. For example, if the two parties are bargaining over how to split a bundle  $e \in R^K$  of commodities, then  $X = \{(x_1, x_2) \in R^{2K} : x_1 + x_2 \leq e\}$ , with the interpretation that the point  $x = (x_1, x_2)$  represents an agreement where the first person gets the bundle  $x_1$ , the second gets  $x_2$ , and  $e - x_1 - x_2$  is left on the table. Or we could imagine two parties bargaining over a labor contract, and then the various components of  $x$  represent the basic wage rate, the level and nature of overtime wages, the level and nature of health benefits, work rules, and so on.

<sup>1</sup> I first heard this argument from Drew Fudenberg.

Each of the two parties derives utility from the various possible splits; we let  $u_i(x)$  be the utility that  $i$  obtains (immediately) from an agreement on  $x$ . If agreement is delayed until time  $t$ ,  $i$  derives net utility  $\delta^t u_i(x)$ , with  $u_i$  normalized so that no agreement gives zero utility. Finally, we imagine alternating offer bargaining; the first player makes an offer (of the form  $x$ ) at date 0, to which the second player can agree or not. If the second player rejects this offer, then the second party can make a counteroffer after a delay of one period, which the first player accepts or rejects, and so on.

As long as the preferences (utility functions) of the two players are common knowledge between them, this situation is not much more complex than the basic model of splitting one dollar. There is a set of feasible utility imputations, the image of  $X$  in  $R^2$  under the vector function  $(u_1, u_2)$  (see section 5.1 if you've forgotten about utility imputations), and the two are (essentially) bargaining over which point in the set of feasible utility imputations to select. We will write  $V$  for the space of feasible utility imputations, and  $V^*$  for the space of feasible utility imputations that are as good as the no-agreement outcome; i.e.,  $v = (v_1, v_2) \in V^*$  if  $v = (u_1(x), u_2(x))$  for some  $x \in X$  and  $v_1, v_2 \geq 0$ .

Assume that  $V^*$  is a compact, convex set. Assume that the Pareto frontier of  $V^*$  is strictly downward sloping. And assume there is some point  $v^*$  in  $V^*$  that is strictly better for both players than disagreement. (You may wish to consider more fundamental assumptions that would justify these assumptions.)

**Proposition 2.** *The (essentially) unique subgame perfect equilibrium of this bargaining game has player 1 offering a point  $x_1$  whenever it is her turn to offer, player 2 offering a point  $x_2$ , and both sides accepting the other's offer, where  $x_1$  and  $x_2$  are Pareto efficient agreements satisfying the equations*

$$\delta u_1(x^1) = u_1(x^2) \quad \text{and} \quad \delta u_2(x^2) = u_2(x^1).$$

We have called this an (essentially) unique equilibrium because the utility imputations  $v^1 = (u_1(x^1), u_2(x^1))$  and  $v^2 = (u_1(x^2), u_2(x^2))$  are unique. But there may be more than one pair of points  $x^1$  and  $x^2$  that give these utility imputations.

The proof is left as an exercise, or see Binmore, Osborne, and Rubinstein (forthcoming). If you take on this exercise, you may find it interesting to consider what happens to the solution as  $\delta$  goes to one. (Assume the Pareto frontier of  $V^*$  is given by a differentiable curve.)

### *Variations*

If it seems remarkable that with these rules we get such definite predictions, let us observe that similar arguments give even more remarkable results. We will give five variations on proposition 1 (so we are back to splitting one dollar), each of which you can prove using the techniques sketched previously.

(1) *Outside options.* Suppose that the bargaining game is just as before, except that player 1 has an outside option. Specifically, she can, after hearing an offer from player 2, choose to accept, to make a counteroffer (after five seconds), or to break off negotiations and take her outside option, which gives her, say, forty-five cents and leaves player 2 with nothing.

Your intuition might be that this strengthens the bargaining position of player 1. But in terms of the theory it does not; the old subgame perfect equilibrium remains the unique subgame perfect equilibrium in this case. (This depends on our choice of  $\delta$ ; in general, it requires that  $\delta/(1+\delta) > .45$ .)

(2) *Different discount rates.* Suppose that the bargaining game is as above, except that player 1 discounts her payoff at an interest rate of 5% per year, and player 2 discounts his payoff at 15% per year. Then the unique subgame perfect equilibrium gives approximately seventy-five cents to player 1 and twenty-five cents to player 2.

This may not seem entirely unintuitive to you. Player 1 is in a stronger bargaining position because she is more patient. But it may be surprising that, when offers are made every five seconds, the difference in levels of patience should have such dramatic results in terms of the outcome.

(3) *Different speeds of response.* Suppose that the bargaining game is as before (with both parties having the same discount rate), but player 1 can make a counteroffer two seconds after hearing player 2's offer, and player 2 takes six seconds to formulate and make a counteroffer. Then the unique subgame perfect equilibrium gives approximately seventy-five cents to player 1 and twenty-five cents to player 2.

(4) *A variation on different speeds of response.* Suppose that the bargaining game is as before, with two changes: Player 1 can respond two seconds after player 2 makes an offer, and player 2 can respond six seconds after player 1 makes an offer. And acceptance is delayed as much as a counteroffer; e.g., if player 1 makes an offer, player 2 cannot respond in any fashion either to accept or to counteroffer for six seconds. Then the unique subgame perfect equilibrium gives approximately twenty-five cents to player 1 and seventy-five cents to player 2.

(5) *Costs of rejecting offers instead of discounting.* Suppose that the players are bargaining to split \$10. In this formulation, they take turns making offers, but there is no discounting. Instead, every time player 1 turns down an offer of player 2, she must pay a fee of \$.10, whereas every time player 2 turns down an offer of player 1, he must pay a fee of \$.11. Then the unique subgame perfect equilibrium is: If player 1 is making the offer, she proposes to take the entire \$10, to which player 2 agrees. If player 2

is making the offer, he offers to take \$.10 and to give her \$9.90, to which she assents.

Although we didn't say it, in each of these variations, all the subgame perfect equilibria have the initial offer accepted with probability one.

(Readers often find the result for variation (5) astounding, so let us sketch the argument here. Let  $x$  be the highest subgame perfect payoff player 2 can get if he makes the first offer. Suppose  $x \geq .11$ . Then if player 1 has the initial offer, she can offer  $x - .11 + \epsilon$  to player 2 and be sure it will be accepted, for any  $\epsilon > 0$ , since if player 2 rejects, the best he can do is  $x - .11$ . If player 2 proposes, in his turn, to take  $x$  and leave player 1 with  $10 - x$ , player 1 will reject the offer, since she can get  $10.00 - x + .11 - \epsilon$  next time, at a cost of only \$.10. This will leave player 2 with  $x - .11 + \epsilon$  next time, which is certainly less than  $x$ . So there is no way that  $x$  can exceed or equal \$.11.<sup>d</sup> Suppose  $x < .11$ . Then player 1, when she has the offer, can be sure of acceptance if she proposes to take the entire \$10.00. Hence she will not agree to take anything less than \$9.90 when it is 2's turn to offer; better to refuse and then pocket the entire \$10.00. The rest of the argument is easy.)

### What lies behind these results

If the result that there is a unique subgame perfect equilibrium is surprising, some of the further results listed above are positively astounding. You might make the case that (1) while perhaps a bit counterintuitive is not outlandish. And there is even some sense to (2), although the differences in discount rates seem to lead to quite extraordinary differences in the outcome. But what are we to make of (3), (4), and (5)? The "solution" bounces around an incredible amount for what seem to be insignificant differences in the bargaining protocol. And, in any case, you might wonder why alternating offer bargaining leads to such different results than simultaneous offer bargaining. Is it the alternation per se?

To understand what is driving the basic Rubinstein result and why we get such extraordinary results in the variations above, consider the following very simple form of bargaining over a dollar, in a situation where players discount their payoffs at a rate of, say, 10% per year. Player 1 makes the first proposal and, at the same time she makes her offer, she is able to "lock" her offer for any length of time she desires up to, say, one hundred years. That is, she can say, "I propose to take ninety-five cents and to leave five cents for you. And I hereby disable myself from amending

<sup>d</sup> This last step is a little sloppy, and we leave it to the careful reader to make it right.

this generous offer or from accepting any counteroffer you might choose to make for ninety-eight years. What is your response?"

Player 2, faced with this offer, is going to accept (if he doesn't get mad, a sentiment with which we will deal in a bit but ignore for now). If he rejects this offer, he will be unable to do anything for ninety-eight years. And even if after ninety-eight years he gets the entire one dollar, it will be worth far less to him (when discounted) than five cents today.

The point is that player 1 by locking her offer puts the onus entirely on player 2. If player 2 wants a better deal, he must wait; he alone can avoid the waiting period by accepting player 1's offer.

This is what is going on in Rubinstein-style bargaining. Once one party makes an offer, the other party alone can avoid the expense of waiting by accepting. If the offer is insufficiently generous, the respondent may indeed wish to suffer the cost of waiting. But then the party making the offer, in equilibrium, makes an offer just sufficiently generous so that the other party is indifferent between accepting or waiting the necessary length of time and then making an offer that is just sufficiently generous so the first party is indifferent between accepting or waiting.... The uniqueness result is not quite intuitive or obvious, but you should be convinced that what drives these equilibria is that each party when it is making an offer is able to put the onus of waiting entirely on the other side.

With this in mind, consider variations (3) and (4) above. In variation (3), player 1 makes an offer that player 2 can accept or reject immediately. But if player 2 wishes to counteroffer, he must wait six seconds. Player 1, on the other hand, must wait only two seconds before counteroffering. Hence, having heard player 1's offer, if player 2 wishes to counteroffer, he must pay three times the waiting cost that player 1 must pay if she wishes to counteroffer. In variation (4), player 1's offer cannot be accepted by player 2 for six seconds, which is the same length of time it takes player 2 to formulate a counteroffer. So the "waiting cost" to player 2 of counteroffering is not six seconds — he must wait that long anyway — but the two additional seconds it takes player 1 to accept or counter. And the waiting cost incurred by player 1 of rejecting an offer of 2 is the six further seconds it will take 2 to respond in any fashion. Hence in (4), player 1 pays the much heavier waiting cost, and her bargaining position is much weaker.

Alternating offers, per se, are not the key here. This point can be made by considering the model of Perry and Reny (1989). In this variation, players can make offers whenever they wish. That is, at any point in time a player is able to make an offer, even if his opponent is simultaneously making an offer. But there is one limitation: If player 1 makes an offer

at time  $t$ , then she cannot change the offer she has made for a set small amount of time, say for six seconds. And player 2 cannot change his offer for some given amount of time, say two seconds. Otherwise, when offers are made is entirely up to the players; there is no alternating structure of anything like that. Bargaining terminates when the two offers made are compatible.<sup>e</sup> Payoffs are discounted at given rates which, for the sake of exposition, we take to be identical.

This game doesn't quite have a unique subgame perfect equilibrium, but it comes close. Specifically, every subgame perfect equilibrium gives players payoffs that are approximately seventy-five cents to player 1 and twenty-five cents to player 2. How does this work? Imagine that the very best equilibrium outcome for player 1 gives her appreciably more than seventy-five cents. Look at some equilibrium that supports this outcome. Player 2 can deviate from this equilibrium by proposing slightly better terms for himself at a time when player 1 is either free to move or is just about to become free to move. By doing so, player 2 freezes his own ability to move for two seconds. *Now everything is up to player 1. She can agree to what player 2 has just proposed, or she can counterpropose, but if she counterproposes, she knows that player 2 is unable to respond for two seconds. The cost of waiting those two seconds is entirely on her shoulders; player 2 is committed for this length of time.* This is just the wedge we had before in alternating offers, and it works virtually the same way here, even though there is no set order of moves and simultaneous offers are feasible, in principle. The key to bargaining power here, in Rubinstein's model, and in the other variations given comes from the ability to put the onus of waiting entirely on the other party.

#### 15.4. The experimental evidence about alternating offers

While the logical key to Rubinstein-style results may now be clear, this isn't to say that we believe these results are empirically valid. To take an analogy, we may understand why the unique subgame perfect equilibrium of the centipede game is for the first player to move to end the game, and at the same time we may give little weight to this analytical conclusion as an empirical prediction. We are motivated, therefore, to seek evidence concerning what happens when individuals do bargain in the fashion of alternating offers.

<sup>e</sup> There are some mathematical niceties to be observed to make sure that everything is well-defined. And the definition of subgame perfection is not entirely straightforward. We give here a synopsis only. The reader should consult Perry and Reny (1989) for details.

Ochs and Roth (1989) give a survey and some fresh results of their own concerning experiments with alternating offer bargaining. In the papers they survey and in their own experiments, the bargaining protocols used are different from Rubinstein's in one important way: In all the experiments, time horizons are very short — sometimes two, sometimes three, and (very rarely) five rounds of bargaining at most. That is, one player makes an offer, the second may counteroffer, and the first may, perhaps, counter-counteroffer, but that is all. If the third offer is not taken, the game is over and neither side wins anything.

This doesn't compromise the theory much at all; if anything, it makes it a good deal simpler. In finite horizon, alternating offer bargaining, if you accept the complete information bargaining model, backwards induction gives you the answer quite quickly.<sup>2</sup> (See problems 6 and 7.) So it is the predictions of backwards induction in the context of bargaining that are being tested.

Ochs and Roth were led by the data they reported to the conclusion that whatever drives the outcomes in alternating offer bargaining is something more complex than the theoretical considerations provided by backwards induction. In particular, notions of equity intrude: In cases where backwards induction yields a very unequal split, proposals for splits that are so unequal are often rejected, *even though by rejecting the offer the rejecting party is giving up money for sure.*

For example, imagine two rounds of bargaining according to the following rules. Player 1 can propose any split she wishes of \$10.00. Player 2 can accept or reject. If player 2 rejects, he can propose any split that he wishes, except that \$9.00 of \$10.00 is "removed"; he can propose a split of \$1.00. (This is like a discount factor of .1.) If player 1 rejects player 2's proposal, the two walk away empty-handed. Backwards induction in this case tells us that player 2, in the event that he is given the opportunity to propose a split, proposes to take \$1.00 or \$.99, to which player 1 accedes.<sup>3</sup> As player 2 can't possibly do better than \$1.00 in the second round, player 1 in the first will propose that she will take \$9.00 or perhaps \$8.99, which he will accept. But experimental results don't conform to this prediction. In cases where player 1 proposes to take \$8.99, player 2 sometimes rejects, presumably on grounds that player 1's "excessive greed" should not be rewarded.

Of course, this sort of consideration puts bargaining in quite a different light from that cast by the theories described above. If player 2 in the

<sup>2</sup> The theory of finite horizon, alternating offer bargaining is developed in Stahl (1972).

<sup>3</sup> Because there are "ties" in the game tree and money is not really perfectly divisible, the predictions are not quite sharp.

situation described has some personal notions of what constitutes equity or excessive greed, and if player 1 is unsure what those notions are, then player 1's initial offer will have as much to do with her assessment about player 2's notions as with the amount of money that will be left on the table if player 2 is allowed to counteroffer. Insofar as one player will not know for sure what his rival views as "greed" or "equity," the simple models of complete information that we have analyzed are missing important pieces of the puzzle.

Moving beyond the experimental evidence so far collected, imagine that we stage infinite horizon bargaining experiments where subjects' discount rates are different. How would we go about controlling discount factors in the laboratory? One experimental procedure that could be followed, which gives discount rates .99 and .97 to the two bargainers, is easily described. First (and for reasons to be explained later), rather than have players bargain over some monetary sum directly, we have them bargain over a "unit of probability." If they split this unit with .643 going to player 1 and .324 going to player 2 (which leaves "probability" .033 on the table), then player 1 is given a .643 chance of winning some monetary prize, and player 2 is given a .324 chance of winning some (other) prize. From the point of view of the theory, it doesn't matter what these prizes are or that they are the same, or even that each player knows what prize the other player is going to get; in essence, as long as winning the prize is better than not, players are bargaining over expected utility, and the theory above applies. But to be definite, we will assume that the players know each others' prizes, know that each knows this, and so on. Players negotiate via computer terminals. Player 1 types in an initial offer, which appears on 2's screen. Player 2 can accept 1's offer or can propose a counteroffer. Player 1 can accept 2's counteroffer or can propose a counter-counteroffer, and so on. And to simulate discount rates, instruct players as follows (for example): Player 1 makes the first offer —  $x_1$  for herself, and  $x_2$  for player 2, where  $x_1 + x_2 \leq 1$ . If player 2 accepts, then player 1 is given an  $x_1$  chance of winning her prize, and player 2 is given an  $x_2$  chance of winning his. Or player 2 can counteroffer, proposing  $y_2$  for himself and  $y_1$  for her, where  $y_1 + y_2 \leq 1$ . If player 1 accepts this, player 2 is given a  $.97y_2$  chance of winning his prize, and player 1 is given a  $.99y_1$  chance of winning hers. Or player 1 can counter with  $z_1$  and  $z_2$ . If this is accepted, player 1 is given a  $.99^2z_1$  chance of winning her prize, and player 2 is given a  $.97^2z_2$  chance of winning his, and so on. (Of course, if players value the time they spend in the laboratory bargaining, then we have not completely controlled for the costs they expend in back-and-forth bargaining. Perfect experimental controls are not going to be possible.)

If we run the experiment just described, with (say) the prize of player 1 being \$20 and the prize of player 2 being \$5, the theory predicts that the players will settle on player 1 getting approximately a .75 chance of her prize and player 2 getting a .25 chance of his.<sup>4</sup> If we change the discount factors so that player 1 "discounts" with .97 and player 2 with .99, but we keep the prizes the same, the theory predicts that the agreed-to probabilities will reverse. Since to my knowledge this experiment hasn't been run, it is not possible to say what will result. But my personal conjecture is that the results will not confirm the theory.

Now consider the following elaboration, in the spirit of Roth and Schoumaker (1983). Have the players bargain via computer and exactly as in Roth and Schoumaker (1983) train the parties to expect a particular split. By having players first play against computer programs, train some players to think that a .2-.8 split is the "right" thing to do (where the .8 goes to player 2 because his prize is smaller), and train others to think that a .5-.5 split is what one does in this case. Do this under a number of alternating move protocols, with discount factors varied (but kept close to one). This elaboration is offered to propose the hypothesis that "equilibrium expectations," developed by the conditioning phase, will organize the data of subsequent bargaining much better than will the theoretical predictions based on models of complete information. "Institutional details" concerning who has which discount rate and what is the precise protocol of bargaining will be swamped by expectations as to what one can expect to get and what one expects must be given in order to come to agreement.

### 15.5. Models with incomplete information

Assuming the conjecture just offered is correct, the theorists' response is clear: Move beyond models of complete information. By moving to models with incomplete information, one hopes that the extreme and un-intuitive sensitivity of equilibrium outcomes to "small" changes in the bargaining protocol that we saw above will disappear.

At the same time, a striking feature of the Rubinstein equilibrium is that agreements are always Pareto efficient; in particular, agreement is

<sup>4</sup> We have players bargaining over probabilities so that they can have different monetary valuations for the marginal unit over which they bargain. We could as well have had them bargaining over one hundred chips, with different dollar:chip exchange rates. That is, each chip could be worth \$.20 to player 1 and \$.05 to player 2. But on theoretical grounds, having them bargain over probabilities is cleaner, because attitudes toward risk are not impounded into the experiment. See Roth (1985) on this general point.

immediate. Recall the "abridgement" from standard textbooks with which we began this chapter and in particular the remark that one might obtain inefficient trades "because of the bargaining process." We see nothing like this in the Rubinstein equilibrium; we see no inefficiencies of any sort. Given the prevalence of labor disputes and other forms of protracted (and, presumably, wasteful) negotiations in reality, it seems that the Rubinstein equilibrium is missing something. The theorist's response to this has been that what is missing is something from the game that is formulated, where the preferences of both sides are common knowledge; what is needed to get delay to agreement and other sorts of inefficiencies is some sort of incomplete information.

Accordingly, much of the recent literature on bargaining and especially on alternating offer bargaining concerns models with incomplete information. In these analyses, it has been typical to base the incomplete information on something tangible about the bargainers. For example, we might imagine the two parties negotiating on a selling price for some asset, where the value of the asset to the potential buyer is not known to the seller. If the buyer values the asset highly, he (the buyer) is more impatient to conclude negotiations; this is meant to make the buyer "weaker" in any negotiations. There has also been some investigation of models with two-sided incomplete information, where in addition to uncertainty about the buyer's valuation of the asset the value of the asset to the seller is unknown to the buyer.

These formulations do not directly address what we see in the experimental literature and what we have conjectured will be found in subsequent experiments. There is no "tangible uncertainty" at work in the Ochs and Roth experiments. Rather, what seems to be at work are the beliefs of one player concerning equity and greed and the assessment of the other player about the beliefs of the first. It is not clear a priori that the forms of incomplete information that have been investigated so far will speak adequately to effects driven by these other sorts of considerations. But any opinions about this are dangerous; this is an area of active and ongoing research, and opinions rendered here will probably be obsolete in very short order.

In any event, the analysis of bargaining with incomplete information very quickly becomes very complex (try the last piece of problem 8 and problem 9 if you want evidence) — a level more complex, for example, than the analysis of the centipede game in chapter 14. (This is so because while one has the same back-and-forth structure as in the centipede game, now what a player can do in his turn, namely make an offer, is a lot richer than the simple binary choice present in the centipede game.) Hence we

will not pursue this topic further here; the literature awaits the interested reader.

### 15.6. Bibliographic notes

The development of noncooperative models of bargaining has been one of the most important and active areas in microeconomics and game theory in the past decade. There are far too many important papers to list them all here. Begin with Nash (1950), who notes the basic problem of indeterminacy in simultaneous offer bargaining. And the classic paper on alternating offer bargaining, Rubinstein (1982), should not be missed. But rather than give references beyond these, I suggest that the reader continue with Osborne and Rubinstein (forthcoming), which provides a comprehensive survey of the field, including work on bargaining with incomplete information. Included also in Osborne and Rubinstein is a topic we have not discussed at all — connections between bargaining theory and the “institutional foundations” of Walrasian (price) equilibrium.

For results in bargaining experiments, the cited papers by Ochs and Roth (1989) and Roth (1983) are good places to start.

Given the lack of sharp predictions in simultaneous offer bidding, much attention has been paid to the axiomatic analysis of bargaining. In a typical analysis, one poses a number of desirable properties for a function that will associate to each bargaining problem out of a class of bargaining problems a “solution,” and then one sees whether those properties (a) can be satisfied and (b) pin down the “solution.” The classic example of this sort of analysis is the Nash bargaining solution, as developed originally by Nash (1953). This form of analysis, which is more in the style of cooperative game theory, is surveyed in Roth (1979).

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### 15.7. Problems

- 1. Consider two firms, one of which is the sole producer of a factor input that is used solely by the second. The producer of this good is characterized by increasing marginal costs of production for the good in question, and the user obtains decreasing marginal benefit (measured in terms of impact on profits) as it increases the scale of its use of this good. Both firms seek to maximize their profits. (You may assume that the marginal cost to produce the first unit of this good is less than the marginal value to the user of the first unit, and for some large enough amount the marginal cost exceeds the marginal value. You may also assume that marginal cost and marginal benefit functions are continuous.)

We asserted at the start of this chapter that this is a special case of an Edgeworth box. How is it special? (Hint: Your first task is to identify the second good.)

- 2. Consider the bargaining protocol described in section 15.1 — one round of simultaneous demands, and then (if a deal is not struck in the first round) simultaneous declarations that players stand firm or accede.

(a) Build a subgame perfect, pure strategy equilibrium for this bargaining protocol in which agreement is not reached on the first round, and the



equilibrium split is  $n$  to player 1 and  $100 - n$  to player 2. Be careful in describing the strategies; this isn't hard, but it isn't trivial either.

(b) Show that for any pure strategy equilibrium that results in agreement, no money is left on the table. Show this as well for variations (a) and (b) in section 15.1. Describe the pure strategy equilibria that leave money on the table (if any exist). Are the equilibria you describe subgame perfect?

(c) Build a subgame perfect equilibrium for this bargaining protocol in which money is left on the table and there is agreement with positive probability. (This will involve mixed strategies, according to part [b].)

■ 3. (a) Provide a proof of proposition 2. Before doing this, you might wish to explore part (c).

(b) Suppose that the Pareto frontier of  $V^*$  is differentiable. What happens to the equilibrium in proposition 2 as  $\delta$  approaches 1? (What happens if the frontier of  $V^*$  is not differentiable?)

(c) In the context of proposition 2, suppose that  $V^*$  is the unit square. That is,  $V^* = \{(v_1, v_2) : 0 \leq v_1 \leq 1 \text{ and } 0 \leq v_2 \leq 1\}$ . What are the subgame perfect equilibria for alternating offer bargaining in this case?

■ 4. For each of the five variations on the basic Rubinstein model outlined in section 5.3, show that there is a unique subgame perfect equilibrium that gives the outcomes claimed. (You were given quite a good start on variation [5]!)

■ 5. Suppose we have the basic Rubinstein model, except that for reasons of feasibility, the only splits of \$1 that can be proposed are those that involve integer multiples of \$.01. Suppose that  $\delta = .9999$ . What happens to proposition 1?

■ 6. Take the simple Rubinstein alternating offer structure where both players have the same discount factor per bargaining period. Imagine, though, that there are only two rounds of bargaining. Player 1 makes an offer, which 2 can accept or reject. If player 2 rejects player 1's original offer, he can make a counteroffer, which player 1 can accept or reject. What is (are?) the subgame perfect equilibrium of this game?

■ 7. Redo problem 6, but assuming that precisely  $N$  rounds of bargaining are permitted. What is the limit of the equilibrium outcome as  $N$  approaches infinity? Repeat this exercise for each of the five variations on the basic Rubinstein model outlined in section 5.3.

■ 8. Suppose we have one-sided bargaining. Player 1, say, proposes a split which player 2 can accept or reject. If player 2 rejects, player 1 can propose another split for 2's consideration, and so on, until an agreement is reached. Each time player 1 makes an offer, the prizes eventually achieved are discounted by  $\delta$ , for some  $\delta < 1$ .

What is (are?) the subgame perfect equilibrium of this game? Do you believe this as an empirical prediction? (A real challenge: Build a model of this sort of bargaining, where player 2 has an "equity threshold"; he won't take any offer below this threshold, and player 1 is unsure what 2's threshold is.)

■ 9. Consider the following incomplete information variation on the Rubinstein model. Two players are bargaining over splitting \$10.00. They take turns making offers, with discount factor  $\delta = .999999$  applied every time an offer is rejected. Player 1 seeks to maximize her expected payoff. There is incomplete information about player 2. Player 1 assesses probability  $\pi$  that player 2 seeks to maximize his expected payoff, and she assesses probability  $1 - \pi$  that he will ask for \$8.00 precisely every time he is able to offer, and he will resolutely turn down any offer that leaves him with less than \$8.00.

(a) Construct a sequential equilibrium for this bargaining game for the case  $\pi = .2$ . (This is hard, but not impossible.)

(b) Construct a sequential equilibrium for this bargaining game for the case  $\pi = .8$ . (This is harder.)

(c) Construct a sequential equilibrium for every possible value of  $\pi$  (and, in particular, for  $\pi$  very close to one). (Good luck!)

*part IV*

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*Topics in information  
economics*

## chapter sixteen

# Moral hazard and incentives

### Prologue to part IV

We turn now to the subject of *information economics*. Information economics is a broad subject with many variations and subtopics, and we will not do anything close to full justice to it here. In this chapter, we consider the problem of *moral hazard*, where one party to a transaction may undertake certain actions that (a) affect the other party's valuation of the transaction but that (b) the second party cannot monitor/enforce perfectly. A classic example here is fire insurance, where the insuree may or may not exhibit sufficient care while storing flammable materials. The "solution" to a problem of moral hazard is the use of *incentives* — structuring the transaction so that the party who undertakes the actions will, in his own best interests, take actions that the second party would (relatively) prefer. For example, fire insurance is often only partial insurance so that the insuree has a financial interest in preventing a fire.

In chapter 17 we look at problems of *adverse selection* where one party to a transaction knows things pertaining to the transaction that are relevant to but unknown by the second party. Here a classic example is life insurance, where the insuree may know things about the state of her health that are unknown by the insurer. The "solution" to problems of adverse selection is *market signaling*, where the party in possession of superior information signals what she knows through her actions. For example, an insurance company may offer life insurance on better terms if the insuree is willing to accept very limited benefits for the first two or three years the policy is in effect, on the presumption that someone who suffers from ill health and is about to die (or has substantial probability of dying) is unwilling to accept those limited benefits.

Whenever there are informational problems of these or other sorts it is natural to ask, What is the best contract that can be devised? We will investigate optimal incentives design in a simple setting in this chapter, but a more general attack on optimal contract and mechanism design stressing cases of adverse selection, which makes use of the *revelation principle*, will be given in chapter 18.

These three topics are all important, but they do not even begin to exhaust the important category of models and concepts from information economics. Particularly noticeable by their absence are models of *optimal search* and *coordination failures*, where parties desirous of making a particular exchange must search for potential trading partners and where the need for search discourages certain otherwise beneficial trading activity, and models of *rational expectations*, where some parties have information that would be useful to others, information that is conveyed at least in part by equilibrium prices themselves.

All these situations and others as well fall under the broad rubric of *information economics* because in each case the driving factor is the lack of information on the part of some market participants, whether about what others are doing, or what others know, or where the best trading opportunities are to be found. This feature was ignored in part II of the book (with a very few exceptions, notably in the analysis of price discrimination). Now this feature becomes the center of attention.

## 16.1. Introduction

As noted above, this chapter concerns transactions taken under conditions of *moral hazard* or, as it is sometimes called, *hidden action*.<sup>1</sup> We have already introduced one example: In the fire insurance business, an insurance company would want the insuree to store flammable materials carefully, keep quality fire extinguishers on hand, etc. To take other examples: If I lease a car from you that I return to you in three years, you want me to have it serviced regularly, to drive it carefully (no redlining), and so forth. If you hire me to do a particular arduous job, you want me to work hard at the tasks that are set for me.

In each of these examples, it is possible to monitor and enforce levels of care, or servicing, or effort. Insurance companies will send out inspectors, and some insurance contracts provide that no benefits will be paid if it can be shown that the insurer did not provide sufficient "due care."

<sup>1</sup> The terms of information economics, such as moral hazard, adverse selection, hidden action, hidden information, signaling, screening, and so on, are used somewhat differently by different authors, so you must keep your eyes open when you see any of these terms in a book or article. For example, we just equated moral hazard with hidden action; you will read elsewhere that there is a serious distinction between them. (My own opinion is that there is a distinction, but it is hardly serious.) I do not wish to subject you to a precise categorization, largely on the grounds that very interesting problems mix more than one form, and then how would we call them? As a consumer of the literature, you should pay less attention to these labels and more to the "rules of the game" the author specifies — who knows what when, who does what when.

## 16.2. Effort incentives: A simple example

A car lease contract may require that routine maintenance is performed. If I am working for you, you might hire monitors to observe my level of effort.

But in each case, perfect monitoring and enforcement may be impossible, and hence the transaction might be structured so that the party taking the action has relatively greater incentive to act in a way the second party prefers. The insurance company may only insure up to 90% of the building. If the leased car is sold after the lease period, the contract may call for the party that leased the car to get a share of the proceeds from the sale.<sup>2</sup> And you might tie my compensation to some observable measure of how hard I work.

From the point of view of providing incentives, we would like to structure the transaction so that the party who is taking the "hidden action" bears fully the consequences of his actions. The insurance company may refuse to give insurance; instead of leasing the car, you may simply sell it to me; you may pay me as a function of the output I produce, using (for example) a piece-rate system, where I am paid a set amount for each piece of output I produce. But in each of these cases there may be "inefficiencies" in such a contractual form: A company owning a warehouse, if closely held by a few individuals, may be less able to bear the risk of having the warehouse burn down than is an insurance company with many shareholders. A leasing arrangement may produce tax savings relative to a direct sale, owing to peculiarities in the tax system.<sup>3</sup> Piece-rate systems may be infeasible because the work involved is machine paced, or because the quality of workmanship as well as quantity may be important, or because the piece-rate system may subject the worker to risks that the worker is less well equipped to bear than is the firm for which he works. So a balance must be struck between providing incentives and exploiting all the other advantages of trade in a particular setting.

## 16.2. Effort incentives: A simple example

All these words and vague generalities may be a bit hard to parse, so let us turn to an illustrative example. Imagine a situation in which one party, called the *principal*, hires a second party, called the *agent*, to perform some task. The agent is drawn from a large population of similar

<sup>2</sup> You don't see this in most car lease contracts in the United States, largely because leasing is motivated by tax considerations, and such a contract would void the tax savings.

<sup>3</sup> You may wonder why such a tax system would be created, but that is well beyond the scope of this book.

agents and is willing to undertake this task as long as his net utility from performing the task is (at least as large as he can get at his next best opportunity; we refer to this level of utility as the agent's *reservation level of utility*). The agent, if and when he is hired, must then decide whether to work hard or not on this particular job. Hard work is not to this particular agent's taste, and so, all other things equal, he would prefer not to work hard. Whether this agent works hard or not determines the value to the principal of having this agent work. If the agent is not going to work hard, then the principal will get very little from the deal — so little that it is not worth her while to pay the agent his reservation wage (a wage high enough so that combined with not working hard the agent's net utility exceeds his reservation level of utility). But if the agent does work hard, then the principal will get enough out of the transaction to make it worthwhile for both sides.

Specifically, suppose that the agent's (reservation level of utility is (completely arbitrarily, as the scale doesn't mean anything) 9. The agent derives utility from how much he is paid,  $w$ , and how hard he works,  $a$ . The level of  $a$  can be "hard" or "high," which we denote  $a = 5$ , or it can be "not hard" or "low," denoted by  $a = 0$ . The agent's overall (von Neumann-Morgenstern) utility from  $w$  and  $a$  is given by

$$U(w, a) = \sqrt{w} - a.$$

If the agent works hard, the accomplished task is worth \$270 to the principal. And if the agent doesn't work hard, the task will be worth only \$70 to the principal.

To get the agent to work at a low level of effort, the principal must offer the agent wages high enough so that  $\sqrt{w} \geq 9$ , or  $w \geq \$81$ . Since the job, if done with low effort, is worth only \$70 to the principal, there will be no deal of this sort.

But if the agent can be persuaded to expend high effort, then the principal must offer the agent wages high enough so that  $\sqrt{w} - 5 \geq 9$ , or  $\sqrt{w} \geq 14$ , or  $w \geq \$196$ . Since the principal values the job done with a high level of effort at \$270, this is a worthwhile deal for her. She should try to arrange this.

How? Perhaps she should write a contract that offers the agent \$197 (be generous!) for performing the task, and trust that the agent will indeed work hard. Trust is nice and can work (although we might try to think why it does), but the title of this chapter is "moral hazard," so we assume this won't work. If the principal offers this agent a fixed fee contract, we

assume that the agent will take the money, put in low effort, and leave the principal paying \$197 for a task that is worth to her only \$70.

Another possibility is to offer a contract that calls for the agent's pay to depend on how much effort he puts in. The contract might read something like "I (the principal) agree to pay the agent \$197 if he works hard, and \$25 (say) if he doesn't." If this contract is enforceable, then the agent will optimally work hard — doing so gives him a utility of (slightly more than) 9, while taking the contract and not working hard would net  $\sqrt{25} - 0 = 5$ , which is less than his reservation utility level.

But is this contract enforceable? Suppose the agent signs it, doesn't work hard, and then claims that he did work hard. The principal will need some tangible evidence that he didn't work hard, evidence that will stand up in some legal proceeding. It may be that no tangible evidence exists, or even that the principal is unable to see any conclusive evidence about how hard the agent worked. (This may seem a bit strange to you, because the principal's valuation of the task depends on how hard the agent worked. But, in a bit, we'll see why there might be no conclusive evidence.) There may be conclusive evidence, but not evidence that a court of law, or whoever is going to enforce this contract, would accept as evidence. Or it may be that court costs are too high to make one side wish to enforce such a contract. For any of these reasons, writing this sort of contract might not work.

We could have the principal monitor the agent's efforts, with a contract that gives the principal the right to fire the agent (at a low level of severance pay) if he doesn't work hard. Of course, the principal might still have to go to the courts to justify a termination, so the problems just mentioned might still be present. And there will be some cost of monitoring the agent; the principal, presumably, has better things to do with her time. The principal, if her time is especially valuable, might think of hiring some third party to monitor the agent. But then she'll have to pay this monitor, and she might be concerned that the monitor and the agent will collude against her; the agent might offer the monitor a bribe if the monitor will say that the agent did indeed work hard. Finally, the agent mightn't sign any such contract, fearing that the principal will fire him just before the task is completed. (In chapter 20, we'll consider how the principal's concern for her reputation among workers in general might reassure this particular worker on this count.)

Even if the principal cannot tie the worker's wage directly to his level of effort, the principal might be able to find some indirect measure of effort to which wages can be tied in a contract that will stand up in court. To give an example of this, we have to be a bit more specific about what this agent